

Health measures for the prevention project

Archived: the search that chose the measure

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Archived exploratory work. This note records the search over candidate health measures, not the measure the project uses. The search ended with **P-LIM3+CC**, reported as θ , h and a ten-deficit index; what that measure is, how it is built and what it delivers is `measuring_health/health_measure_construction.tex`. Read this note for the alternatives that were considered and why they lost — 85 candidates, in `redo_concepts/baseline_measure_comparison.xlsx`. Its build scripts have moved to `data_cleaning/archive/` and still run; nothing in it has been rewritten to match the choice, so where the two notes differ the construction note is current.

Built and validated by `data_cleaning/archive/03--07` and `descriptives/01--07`; every number here is printed by those scripts’ self-validation output. This note records which UKHLS health variables the project uses, how each measure is constructed and why, what the new graded-response measures deliver against everything that came before them (the original physical GRM, PCS/MCS, the physical-only composite, SF-6D, GHQ), and how the ever-diagnosis, dimensionality and ceiling questions resolve.

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1 The variable inventory

Measurement models are restricted to items asked in (nearly) every wave, so a score means the same thing wherever it lands in the panel. Presence was verified against the raw headers of all fifteen `indresp` files, labels against the UKDA-6614 dictionaries.

Variables	Waves	Used
SF-12, all 12 items (self-completion preferred)	all 15	both GRMs; PCS/MCS re-build; SF-6D
GHQ-12 items (<code>scghqa..1</code> , 4 categories)	all 15	mental GRM
Impairment list <code>disdif1..12</code> , gated on <code>health</code>	all 15	physical GRM (FUNC testlet)
Long-standing illness (<code>health</code>)	all 15	gate for <code>disdif</code> ; descriptive
Diagnosed conditions (<code>hcond*</code> families, codes 1–17)	all 15 (§2)	condition items; sensitivities
Health satisfaction (<code>sclfsat1</code>)	all 15	validation outcome only
SWEMWBS; ADL module; loneliness	1/4/7/10/13; 7/9/11/13; 9–15	excluded: fail the every-wave rule
GP visits, out-patient attendance, in-patient stays (<code>h12gp</code> , <code>h12hop</code> , <code>hosp/hospd</code>)	7–15	criterion-validity outcomes
Height/weight (self-report)	wave 1 only	excluded (biomarkers are UKDA-7251, not held)

2 Chronic conditions

2.1 Construction

UKHLS asks the full “ever diagnosed” inventory (`hcond1..17`) at a person’s *first* interview; later waves record conditions newly reported since the last interview under a name that changes with questionnaire redesigns — `hcondn{i}` in waves 2–9; from the wave-10 redesign, `hcondcode{i}` and `hcondncode{i}`, both present in waves 10–15 and both used — all dictionary-verified. The running indicator (*ever had condition i by wave w* $\Leftrightarrow$ *reported in any family at any wave $\leq w$*) is monotone and defined from the first inventory onward (76% of person-waves). Only the wave-1 code list is used, so the definition is identical in every wave. Per-condition indicators, diagnosis ages and group counts live in `chronic_conditions.parquet`; nothing is collapsed before modelling.

This corrects two defects in the earlier EIT count, which used `hcondns*` (“still has *newly diagnosed* condition”) as its new-report family: that family has no codes 6 and 7 at all — every in-panel heart attack and stroke was missed — and it drops new diagnoses reported as already resolved. Our count is $\geq$ the EIT count on 100.000% of 458k person-waves (exact on 94.2%; 26,516 documented additions, concentrated in MI, stroke and cancer).

2.2 Why “ever”, not “current”

The “still has” family (`hconds##`) is asked *only at the wave a condition is reported*: at wave 6, of 5,090 ever-asthma respondents, ~ 570 were routed to it. A per-wave “current” variable would be $\sim 90\%$ carry-forward imputation. Beyond measurability, self-reported remission conflates control with cure — managed hypertension and diabetes read as “no longer have”, the mechanism behind the under-reporting found against measured disease (Johnston, Propper & Shields 2009) and GP records (Kriegsman et al. 1996: good agreement for diabetes, poor for heart disease and arthritis). The monotone ever-indicator matches the deficit-accumulation tradition (Mitnitski, Mogilner & Rockwood 2001; Searle et al. 2008). Its genuine cost — old diagnoses shadowing current health — is quantified in §5, not assumed away.

2.3 Why grouped, not seventeen indicators

Grouping is the testlet logic applied to diagnoses. CHD, angina, MI, CHF and stroke are one disease process reported several ways; as separate items they would violate local independence exactly as the `sf2a/sf2b` pair did before the original GRM collapsed them (Wainer & Kiely 1987; Yen 1993). Rare conditions (1–2% prevalence) also make weak, unstable IRT items — minimum prevalence is an explicit deficit-inclusion criterion in Searle et al. (2008) — and groups keep a stable clinical meaning while the code list grows (cf. Barnett et al. 2012). An ordinal within-group count (0/1/2+) keeps a severity gradient the way `PF = sf2a + sf2b` does. The groupings follow the disease process rather than the questionnaire. **CVD** collects the several names UKHLS gives to one underlying process — atherosclerosis and its consequences — which is why a respondent can plausibly report three of its five items at once. **METAB** keeps the two long-term managed conditions apart from those acute events: hypertension and diabetes are treated rather than survived, and both are far commoner. **RESP** spans the chronic airway diseases; **MSK** is the single commonest cause of physical limitation in the data; **CANCER** stays alone because it is a distinct process and the one most relevant to mortality. **OTHER** pools the remainder, none of which reaches 5%, so that no item has to be estimated on a prevalence near 1%.

The seventeen conditions, their groups, and what each one is:

Group	Condition	What it is	Ever
CVD	angina	chest pain when the heart's blood supply cannot meet demand, typically on exertion	2.9%
	heart attack	a blocked artery kills part of the heart muscle	2.7%
	coronary heart disease	the arteries feeding the heart narrow with fatty deposits	2.4%
	stroke	blood supply to part of the brain is cut off, damaging brain tissue	2.1%
	congestive heart failure	the heart pumps too weakly, causing breathlessness and fluid retention	0.8%
	<i>any of the above</i>		7.4%
METAB	high blood pressure	blood pressure persistently above the healthy range	21.4%
	diabetes	the body cannot regulate blood sugar properly	7.2%
	<i>either</i>		24.7%
RESP	asthma	airways narrow and inflame in episodes, causing wheeze and breathlessness	15.4%
	chronic bronchitis	long-term airway inflammation with persistent cough and phlegm	2.0%
	emphysema	the lungs' air sacs are permanently damaged, so breathing is laboured	0.8%
	<i>any of the above</i>		16.7%
MSK	arthritis	painful inflammation and stiffness of the joints	16.2%
CANCER	cancer or malignancy	any malignant tumour	5.1%
OTHER	hypothyroidism	an under-active thyroid gland, slowing the body's metabolism	4.2%
	liver condition	any long-term liver disease, such as cirrhosis or hepatitis	2.2%
	hyperthyroidism	an over-active thyroid gland, speeding the metabolism up	1.3%
	epilepsy	a tendency to recurrent seizures from abnormal electrical activity in the brain	1.3%
	<i>any of the above</i>		8.1%
DEPR[†]	clinical depression	a diagnosed depressive disorder, as distinct from transient low mood	8.2%

“Ever” is the share of person-waves at which the condition has been reported at or before that wave, among the 458,468 person-waves with an inventory. Conditions are listed within group by prevalence. [†]DEPR is the one group assigned to the mental bank rather than the physical one.

2.4 Diagnosis timing

Every inventory report carries `hconda##` (age told); every in-panel report is dated by its wave: **100.0% of ever-flags carry a usable diagnosis age**, so recency-restricted variants are exact.

The stock is old: median time since diagnosis 11 years ($p_{25} = 5$, $p_{75} = 21$); at ages 65–80, 54% of flags are >10 years old, 26% >20.

3 The graded-response measures

3.1 Why separate physical and mental banks

The measurement model must stay a static map from responses to a score (dynamics are estimated afterwards), and PROMIS practice is separate unidimensional banks per domain (Cella et al. 2010). Two clean banks also make the physical–mental correlation an *estimate* rather than an artifact: the SF-12’s orthogonal scoring forces $r(\text{PCS}, \text{MCS}) = 0$; our corrected promax on the subscales gives 0.59; published correlated-scoring estimates are 0.62 (Farivar, Cunningham & Hays 2007) and 0.71–0.73 (Tucker et al.); the two GRM thetas correlate **0.457**. Cross-loading SF-12 items are assigned by this repo’s own UK wave-1 varimax loadings (phys/ment): PF .86/.14, RP .85/.25, BP .75/.17, GH .72/.27 → physical; MH .09/.89, RE .26/.82, SF .48/.64 → mental. **VT (energy, .55/.44) enters neither bank** (PROMIS likewise treats fatigue as its own domain); cognitive and sensory disdif items are separate WHODAS-type constructs and are excluded from both.

3.2 The banks

Every item is coded so category 1 is worst health.

Spec	Items
P-FUNC	GH, PF, RP, BP (the validated original testlets) + FUNC: count of physical disdif limitations {mobility, lifting, dexterity, coordination, personal care}, 0/1/2/3+; health =“no” is a valid structural zero (module skipped by design)
P-FULL	P-FUNC + six ever-diagnosis group items (CVD/METAB/RESP 0/1/2+; MSK/CANCER/OTHER binary)
P-REC	as P-FULL, counting only diagnoses within the last 10 years
P-TIMED	as P-FULL with each group split into recent ($\leq 10y$) and stale ($>10y$) items — the model <i>estimates</i> the recency weighting (§5)
MENT	MH ((6 – sf6a) + sf6c – 1, 9 levels), RE, SF, GHQ-positive and GHQ-negative wording testlets (19 levels each), ever-depression
COMBINED	P-FULL $\cup$ MENT: all seventeen good-coverage items in one bank (§6)

GHQ-12 is validated internationally (Goldberg et al. 1997) but its factor-analytic “dimensions” largely reflect negative-wording method effects (Hankins 2008, contra Graetz 1991). The pre-specified rule — fit item-level; collapse by wording if Yen’s Q_3 shows positive within-wording dependence — *triggered* (within-wording Q_3 max +0.26), the same cure the original GRM applied to the PF/RP pairs.

3.3 Estimation and diagnostics

Same machinery as the exactly-replicated original (Samejima 1969; marginal ML by EM over response patterns, 121-point quadrature, pooled latent $N(0, 1)$; multigroup by single year of age 20–90 for μ_a, σ_a). Samples: P-FUNC 497,423 person-years; P-FULL/P-REC/P-TIMED 387,599 (condition items exist from first inventory); MENT 379,462; COMBINED 376,019.

	P-FUNC	P-FULL	P-REC	MENT	COMBINED
max positive Q_3	-0.048	+0.144	+0.080	+0.153	+0.582 [†]
EAP identity (var + $E[\text{psd}^2]$)	1.000	1.000	1.000	1.000	0.999
r with original GRM θ	0.994	0.985	—	—	0.917

[†]the mental-pair clustering that is itself the two-dimensionality evidence; the physical side stays clean (§6).

Discriminations: the functioning items dominate — FUNC $a = 3.3$ (P-FUNC: GH 2.0, PF 3.7, RP 3.4, BP 2.4) — while the condition-group items are weak (P-FULL: MSK 1.30, CVD 1.23, METAB 0.86, OTHER 0.65, CANCER 0.60, RESP 0.41): once functioning is measured well, a diagnosis adds comparatively little about *current* physical health — with arthritis, a condition that *is* impaired functioning, the strongest of them. On the mental side, the GHQ negative-wording testlet ($a = 3.15$) and MH (2.75) carry the bank; the ever-depression diagnosis is again weak (0.97).

3.4 Routing: who is asked what

Two items in the physical bank are asked conditionally, and both deserve stating plainly.

The functioning testlet is gated on long-standing illness. The `disdif` impairment list is only put to respondents who report a long-standing illness, so FUNC takes its best category structurally for the 66.2% of scored rows with `health == 2`. Only 33.8% of rows actually answer the module, and among those FUNC is well spread (17/14/18/51% across its four levels). The item is therefore 83.4% top category overall: it says nothing about differences among the healthy and everything about gradations among the ill — which is exactly what its estimated hurdles (-1.83 , -1.42 , -1.05) and its high discrimination ($a = 3.4$) encode. This is a feature for our purposes, because the ill tail is where the other items have run out (§7), but it does mean FUNC is partly a re-reading of the long-standing-illness question. It is not double counting in a way the model minds: FUNC’s largest residual correlation with any other item is $+0.031$. A further 6,451 rows report an illness but leave the module unanswered; those are treated as missing, not as zero limitations.

The diagnosis items exclude the BHPS sample entirely. The condition inventory is asked at a person’s first interview, and wave 2 — where the BHPS continuers joined UKHLS — has no inventory at all. They are then never routed back to it: only 4.4% of the 16,204 people first interviewed at wave 2 ever receive one, against 94.8% of wave-1 entrants and 70.7% of wave-3 entrants. P-FULL therefore drops 14,721 people outright (17.4% of persons, 109,470 person-waves), and this is a survey-design exclusion rather than attrition.

Whether that matters depends on the use. On health it looks benign: the dropped people are indistinguishable from the kept on every measured dimension (mean GH 3.32 vs 3.32, PF 4.23 vs 4.24, FUNC 3.66 vs 3.67). On *panel structure* it is not benign at all: the excluded BHPS members are the longest-observed people in the study (7.4 observations each against 5.5 for those kept). Any diagnosis-bearing measure therefore discards precisely the respondents with the richest longitudinal information, which is a poor trade for trajectory work (§11).

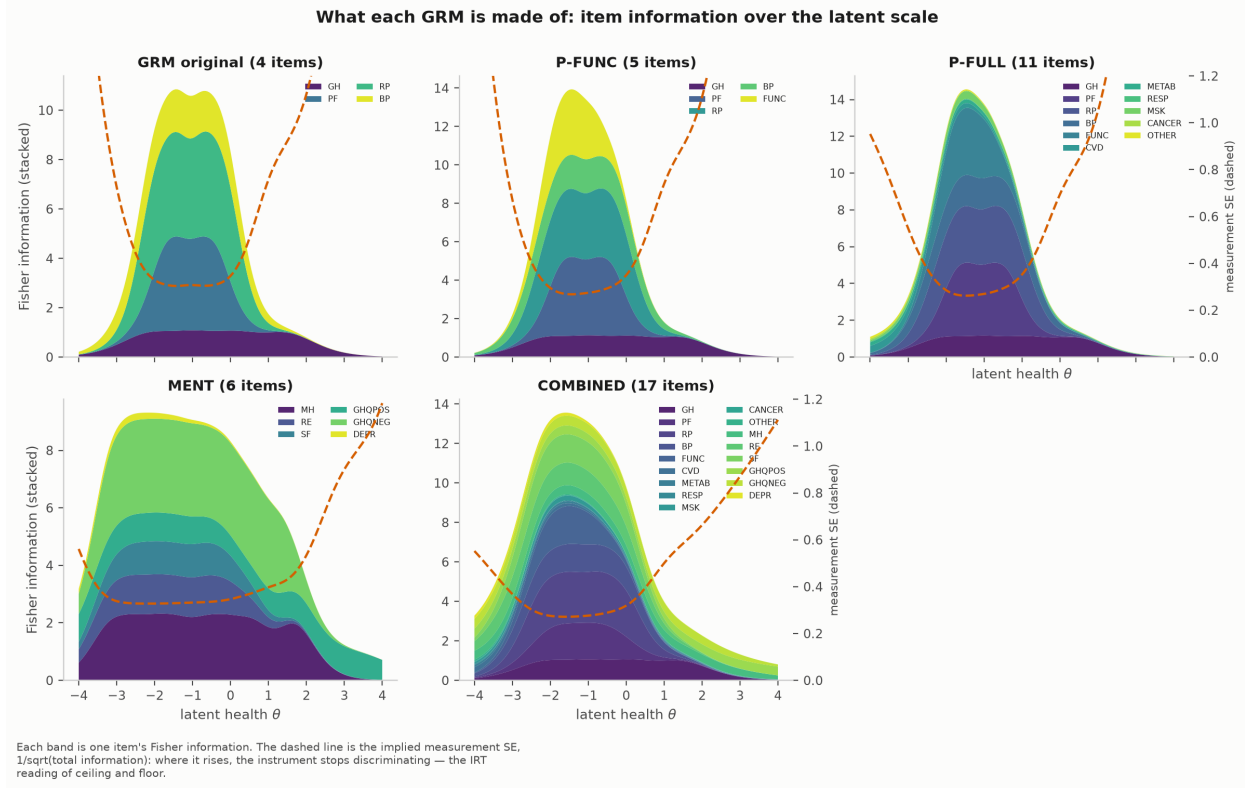


Figure 1: What each GRM is made of: stacked item information over the latent scale, with the implied measurement SE (dashed, right axis); each band is one item, ordered as in the legend. *Panels:* **GRM original** is the four-testlet physical measure replicated from the EIT note; **P-FUNC** adds the functioning testlet; **P-FULL** adds the six diagnosis groups; **MENT** is the mental bank; **COMBINED** is all seventeen items at once. *Physical items, from the SF-12:* **GH** general health (the single self-rated-health question), **PF** physical functioning (moderate activities, climbing stairs), **RP** role-physical (cutting down or being limited at work because of physical health), **BP** bodily pain; **FUNC** is the count of Equality Act physical limitations (mobility, lifting, dexterity, coordination, personal care). *Diagnosis groups:* **CVD** cardiovascular events, **METAB** diabetes and high blood pressure, **RESP** chronic airway disease, **MSK** arthritis, **CANCER**, **OTHER** the pooled remainder of thyroid, liver and epilepsy (§2.3). *Mental items:* **MH** mental health (calm; downhearted), **RE** role-emotional (cutting down or working less carefully because of emotional health), **SF** social functioning (health interfering with social activities), **GHQPOS** and **GHQNEG** the positively- and negatively-worded halves of the GHQ-12, **DEPR** ever-diagnosed clinical depression. **FUNC** adds a broad band of precision on the ill side; the condition items are visibly thin; GHQ-negative dominates the mental bank; the combined bank is tallest and widest but its SE still rises sharply above $\theta \approx +1$.

4 How the measures relate

Correlations over all person-years (descriptives/01):

	θ phys (FULL)	θ ment	θ combined
original GRM (4 testlets)	0.985	0.48	0.92
UKHLS PCS	0.904	0.23	0.76
physical-only composite	0.946	0.48	0.90
UKHLS MCS	0.27	0.882	0.58
GHQ-12 Likert	-0.38	-0.904	-0.64
SF-6D utility	0.71	0.799	0.88
chronic count	-0.58	-0.21	-0.51
θ phys $\times$ θ ment		0.457	

Three reads. The physical GRM is the object the project already validated, with functioning attached. The mental GRM is the MCS/GHQ construct without the orthogonality artifact. SF-6D correlates *more* with the mental theta (0.80) than the physical one (0.71) — and its closest cousin is the *combined* bank (0.88): the SF-6D was a mixed measure all along.

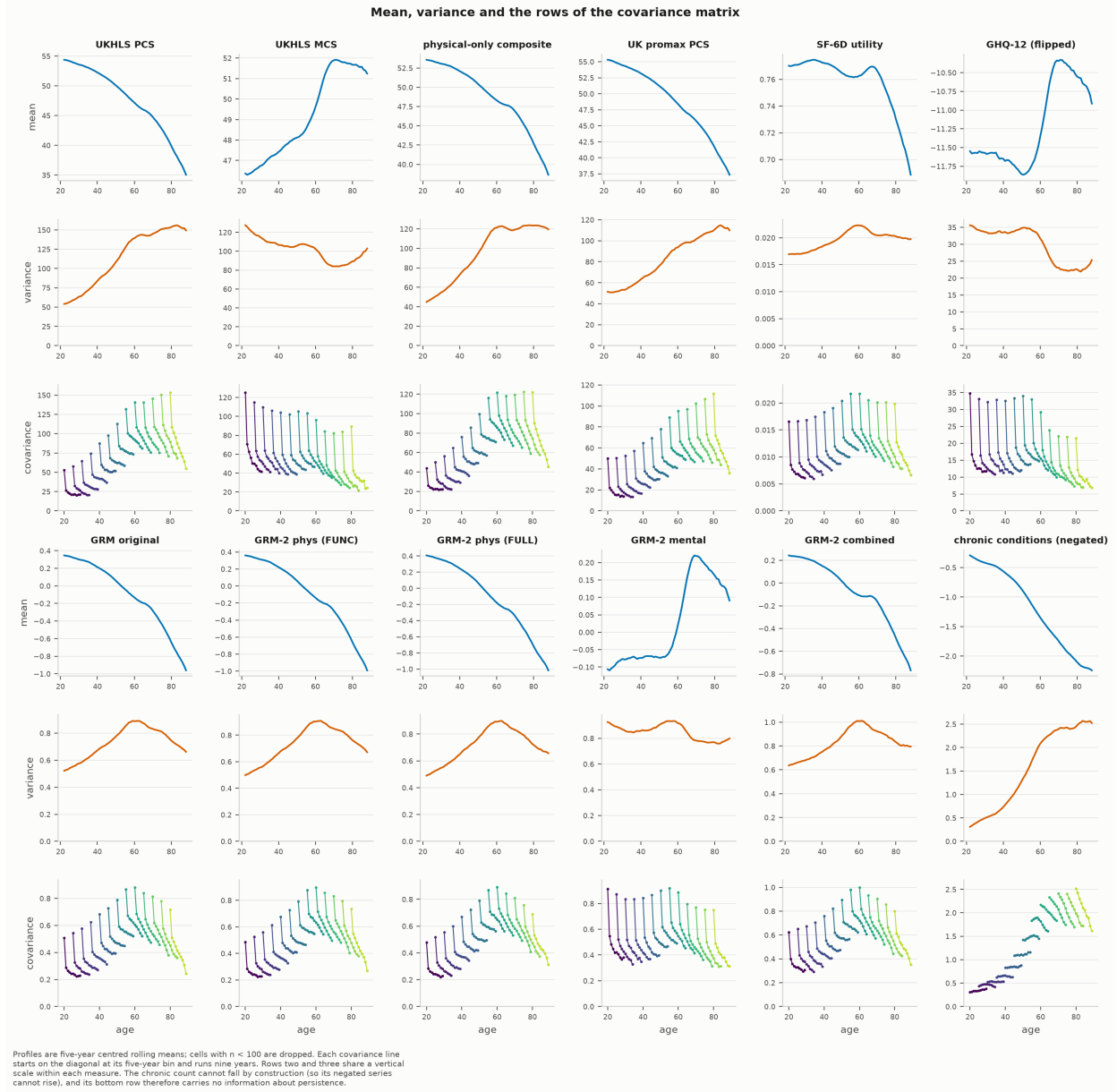


Figure 2: Mean, variance and the rows of the covariance matrix (the empirics note’s anatomy) for the full metric suite, two banks of six. GHQ and the chronic count are *negated*, so on every panel higher means better health. The mental family (MCS, GHQ, mental GRM) shows the textbook late-life wellbeing improvement; the physical measures share one anatomy — accelerating mean decline, rising-then-plateauing dispersion, fanning covariance rows; the chronic count gives the physical mean and variance reads from items fully disjoint from the SF-12, while its covariance row carries no persistence information because the count cannot fall (so its negated series cannot rise).

Two cautions on the mental hump, because on an auto-scaled axis it looks dramatic. It is modest in size: mean GHQ falls from 11.75 at age 55 to 10.24 at 70, which on a 0–36 scale with a standard deviation of 5.62 is 0.27 sd; the axis spans less than two points, which is what makes it read as a step change. And *most of it is composition rather than ageing*: among the 1,461 people observed in

both windows, mean GHQ improves by only 0.44 points between ages 53–57 and 68–72, against a cross-sectional gap of 1.42 over the same ages, so under a third of the apparent improvement survives holding the people fixed. The remainder is selective non-response and attrition — GHQ completion falls from 91% at age 70 to 77% at 90 — together with cohort differences. This applies to every panel of the figure: these are pooled cross-sections in which age, cohort and period are confounded, and they are descriptive scaffolding rather than within-person trajectories. The mixture models of §9 do use within-person information and are not subject to it.

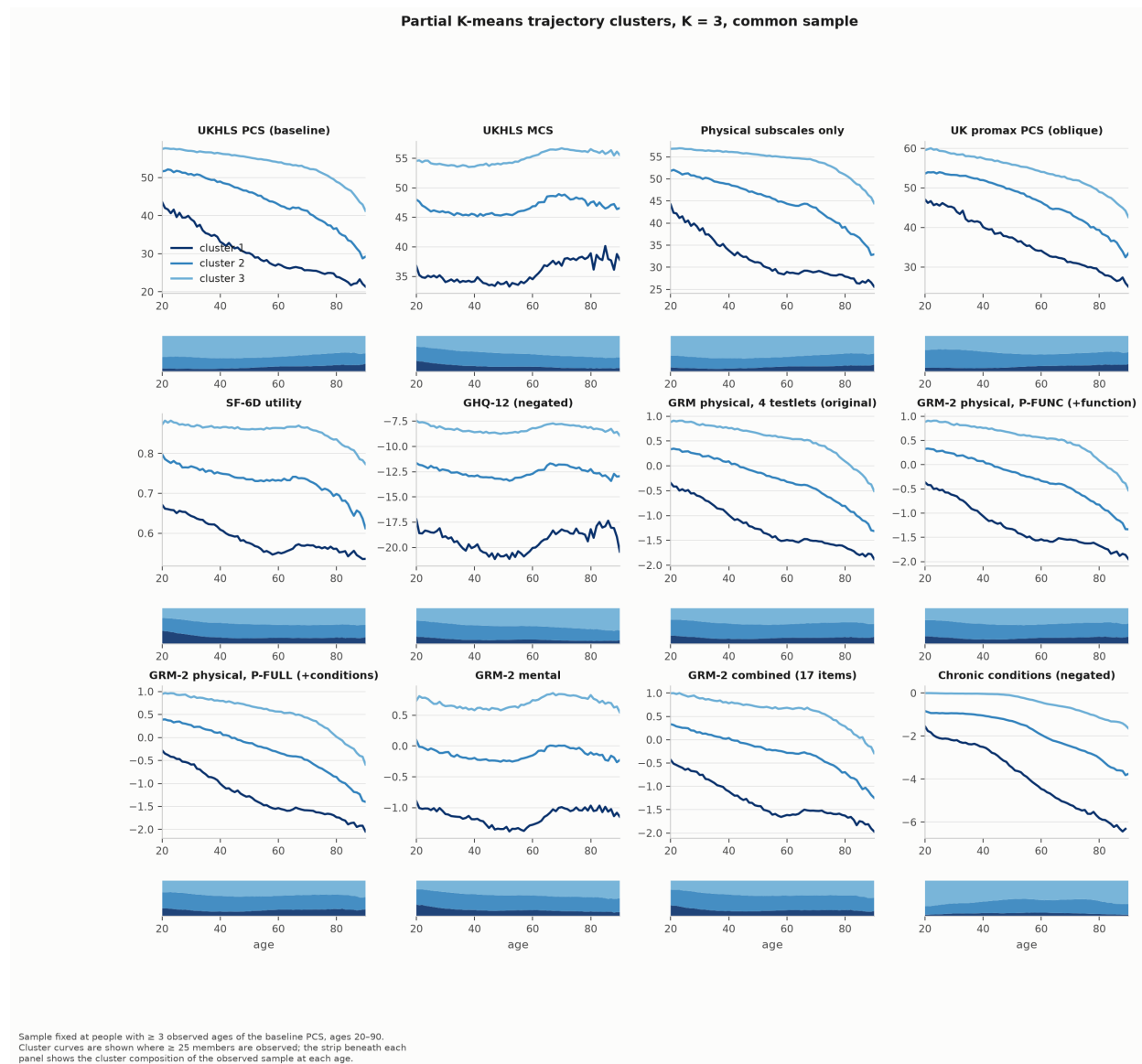


Figure 3: Partial K-means trajectory clusters ($K = 3$, 50 seeded inits, sample fixed at people with ≥ 3 observed baseline-PCS ages, ages 20–90), across the same twelve measures as Figure 2. GHQ and the chronic count are negated so every panel runs higher = better health. The chronic count is monotone by construction, so its three bands can only fall and never cross — a property of the measure, not a finding about people.

Partial K-means (Figure 3): changing the *scoring* moves the typology modestly while changing the *construct* redraws it. Against the PCS baseline, on the common sample:

Measure	ARI	% same cluster	chance
Physical subscales only	0.667	87.9	43.0
UK promax PCS (oblique)	0.464	79.4	39.5
P-FUNC	0.442	76.9	38.1
GRM original (4 testlets)	0.425	75.6	37.7
P-FULL	0.409	74.9	37.5
GRM combined (17 items)	0.266	66.1	36.8
SF-6D utility	0.185	60.0	37.4
Chronic conditions (negated)	0.120	56.1	44.3
GHQ-12 (negated)	0.079	51.4	40.8
GRM mental	0.074	49.2	37.1
UKHLS MCS	0.053	48.0	40.0

The physical measures agree substantially with the baseline whatever the scoring — 0.41 to 0.67 — and the ordering is instructive: the composite that merely re-weights the same subscales agrees most, the IRT re-spacings sit in the middle, and the further a measure moves from the PCS’s item set the lower it falls. The mental measures sit at the bottom, essentially at chance: MCS at ARI 0.053 and the mental GRM at 0.074, with same-cluster rates of 48% and 49% against chance rates of 40% and 37%. The combined bank lands between the two families at 0.27, which is what a physically-dominated mixture of them should do. So the physical typology is robust to how physical health is scored, and the mental dimension partitions people almost independently of it — the same conclusion the two-bank correlation of 0.457 reaches by a different route.

5 The ever-diagnosis question, resolved

The misgiving: historical diagnoses accumulate mechanically with age, so a diagnosis-bearing measure could manufacture “decline” out of stale paperwork. Four reads.

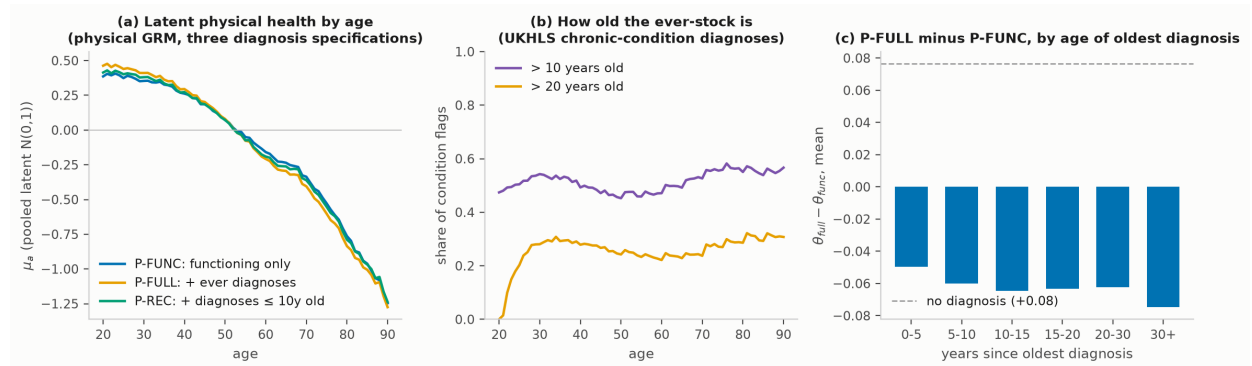


Figure 4: The ever-diagnosis question, all three panels on the physical GRM. (a) latent age profiles μ_a under P-FUNC (functioning only), P-FULL (plus ever-diagnoses) and P-REC (plus diagnoses at most ten years old); (b) the staleness of the UKHLS chronic-condition stock; (c) the per-person gap $\theta_{P-FULL} - \theta_{P-FUNC}$ against the age of that person’s oldest diagnosis.

- (1) **The aggregate effect is real but bounded.** $\mu(80) - \mu(30)$ is -1.117 SD functioning-only, -1.247 with ever-diagnoses, -1.170 with only diagnoses ≤ 10 years old. Diagnoses steepen the measured lifecycle decline by 0.130 SD (12%), and 59% of that extra steepness comes from diagnoses more than a decade old — leaving the stale-accumulation component at $\sim 6\%$ of the total decline.
- (2) **The potential was much larger than the realised effect.** Half the ever-stock is >10 years old at every age past 40 (§2.4). The damage stays small because the estimator protects itself: items that cannot predict the functioning responses get low discriminations, so stale-heavy condition items simply carry little weight.
- (3) **The items cannot tell stale from fresh — which cuts both ways.** The person-level penalty $\theta_{\text{full}} - \theta_{\text{func}}$ is flat in the age of the oldest diagnosis (-0.05 SD at 0–5 years, -0.075 at 30+). Read as measurement of *current* health, that is exactly the mechanism feared: no within-item discounting exists. It is an argument *against trusting ever-items as current-state measures*, and simultaneously the reason their damage is capped.
- (4) **Asked directly, the data mostly decline to discount.** P-TIMED splits every group into recent (≤ 10 y) and stale (>10 y) items and lets the model estimate their separate discriminations — the in-framework analogue of an estimated recency-weight function:

Group	a recent	a stale	ratio
CVD	1.19	1.21	0.99
METAB	0.59	0.81	0.72
RESP	0.68	0.29	2.31
MSK	0.83	1.22	0.68
CANCER	0.59	0.56	1.05
OTHER	0.63	0.63	1.00

Only respiratory disease wants a discount (remitting childhood asthma). For arthritis and the metabolic conditions the *stale* item is more informative — progressive diseases, where an old diagnosis means longer duration and worse current health. The mechanistic-accumulation story is really a story about *remitting* conditions; for progressive conditions the old diagnosis is legitimately informative.

Related literature. Estimating recency weights on past exposures is the weighted-cumulative-exposure literature (Sylvestre & Abrahamowicz 2009; review in Kelly et al. 2024; R package WCE), with spline-estimated weight-by-recency functions — P-TIMED is its two-knot IRT analogue. Letting item parameters vary continuously with a covariate such as time since diagnosis is moderated nonlinear factor analysis (Bauer & Hussong 2009; regularised variants for DIF detection), the natural next refinement if a smooth weighting is ever needed. The simpler administrative-data practice is a fixed lookback window (Preen et al. 2006), which P-REC implements. Health-state valuation handles the same issue with acute vs. post-acute utilities for events like stroke.

Verdict. For lifecycle *dynamics*, use P-FUNC (or the original GRM, $r = 0.994$): a pure current-state instrument. P-FULL is a sound descriptive companion whose accumulation bias is now characterised and bounded (~ 0.08 – 0.13 SD over the lifecycle, concentrated in remitting conditions); P-REC and P-TIMED are its sensitivity band. Where conditions matter substantively, model them as separate channels in the Bayesian mixture, where “ever” is the correct state variable.

6 One dimension or two? The combined GRM

The combined bank (all seventeen items, one latent dimension) fits, scores, and correlates sensibly ($r = 0.92$ with the physical bank, 0.75 with the mental, 0.88 with SF-6D). But it fails the assumption the others satisfy: after conditioning on its single dimension, the mental items remain strongly positively correlated — $Q_3(\text{MH}, \text{GHQ}^-) = +0.58$, $Q_3(\text{GHQ}^+, \text{GHQ}^-) = +0.54$, $Q_3(\text{MH}, \text{RE}) = +0.33$ — while the physical side stays clean (max phys-phys $Q_3 +0.20$). That residual clustering is the data saying the mental items share a dimension the combined score cannot represent: with $r(\theta_P, \theta_M) = 0.457$, one dimension is a genuine misspecification, not a summary. The same criteria table shows the cost in substance: the combined score reintroduces a mental-health artifact of -0.82 SD at a fixed physical profile (the physical banks: 0.00), and its cohort spread collapses to 0.06 SD not because it is clean but because physical health improving and mental health deteriorating across cohorts *cancel* inside it — precisely the mixing the EIT note warned about. Use it only where a single index is externally imposed; a bifactor model (general + physical/mental group factors) would be the defensible one-number construction if that need becomes real.

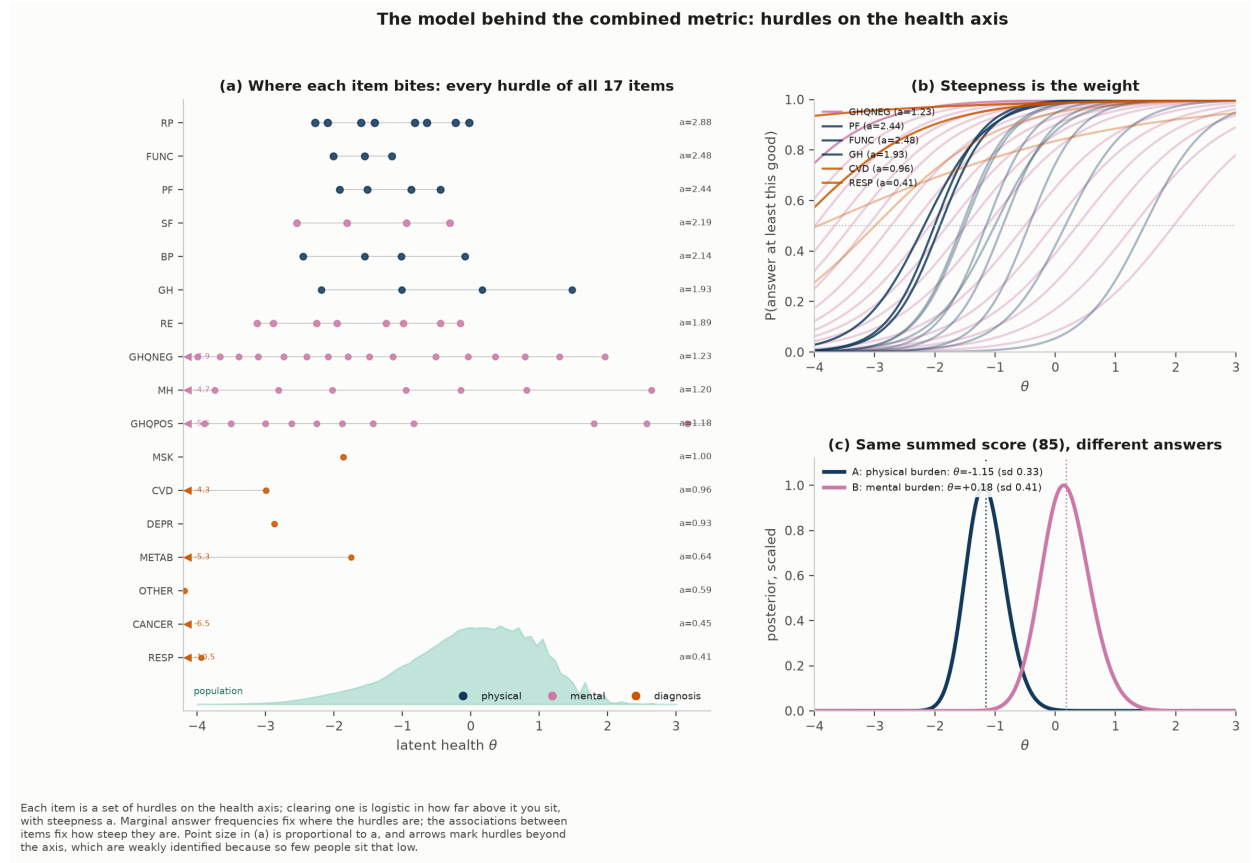


Figure 5: The model behind the combined metric, in the empirics note’s hurdles-on-the-health-axis form, for the most all-encompassing bank. (a) every hurdle of all seventeen items against the population distribution of θ ; (b) the hurdle curves for items spanning the weight range; (c) two people whose category numbers sum to the same total.

Figure 5 draws the combined bank as a set of hurdles. Panel (a) makes the division of labour

visible: the physical items (RP, FUNC, PF, BP, GH; $a = 1.9$ – 2.9) cluster their hurdles between -2.5 and 0 , the mental items (GHQNEG, MH, GHQPOS; $a = 1.2$) spread theirs across the whole axis and beyond it, and the diagnosis items ($a = 0.4$ – 1.0) sit far to the left. Panel (c) then shows what that means for a score. Two people whose answers sum to exactly the same total — one carrying the deficit on the physical items, the other on the mental ones — receive $\theta = -1.15$ and $\theta = +0.18$, a gap of 1.3 standard deviations. The combined bank is not a balanced physical–mental index; it is physically dominated, because the physical items are the steeper and hence the more reliable ones. That is a second, independent reason (alongside the residual clustering of §6) to prefer the two-bank design.

Figure 6: The measure-choice panel across measures (the empirics note’s Figure 30, rebuilt): mean age path, dispersion, how much room each measure has left at its two limits, and the two contamination reads. Seven measures: UKHLS PCS, SF-6D utility, the original four-testlet GRM, P-FUNC, P-FULL, the mental GRM and the combined GRM. Panel (c) now carries both limits — the solid line is the share of person-years at the measure’s exact maximum, the dashed line, mirrored below zero, the share at its exact minimum, and both are read as positive percentages. The asymmetry is the point: ceiling mass runs to 15% at the young end while floor mass never exceeds about 1.5% anywhere, so these instruments run out of room among the healthy long before they do among the ill.

7 Ceiling and floor

Two different questions hide under “ceiling effects”: how much probability mass sits at the best attainable response profile, and whether the instrument can *discriminate* among the healthy. More items help enormously with the first and only marginally with the second.

Measure	ceiling %	floor %	range (sd units)	psd at top	psd at bottom
UKHLS PCS (12 items)	0.89 ^a	~0	6.5	—	—
Physical-only composite	9.95	0.91	4.1	—	—
SF-6D utility	2.98	0.18	4.7	—	—
GHQ-12 Likert	0.20	0.31	6.4	—	—
GRM original (4 testlets)	9.55	0.97	4.4	0.65	0.45
P-FUNC	9.53	0.76	4.5	0.65	0.44
P-FULL	7.16	0.00	5.3	0.64	0.56
MENT	0.11	0.03	6.5	0.60	0.45
COMBINED	0.05	0.00	7.5	0.66	0.55

^ashare at the all-best 12-item profile (the exact-maximum read is degenerate for a continuous derived score). “psd” is the EAP posterior sd at the all-best / all-worst response pattern.

The optimistic half of the hypothesis is confirmed: ceiling mass collapses from 9.5% (original GRM) to 7.2% with conditions and to 0.05–0.11% for the mental and combined banks, and the floors of P-FULL and COMBINED empty entirely. Panel (c) of Figure 6 shows where it binds: the young end, where the physical banks still park 12–15% of person-years at the maximum.

The pessimistic half is also right, and Figure 1 shows why: the posterior sd at the very top is ~0.6–0.66 *for every version* — seventeen items buy no more precision among the healthy than four, because every added item has strongly negative thresholds (all these instruments measure ill-health). Extra items *spread* the top mass into a finer ranking (ties broken by GHQ and diagnoses) without adding information where the healthy live. Genuinely fixing the ceiling requires items with positive thresholds — vigorous-functioning items of the PROMIS physical-function type — which UKHLS does not carry. For the growth mixtures of §9 onwards this matters more than it might seem. They do not see the ordered item categories; they take each person-wave’s EAP score as a Gaussian observation, so an all-best response pattern enters as a single value at the maximum, with nothing to indicate that the person’s true level may lie above it. Section 10 finds the consequence in the estimated class structure.

The floor, in depth

The floor is the mirror image, and here the added items genuinely work — their negative thresholds sit exactly where the severely ill live (`descriptives/06, floor_depth.csv`):

	measurement SE at $\theta =$			min	% < -3	% at worst	distinct θ in
	-4	-3	-2	θ		pattern	bottom 1%
GRM original	2.15	0.73	0.33	-2.61	0.00	0.97	2
P-FUNC	2.12	0.67	0.30	-2.68	0.00	0.76	3
P-FULL	0.95	0.55	0.29	-3.38	0.03	0.00	575
MENT	0.56	0.34	0.33	-3.72	0.30	0.03	1,524
COMBINED	0.55	0.37	0.28	-4.43	0.24	0.00	3,708

A note on reading Figure 1: P-FULL carries visible information out to the left edge of the axis where the other physical banks have none. That is real measurement, not an artefact of where the axis stops. Total test information at $\theta = -4$ is 1.10 for P-FULL against 0.22 for P-FUNC, and at $\theta = -6$ it is 0.37 against 0.00; the axis is not hiding a cliff, the curve simply keeps declining. The information there comes almost entirely from the diagnosis items — CVD (34% of it), METAB (18%), OTHER (10%) — whose hurdles sit at -3.6 and -4.1 , precisely in that region. It is the opposite of a floor effect: a floor effect is information *collapsing* at the bottom, which is what the functioning-only banks do. The caveat is that a few of these thresholds are weakly identified because so few people sit that low (RESP’s first hurdle is estimated at -10.5 , CANCER’s at -6.5 in the combined bank), so the far tail should not be read as precisely located — but at $\theta = -4$ the contributing hurdles are all within the range the data actually cover.

The original GRM (and P-FUNC) have a real floor problem: the instrument goes dark below $\theta \approx -2.5$ (SE 0.7 at -3 , 2.1 at -4), the scale simply ends at -2.6 , and the entire bottom percentile collapses onto two or three response patterns — the worst-off are a spike, not a distribution. The condition items repair precisely this: their thresholds (-2.6 to -4.6) are the only ones deep in the ill tail, so P-FULL keeps SE at 0.55 at $\theta = -3$, extends the scale to -3.4 , empties the worst-pattern spike entirely, and resolves the bottom percentile into 575 distinct positions. Among the P-FUNC bottom *decile* (37.6k person-years — the population a prevention analysis targets), P-FUNC distinguishes 1,252 values, P-FULL 16,522, COMBINED essentially one per person-year. The mental bank has the best floor of all (GHQ thresholds reach -3.5 ; SE 0.34 at -3).

This sharpens the ever-diagnosis trade-off of §5: the diagnosis items’ cost is a bounded accumulation bias in the age gradient; their benefit is measurement depth among the severely ill, which functioning items alone cannot provide because severe limitation saturates them. For analyses that turn on the worst-health group — describing the bottom class, targeting — P-FULL is the better physical instrument; for population dynamics, P-FUNC remains the clean choice.

8 Convexity against healthcare use

Is healthcare use convex in health — falling steeply among the ill, flat among the healthy — and does each metric’s cardinalisation carry that shape? Following the construction-note convexity exercise, each outcome is regressed on the standardised measure and its square on a common sample (waves 7–15, all fifteen measures observed; $n = 209,042$ in-patient, 208,936 GP, 208,951 out-patient), with measures oriented so higher = better: on a declining relationship, a positive quadratic = convex.

One data correction first, and it runs the other way from the one recorded in an earlier draft of this note. UKHLS gives *three* tiers of contact, not two: `h12gp` is “Visited GP in last 12 months” and `h12hop` is “Hosp or clinic out-patient last 12 months”, both banded 0–4 and both running waves 7–15. An earlier draft asserted the opposite — that `h12gp` was the out-patient item and that the mainstage carried no GP-visit count — on a faulty dictionary parse that matched an adjacent variable block. The UKDA labels were re-checked directly in waves g, j and o. Everything previously reported here as “out-patient” was in fact GP visits; the table below separates the two, and the genuine out-patient column is new. `servuse1` (“service use: your local doctor”, yes/no, five waves) is kept as a supplementary binary.

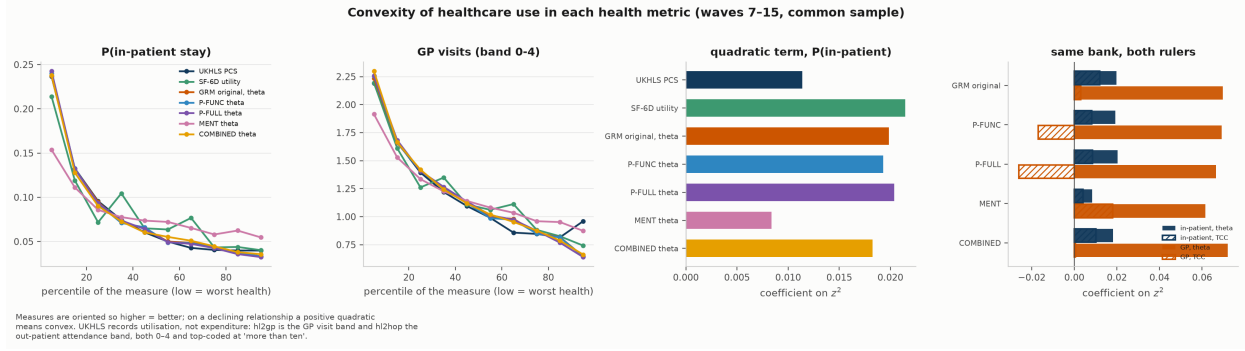


Figure 7: Healthcare use across the distribution of each measure, and the quadratic (convexity) term for in-patient admission. All fifteen measures are tested (Table above); the seven plotted in the first two panels are UKHLS PCS, SF-6D utility, the original GRM, P-FUNC, P-FULL, the mental GRM and the combined GRM, each oriented so higher means better health. The fourth panel sets each bank's θ (solid) against its TCC (hatched) for in-patient stays and GP visits: identical people, identical ranking, different units. Note that **h12gp** is the GP visit band and **h12hop** the out-patient band; an earlier draft had these swapped.

Measure	quadratic term (t)			$P(\text{in-patient})$	
	in-patient	GP	out-patient	bottom 1%	top 1%
UKHLS PCS	+.011 (24)	+.032 (18)	+.032 (20)	.312	.042
UKHLS MCS	+.004 (8)	+.038 (22)	+.020 (13)	.128	.134
Physical-only composite	+.011 (22)	+.002 (1)	+.025 (14)	.331	.033
SF-6D utility	+.021 (40)	+.119 (61)	+.111 (60)	.289	.039
GHQ-12 (flipped)	+.003 (8)	+.007 (5)	+.004 (3)	.189	.059
<i>each bank on both rulers</i>					
GRM original, θ	+.020 (43)	+.070 (42)	+.081 (52)	.331	.033
GRM original, TCC	+.012 (23)	+.003 (2)	+.023 (13)	.331	.033
P-FUNC θ	+.019 (42)	+.069 (42)	+.083 (54)	.322	.033
P-FUNC, TCC	+.008 (17)	-.017 (-9)	+.007 (4)	.322	.033
P-FULL θ	+.020 (45)	+.067 (40)	+.083 (54)	.349	.033
P-FULL, TCC	+.009 (17)	-.026 (-15)	+.001 (0)	.349	.033
MENT θ	+.008 (19)	+.062 (37)	+.039 (25)	.206	.056
MENT, TCC	+.004 (10)	+.018 (12)	+.014 (9)	.206	.056
COMBINED θ	+.018 (44)	+.072 (49)	+.078 (56)	.331	.032
COMBINED, TCC	+.010 (24)	+.001 (0)	+.023 (16)	.331	.032

Every bank now appears on both rulers, and the last two columns are the control: they are rank statistics, so within a pair they are *identical* across θ and TCC to three decimals. Anything that moves between the two rows of a pair is cardinalisation and nothing else.

Four reads.

Utilisation is strongly convex in every latent-scale metric. The in-patient rate runs from $\sim 24\%$ in the worst decile to $\sim 3\text{--}4\%$ in the best (base rate 8.1%), with the quadratic terms at $t = 40\text{--}45$ for the GRM thetas. The gradient concentrates exactly where §7's floor analysis lives: the region in which measures differ in resolution is the region that carries the utilisation risk.

Cardinalisation, not construct, drives the convexity — and the three tiers part company. Each θ /TCC pair ranks people identically, so every difference below is units and nothing else.

In-patient admission is convex on both rulers. Every TCC row keeps a t between 10 and 24, roughly half the θ magnitude. Re-ruling moves the size of the effect and not the conclusion. This is the tier to trust: it is also the only one measured as a true count rather than a band.

GP visits are where the ruler decides the answer. On θ every bank is strongly convex, $t = 37$ –49. On the TCC the same banks give $+0.003$ ($t = 2$) for the original, $+0.001$ ($t = 0$) for COMBINED and $+0.018$ ($t = 12$) for MENT — and turn *concave* for both physical banks, -0.017 ($t = -9$) for P-FUNC and -0.026 ($t = -15$) for P-FULL. A t of -15 is not a null result; it is a confident statement of the opposite shape.

Out-patient attendance sits between the two. On θ it is the most convex of the three tiers ($t = 52$ –56 for the physical and combined banks). On the TCC it attenuates hard — to $+0.007$ ($t = 4$) for P-FUNC and $+0.001$ ($t = 0$) for P-FULL — but it does *not* change sign. So the outright reversal is specific to GP contact, the tier whose gradient is shallowest and whose top band saturates soonest; for out-patient care the honest summary is that re-ruling collapses the curvature to nothing rather than inverting it.

Convexity is therefore a property of the chosen units, and a “convex health effects” claim means nothing until the scale is fixed. Conversely `grmh` is the natural unit when something close to linear-in-utilisation spacing is wanted — the near-zero TCC quadratics are that property, stated — and θ the natural latent-normal unit for the mixture models.

The floor resolution pays off in real outcomes. P-FULL’s bottom 1% has a 34.9% admission rate against 32.2% for P-FUNC and 31.2% for the PCS: the diagnosis items isolate a genuinely sicker extreme tail, the outcome-side counterpart of §7’s 575-fold floor de-clumping.

The supplementary GP binary is uninformative about shape. The `servuse1` yes/no item adds nothing over the `h12gp` band. At a 77.8% base rate ($n = 23,953$ on its five waves) it saturates: the θ quadratics are ≈ 0 for every bank. The TCC rows are uniformly negative and significantly so ($t = -5$ to -8), the same units effect seen in the banded tiers. Bottom-1% GP use is $\sim 97\%$ against 56–74% at the top, by measure — a gradient, but the outcome’s own ceiling swamps any curvature. The banded `h12gp` count is the more informative of the two; an exact visit count would need linked administrative data.

8.1 How much of that convexity is real?

The figure above plots utilisation against *percentiles* of each measure, and the claim of convexity deserves more scrutiny than a percentile axis can give it. Two separate things inflate it.

The horizontal scale. Percentiles are a rank transform, so equal horizontal steps are equal shares of people rather than equal amounts of health. For a roughly normal θ the bottom decile spans far more of the health scale than the fifth does, and compressing it to the same width bends the left-hand end upward. Panels (a) and (b) of Figure 8 show the same relationship on the two axes: the sharp elbow on the percentile axis becomes a smooth decay on θ . The binning rule itself turns out not to matter — equal-count and equal-width bins trace the same curve — so this is the axis, not the aggregation. The formal test in the table above was already run on the measure’s own scale, so only the picture was affected; but the picture is what carried the claim.

The vertical scale, which matters more. In-patient stays run at 8.2%, and the logistic curve is convex everywhere below one half. A relationship that is exactly *linear in the log-odds* therefore

appears convex when plotted as a probability. Panel (b) draws a logistic with no quadratic term at all against the observed decile means: it reproduces the curvature almost exactly. Panel (c) puts the same data on the log-odds scale, where it is close to a straight line.

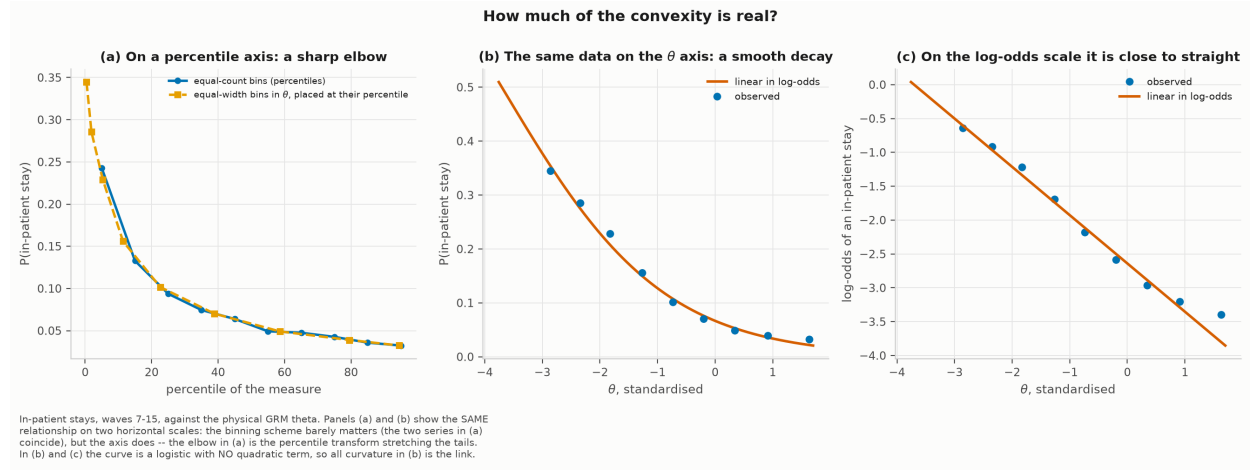


Figure 8: In-patient stays against the physical GRM θ . The fitted curve in (b) and (c) is a logistic with no quadratic term, so every departure from a straight line in (b) comes from the link function alone.

Two axes, not one. These are separate choices and it is easy to run them together. Convexity is $\partial^2 y / \partial x^2$, so it depends on how *both* axes are cardinalised, and the two questions are independent:

	what is being chosen	options here
horizontal	the health measure	θ , <code>grmh</code> , PCS, SF-6D
vertical	the link for the outcome	probability, log-odds

The θ versus `grmh` distinction is horizontal: two cardinalisations of the same latent health. The probability versus log-odds distinction is vertical: two ways of writing the same risk. In the table below, differences *down* a column are the horizontal effect and differences *across* the columns are the vertical one.

The cleanest demonstration that neither is innocuous is the pair θ and `grmh`. `grmh` is the test characteristic curve applied to θ , a monotone transform, so the two rank every person identically — their Spearman correlation is 1.000000 to six decimal places. They differ only in spacing, and they differ a great deal: the bottom quarter of the distribution spans 1.12 units of θ but only 0.33 of `grmh`, while the top quarter spans 0.77 of θ and just 0.03 of `grmh`, because the TCC runs into its ceiling. On the log-odds scale θ is convex (+.051) and `grmh` concave (−.056).

Two measures that order people identically disagree about the sign of the curvature. Convexity is therefore not a property of the health ordering at all; it is a joint property of the ordering and the units chosen to express it. That is the strongest form of the caution for the framework: no amount of better ranking, more items or finer measurement will settle a convexity question, because ranking is not what is at issue.

Re-running the quadratic test on both scales makes the point numerically:

Measure	probability scale		log-odds scale	
	z^2 coef	t	z^2 coef	t
GRM physical, θ	+.020	44.7	+.051	8.6
SF-6D utility	+.021	39.8	+.112	18.1
UKHLS PCS	+.011	23.7	−.028	−4.8
GRM physical, TCC scale	+.009	17.4	−.056	−10.2

On the probability scale every measure is emphatically convex; on the log-odds scale two of the four change sign. The PCS and the TCC cardinalisation are mildly *concave* in log-odds. Only θ and the SF-6D remain convex there, and θ ’s curvature is small against its linear slope of -0.71 per standard deviation.

What this means for the framework. The framework asks whether the return to preventing a unit of health loss is larger for the already-unwell, which is a question about the second derivative of a structural relationship — and a second derivative is not invariant to how either axis is cardinalised. Three choices have to be made explicit before the question has an answer: the health scale (θ , its TCC transform, and the PCS rank people almost identically but disagree about curvature), the outcome scale (probability, log-odds, or eventually cost), and, if cost is the target, the fact that we observe utilisation counts rather than spend at all. The honest summary of §8 is therefore narrower than it first appeared: *utilisation falls steeply with health among the unwell and flattens among the healthy, and on the natural multiplicative scale for a risk that relationship is close to log-linear.* A convexity claim strong enough to carry a policy conclusion needs a cost outcome and a defended cardinalisation, neither of which UKHLS supplies on its own.

8.2 If the object is expected cost

The binary outcome is part of the problem but not the whole of it, and naming which part matters. A binary outcome forces a *bounded* vertical scale, and on a bounded scale curvature is partly mechanical: below one half the logistic is convex whatever the data do. That is the vertical half of §8.1. The horizontal half is untouched by it — θ and `grmh` rank people identically and still disagree about the sign of the curvature, and no change of outcome fixes that.

But if the motivation is *an individual’s expected cost to the health service*, the vertical problem largely dissolves. Money is a ratio quantity with a meaningful zero and a meaningful unit, so “convex in pounds” is a substantive claim rather than a convention. Probability and log-odds are two conventions for writing the same risk and neither is privileged; pounds are what the system actually spends.

UKHLS carries no spend, but it carries what spend buys: nights as an in-patient (`hospd`) and out-patient attendance bands, waves 7–15. Unit costs that do not themselves vary with health only rescale the curves, so the *shape* of expected cost can be studied in physical units without inventing prices.

The decomposition that matters. Expected nights is $P(\text{admitted}) \times E[\text{nights} \mid \text{admitted}]$, and §8 measured only the first factor. Both move with health:

Decile of θ	$P(\text{admitted})$	nights if admitted	expected nights
1 (least healthy)	.240	13.7	3.28
5	.062	4.9	0.30
10 (healthiest)	.032	3.3	0.10
bottom vs top	$7.4\times$	$4.2\times$	$31.9\times$
share of the log gap	58%	41%	—

The least healthy decile is admitted seven times as often *and* stays four times as long, so it consumes about **thirty-two times** the bed days of the healthiest. The intensive margin contributes 41% of the gradient and was invisible to the binary analysis.

That $31.9\times$ is itself an understatement, because 12.7% of admissions are childbirth (`hospch`), at a mean age of 31.7 against 57.9 for every other admission. Maternity is not health deterioration, and it lands almost entirely on healthy people in their early thirties, so it inflates the denominator. Excluding it the ratio is $52.7\times$. Any costing of these data should separate maternity first.

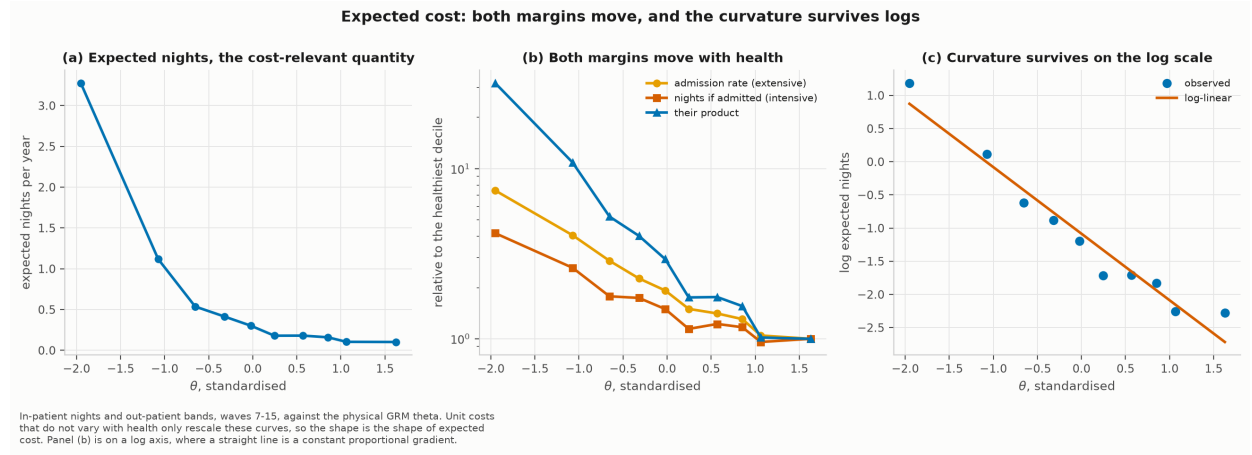


Figure 9: In-patient nights and out-patient bands against the physical GRM θ . Unit costs that do not vary with health only rescale these curves.

And the curvature survives the log. On the decile means the quadratic in $\log E[\text{nights}]$ is $+0.175$, against $+0.087$ for $\log P(\text{admitted})$ and $+0.016$ for out-patient bands. So where the binary outcome was close to log-linear — and flipped sign under other cardinalisations — expected bed days is convex *even in logs*, because the two margins compound. The out-patient channel is log-linear and contributes none of it; the entire result comes from in-patient nights.

So the answer to “what would make sense” is threefold. Use expected cost, not a binary event: it fixes the vertical axis on a scale that means something, and it is the quantity the framework is actually about. Model both margins, because the intensive one carries 41% of the gradient and a probability model discards it. And define the intervention in units that survive re-cardinalisation — a condition prevented, an admission averted, a year of health — rather than “a standard deviation of θ ”, because that is the one problem a better outcome variable cannot solve. On that footing the framework’s convexity assumption looks defensible in a way §8.1 alone would not suggest: a factor of thirty-two, with curvature that survives logging, is a large and robust gradient.

What would still be needed. Nights are a proxy for cost with fixed prices per night, which understates the cost of complex admissions and ignores primary and community care and prescribing

entirely. Linked HES cost data would settle it. And §8.4 applies here too: some of this gradient is proximity to death rather than health level, which needs mortality in the model to separate.

8.3 How far UKHLS can go, and what linkage would add

There is more in UKHLS than nights. Beyond `hosp` (any admission) and `hospd` (nights), waves 7–15 carry `hospch`, a childbirth flag, and from wave 2 onwards a condition-level breakdown: `hospci` / `hospci` record whether an admission and how many of its nights were due to each *newly reported* condition, and from wave 11 `hospcpi` / `hospcpi` do the same for *previously diagnosed* ones. The condition list runs to about forty codes in the later waves, well beyond the seventeen of §2.3 — COPD, chronic kidney disease, dementia, the individual cancers and the individual mental health conditions all appear separately.

Variable	Waves	Coverage
<code>hosp</code> , <code>hospd</code>	7–15	all respondents; nights for all admitted
<code>hospch</code> childbirth	7–15	12.7% of admissions
<code>hospc</code> / <code>hospc</code> new condition	2–15	21% of admissions
<code>hospcp</code> / <code>hospcp</code> prior condition	11–15	11% of admissions
either attribution		29% of admissions, 65% of nights
<code>h12gp</code> GP, <code>h12hop</code> out-patient	7–15	all respondents, banded 0–4
<code>servuse1/2</code> doctor, hospital	4, 6, 10, 14, 15	five waves, binary

So the ceiling for UKHLS alone is: expected bed days, with maternity separable, and with roughly two-thirds of nights attributable to a named condition. That is enough to weight nights by condition using published reference costs and to build a considerably better cost proxy than the flat per-night figure of §8.2 — the attributed nights concentrate in stroke, COPD, asthma, osteoarthritis, MI and chronic kidney disease, which have very different unit costs. It is not enough for real costing. Primary care, prescribing, A&E attendance, elective versus emergency status and procedure codes are all absent, and out-patient use is only a four-band count.

Linkage is feasible, with real conditions. Understanding Society participates in the UK Longitudinal Linkage Collaboration (UK LLC), which links consenting participants to NHS primary and secondary care records, including HES, and holds them in a Trusted Research Environment at Swansea. Access is free to UK-based researchers for public-good, non-commercial work, but requires ONS Accredited Researcher status (or enrolment on the Safe Researcher training), an expression of interest followed by a full application, and — the practical constraint for us — analysis inside the TRE rather than locally. Nothing in the current pipeline could be run on a laptop any more.

The binding question is sample size. Linkage covers only consenting participants: the COVID-19 study contributed roughly 8,000, against the 84,679 people in our P-FUNC sample, and Understanding Society has stated an intention to extend linkage to all main-survey participants — worth checking the current position before planning around either figure. At 8,000 the latent-variable machinery in this note would be badly underpowered; at full coverage it would be transformative, since HES supplies exactly what is missing: every admission with its diagnosis and procedure codes, elective versus emergency, and a route to HRG-based costs rather than proxies.

The sensible order. Exhaust UKHLS first — separate maternity, weight the attributable two-thirds of nights by condition, and see whether the $52.7\times$ gradient and the log-scale curvature survive a better cost proxy. That is a few days’ work on data already on disk, and it would tell us

whether linkage is worth the overhead. If it is, the linkage application is a months-long process whose main risk is the consented sample being too small for this kind of model. The staged plan for that work, with the unit-cost sources and the validation targets, is in `cost-proxy-plan.pdf`.

8.4 What the literature actually establishes

Convexity of healthcare costs in health is widely assumed in prevention arguments, so it is worth separating what is *accepted*, what is *measured*, and what is actually *known* about the functional form. The three come apart.

Firmly measured, and about the distribution rather than the shape. Spending is extremely concentrated: the top decile of spenders accounts for more than half of total expenditure, and about a third of them are still in the top decile a year later. This is robust and uncontroversial, but it is a fact about the *marginal distribution of spending*, not about the derivative of cost with respect to health. A concentrated outcome is consistent with many shapes, including a linear one applied to a skewed health distribution.

Firmly measured, and genuinely about shape — but on the count scale. The multimorbidity literature reports a “near-exponential” relationship between the number of chronic conditions and cost, with expenditure roughly doubling per additional condition and specific combinations costing more than the sum of their parts (superadditivity). Here the scale is defensible in a way ours is not: “one more condition” has a meaning that “one more standard deviation of θ ” does not. But note what the finding *is*: exponential in the count is exactly linear in log cost. The headline result of that literature is a multiplicative model, which is the same shape §8.1 found in our data — not evidence of curvature beyond multiplicative.

Assumed by the standard specification, and therefore weakly tested. The default model for healthcare costs in health economics is a GLM with a log link (Manning & Mullahy 2001; Jones 2010), under which covariates act multiplicatively by construction. A log-link model is convex in levels automatically, whatever the data say. So a large share of the empirical literature cannot be evidence *for* convexity in levels, because it imposes it; the interesting question — is there curvature on the log scale? — is rarely posed. We found none of the reviewed work testing convexity against a continuous health measure as a functional-form question.

A rival explanation for the shape. The “red herring” literature (Zweifel, Felder & Meiers 1999, and the twenty-year reappraisals) argues that costs track *proximity to death* rather than age or health level, with spending tripling in the final quarters of life. If most of the steepness at the bottom of the health distribution is people close to death, then the apparent convexity in health is partly a survival gradient wearing a health costume — which this analysis does not separate. The cost gradient is estimated without reference to survival, and the mortality channel later added to the multidimensional model (§12) describes hazard by class, not proximity to death within the cost data.

And a caution about the policy inference. Even granting convexity, Rose’s prevention paradox (1981, 1985) blocks the automatic move from “the return per person is highest at the bottom” to “target the bottom”: most cases arise from the large moderate-risk middle rather than the small extreme tail, so aggregate returns can favour population-wide action even when individual returns are convex. Convexity in the individual return function and optimality of targeting are different claims.

Where this leaves us. The proposition that costs rise steeply as health deteriorates is well

measured and not in doubt. The stronger proposition the framework needs — that the second derivative is positive on a defended scale, so that a unit of prevention is worth more to the already unwell — is much less established than its ubiquity suggests: it is usually assumed through a log link, demonstrated on a count scale where it amounts to log-linearity, or inferred from concentration statistics that do not identify a shape. Our own finding of near-log-linearity (§8.1) sits comfortably within the literature rather than against it. Testing the framework’s version of the claim needs a cost outcome rather than utilisation counts, a health scale defended on grounds other than convenience, and mortality in the model so that proximity to death is not doing the work.

9 Latent-class trajectories under AR(1) persistence

The measures feed the Bayesian clustering layer: $K = 3$ quadratic growth mixtures (the consolidated Stan model, partition inits with jitter, 4 chains $\times$ 1000+1000) on each channel’s frozen lifecycle contract, in four specifications — no AR term; AR(1) persistence in the class residual; AR(1) with *within-person holdout*, where each person’s last two observations are excluded from the likelihood and scored as a conditional predictive density, with the sample restricted to people observed at least five times; and a state-space specification in which an AR(1) latent health state carries the persistence and each observation adds an independent one-period *measurement error* on top (*Separating persistence from measurement error*, below). Class 1 is worst health throughout (the anchor convention). All fifteen fits converged. The first batch of eight took 15.9 h, the three state-space fits 14.9 h, and their three holdout twins together with the instrument control 17.0 h, each on two workers.

Channel	Spec	Persons	θ (shares)	ρ (persistence)	max $\hat{R}$	wall
PCS	AR(1)	50,194	.165 / .273 / .561	.67 / .31 / .29	1.004	8.3 h
PCS	AR(1) + holdout	39,349	.146 / .234 / .620	.67 / .26 / .30	1.006	6.2 h
GRM phys (P-FULL)	baseline	38,963	.159 / .437 / .404	—	1.003	2.0 h
GRM phys (P-FULL)	AR(1)	38,963	.328 / .360 / .312	.78 / .27 / .39	1.001	4.1 h
GRM phys (P-FULL)	AR(1) + holdout	29,901	.313 / .362 / .325	.78 / .26 / .37	1.002	2.5 h
GRM combined	baseline	38,181	.163 / .454 / .383	—	1.002	1.9 h
GRM combined	AR(1)	38,181	.290 / .379 / .331	.77 / .34 / .44	1.002	3.6 h
GRM combined	AR(1) + holdout	29,280	.271 / .377 / .352	.77 / .32 / .44	1.003	3.1 h
GRM original	ME	50,042	.267 / .497 / .236	.95 / .89 / .81	1.005	11.1 h
GRM original	ME + holdout	38,971	.248 / .498 / .254	.95 / .89 / .80	1.002	10.3 h
GRM phys (P-FUNC)	ME	50,001	.254 / .487 / .259	.95 / .89 / .82	1.004	9.9 h
GRM phys (P-FUNC)	ME + holdout	38,923	.243 / .487 / .270	.95 / .89 / .81	1.007	10.5 h
GRM phys (P-FUNC)	ME, P-FULL roster	38,963	.261 / .489 / .250	.95 / .89 / .82	1.002	6.7 h
GRM phys (P-FULL)	ME	38,963	.261 / .478 / .261	.95 / .90 / .86	1.002	5.0 h
GRM phys (P-FULL)	ME + holdout	29,901	.249 / .480 / .271	.96 / .90 / .86	1.008	6.1 h

ME: AR(1) latent state plus one-period measurement error, with or without holdout. “P-FULL roster” is the instrument control: P-FUNC fitted to the exact person-age cells of the P-FULL contract.

Persistence is strongly class-graded, on every channel. The worst-health class carries $\rho_1 = 0.67$ – 0.78 against 0.26 – 0.44 for the other two, with posterior standard deviations of 0.003 – 0.007 : shocks to the least healthy class decay several times more slowly than shocks elsewhere, and the ordering is identical across three different instruments. This is the mixture-model counterpart of the EIT note’s persistent-layer anatomy, estimated jointly with the class structure rather than imposed. It is also, as the state-space subsection below shows, a statement about an *attenuated*

parameter: once a one-period measurement error is allowed, the ordering survives but the grading largely does not, and every ρ here should be read with that section in view.

The AR term, not the scale, redistributes the classes. The interim reading of these fits was that the GRM channels produce near-even class shares where the PCS stays bottom-heavy. The baselines refute the measurement-scale half of that: without an AR term, *both* GRM channels put only 16% in the worst class (.159 and .163), close to the PCS convention. Adding persistence roughly doubles that share to .29–.33. The mechanism is visible in the intercepts: the baseline worst class sits at $\alpha_1 \approx -1.6$ on all three channels, while under AR(1) the GRM worst class moves up to $\alpha_1 \approx -1.0$ and takes in many more people. Without persistence, a small extreme class must absorb the autocorrelation of repeated low observations; once ρ can do that work, the class boundary moves and a broader, less extreme worst class emerges. The PCS resists this (θ_1 .165 under AR(1)), which is consistent with its coarser floor (§7): it cannot separate the severely ill finely enough for the boundary to move.

Class trajectories and shares

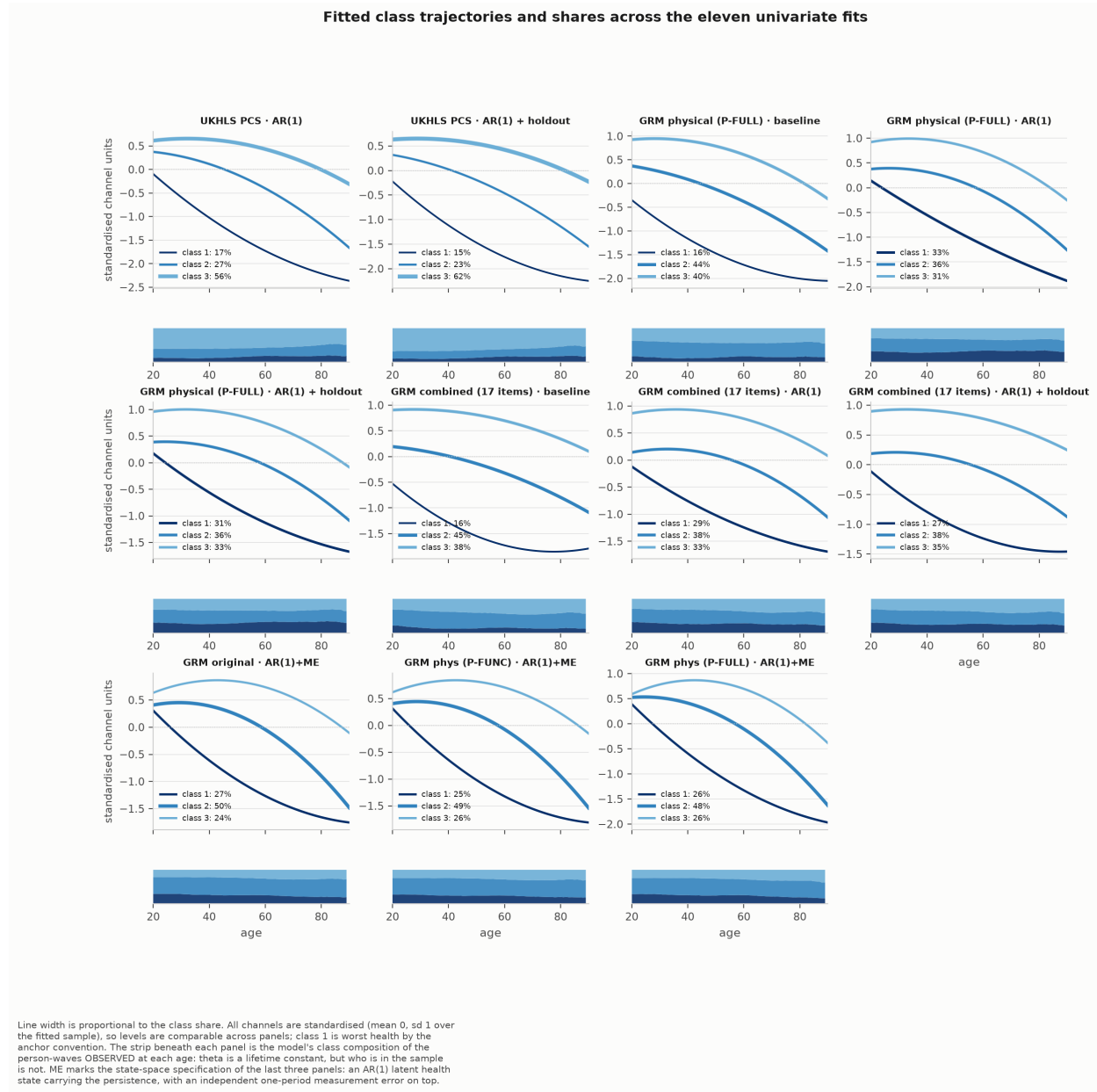


Figure 10: Fitted class mean trajectories $\alpha_k + \beta_k a + \gamma_k a^2$ (with $a = (\text{age} - 55)/10$) for all eleven fits, on each channel's standardised scale. Channels: the UKHLS PCS, the physical GRM in its P-FULL specification (`theta_phys_full`: functioning plus diagnosis groups), the combined seventeen-item GRM (`theta_combined`), the original GRM (`grm.theta`) and the functioning-only physical GRM (`theta_phys_func`). Line width is proportional to the class share. “ME” in the last three titles marks the state-space specification described below: an AR(1) latent state plus an independent one-period measurement error. The strip beneath each panel is the class composition of the person-waves observed at each age.

Figure 10 puts every fit on one standardised axis. Three regularities hold everywhere: the three classes are ordered in level at every age and never cross; all three decline; and the worst class declines *fastest*, so the classes fan apart over the life course. The gap between the best and worst class widens from 0.9–1.9 sd at age 30 to 1.8–2.2 sd at age 80, with the narrowest age-30 gaps and the widest fanning in the three state-space fits.

Fit	worst-class share	level gap (best – worst) age 30	age 80	decline, age 30–80 worst	best
PCS, AR(1)	.165	1.24	2.20	–1.62	–0.66
PCS, AR(1) + holdout	.146	1.35	2.19	–1.45	–0.61
GRM phys, baseline	.159	1.74	2.06	–1.22	–0.89
GRM phys, AR(1)	.328	1.21	1.81	–1.44	–0.84
GRM phys, AR(1) + holdout	.313	1.22	1.79	–1.32	–0.75
GRM comb, baseline	.163	1.86	2.19	–0.90	–0.57
GRM comb, AR(1)	.290	1.38	1.95	–1.12	–0.56
GRM comb, AR(1) + holdout	.271	1.41	1.91	–0.96	–0.46
GRM original, AR(1) + meas. err.	.267	0.98	1.92	–1.47	–0.53
GRM phys P-FUNC, AR(1) + meas. err.	.254	0.97	1.94	–1.52	–0.55
GRM phys P-FULL, AR(1) + meas. err.	.261	0.91	1.90	–1.69	–0.70

Standardised channel units. Class 1 is worst health.

The specifications differ in *how* they separate the classes, and the difference is systematic. Adding the AR term to a GRM channel moves the worst class up in level (its age-30 gap to the best class narrows from 1.74 to 1.21 on the physical channel) while making it decline more steeply ($-1.22 \rightarrow -1.44$) and roughly doubling its membership. In other words the baseline finds a *small, extreme, comparatively flat* worst class; with persistence available the model instead finds a *larger, less extreme, faster-declining* one. The ratio of worst-class to best-class decline — a compact summary of how much of the story is trajectory divergence rather than fixed level differences — rises from 1.37 to 1.72 (physical) and 1.57 to 2.00 (combined) when persistence is added, while the PCS sits at 2.4–2.5 in both its specifications.

The holdout fits track their full-sample siblings closely on every one of these quantities (shares within .02, level gaps within .11 sd, declines within .18 sd), so the restriction to people with five or more observations changes the estimated typology very little.

Separating persistence from measurement error

Under the AR(1) specification the observation *is* the state, so ρ has to absorb both genuine persistence and any transient noise in the instrument. If a GRM score wobbles year to year for reasons that have nothing to do with health — a different interviewer, a bad week, the coarseness of a five-point item — that wobble enters as a fast-decaying shock and drags ρ towards zero. The state-space specification separates them: a latent health state follows the AR(1), and each observation adds independent noise that does not persist. In reduced form this is ARMA(1,1). Identification comes from the autocorrelation function, since $\text{corr}(k) = \rho^k \times \text{signal share}$ for $k \geq 1$, so ρ is fixed by the *ratio* of successive autocorrelations and the signal share by their *level*. The state is marginalised by a Kalman filter rather than sampled, so no latent series enters the parameter vector. Measurement error is one scale per channel, shared across classes: it is a property of the instrument, not of the person’s class.

The specification was validated on simulated data first. On 1,500 simulated people with $\rho = 0.75$, $\sigma_{\text{state}} = 0.45$ and $\sigma_{\text{meas}} = 0.35$, the state-space fit recovers $\rho = 0.738$, $\sigma_{\text{state}} = 0.384$ and $\sigma_{\text{meas}} = 0.280$ against truths of 0.750, 0.375 and 0.292 on the standardised scale. Fitting AR(1) to the *same* simulated panel returns $\rho = 0.564$, an attenuation of 24.7%, and inflates σ from 0.375 to 0.505. The misspecification the model exists to fix is visible in the recovery exercise before it is asserted of the data.

In the real data the attenuation is far larger than that. Comparing P-FULL under the two specifications on the *identical* sample — the same 38,963 people and 334,194 person-waves, one specification change:

Specification	ρ by class			σ
	1	2	3	
AR(1)	.780	.274	.390	.532
AR(1) + measurement error	.954	.902	.862	.229 (+ .418 meas.)

Classes 2 and 3 move from looking close to transient to being nearly as persistent as class 1. What the AR(1) fits read as health rebounding quickly was largely the instrument.

The variance is reallocated, not invented. This is the check that matters, because a model that simply added a free noise parameter could manufacture persistence. Total implied observation sd is preserved to 2–3%: $0.850 \rightarrow 0.871$ for class 1 and $0.577 \rightarrow 0.616$ for class 3. The innovation sd of 0.532 splits into a state innovation of 0.229 and a measurement scale of 0.418. In signal-share terms — the fraction of observation variance that is persistent state — the physical GRM runs .77/.62/.54 across classes. *Between a quarter and a half of the year-to-year movement in these scores is measurement.*

The instruments differ, and in the expected direction. σ_{meas} is .466 for the original GRM, .442 for P-FUNC and .418 for P-FULL. The first comparison is clean: the original GRM and P-FUNC run on essentially the same sample (50,042 and 50,001 people), so the GRM-2 item work bought a real reduction in measurement noise, not just a re-scaling. The second runs on a different sample and so is not clean, but the nature of the difference can be pinned down exactly, and it does not point the way the caveat would suggest.

P-FULL’s roster is a strict *subset* of P-FUNC’s: nobody is in P-FULL and not in P-FUNC. The loss is 11,038 whole people, which accounts for 99.8% of the 101,454 missing rows; window truncation is negligible, as 99.9% of the retained people keep an identical observation window (mean 8.58 waves under both). So this is a person-selection difference, not a shorter panel. The mechanism is the one anticipated in §2: 75.7% of the dropped people are first observed at wave 2 against 0.1% of the retained, i.e. they are the BHPS entrants who never receive the condition inventory.

On every quantity that ought to matter they are the same people. Mean $\theta_{\text{P-FUNC}}$ is +0.004 against –0.001 for the retained, with standard deviations of 0.908 and 0.902; mean age 51.7 against 50.8; mean birth year 1964 against 1965. Two differences are real and both cut *against* reading the noise gap as a sample artefact. Their mean absolute year-to-year change in $\theta_{\text{P-FUNC}}$ is 0.4349 against 0.4491, so they are slightly *less* volatile than the people P-FULL keeps — removing them should if anything raise the measured noise, not lower it. And they are observed *more* often (9.17 waves against 8.58), so their removal mildly weakens the identification that separates state from noise.

The direct test has since been run: P-FUNC refitted on the exact person-age cells of the P-FULL contract (38,963 people, 334,194 rows), so that the measure on the channel is the only thing that

differs. It returns $\sigma_{\text{meas}} = 0.4453$ (max $\hat{R}$ 1.003) against P-FULL’s 0.4184, a gap of 0.027 — slightly *larger* than the 0.023 across the two rosters, as the diagnostic above predicted it would be. The diagnosis items genuinely make the score a less noisy instrument, and the reading that conditions anchor it, being stable and consistently reported, survives the test it could most easily have failed.

The typology itself changes, not just the persistence. Class shares become markedly more balanced, close to .26/.49/.26 on all three channels, against the AR(1) fits’ .29–.33 in the worst class and the baselines’ .16. More interesting is *how* the worst class is now identified. Its age-30 gap to the best class is the narrowest of all eleven fits (0.91 sd on P-FULL, against 1.21 under AR(1) and 1.74 at baseline), while its decline is the steepest (–1.69 sd from 30 to 80, against –1.44 and –1.22). The fanning ratio — the age-80 gap over the age-30 gap — is $2.1\times$, the largest in the table. Once transient noise is removed the worst class is defined by its *trajectory* rather than by sitting low throughout: these are people who start close to everyone else and fall away.

And the composition strips reverse direction. Under AR(1) the worst class’s share among person-waves observed at each age drifts *upward* with age (.298 at 30 to .323 at 85 on P-FULL). Under the state-space specification it falls, and substantially: .279 at 30 to .177 at 85. This is what a fast-declining worst class should look like in an unbalanced panel that loses people — the class is progressively thinned by attrition and death — and it is a consequence of the reclassification above rather than an independent finding.

What this costs the earlier results. The class-graded persistence claim above — $\rho_1 = 0.67\text{--}0.78$ against 0.26–0.44 elsewhere — is a statement about an attenuated parameter. The *ordering* survives (class 1 remains the most persistent on all three state-space fits) but the *grading* largely does not: .95/.89/.81 is a far flatter profile than .78/.27/.39. Anything downstream that used ρ as a substantive quantity rather than a nuisance parameter — including the AR-conditional held-out densities below — should be read as conditional on the AR(1) specification.

Holdout twins of the three state-space fits were run on the same last-two-observations rule as the AR(1) holdout fits. The two sets of stored held-out densities are different estimands — the AR(1) ones ignore each person’s history, the state-space ones condition on it — so the comparison is made on matched estimands under *Out-of-sample prediction* below. Separating measurement error does improve the forecast, but by 0.084 nats per person rather than the 0.242 that setting the stored numbers side by side would suggest. Section 10 repeats the typology analysis of this subsection with four classes.

Out-of-sample prediction

What the model’s held-out density conditions on. As the eight fits were run, the Stan generated quantities restarted the AR(1) recursion at the first held-out row with the *stationary* variance, so the person’s own last fitted value was not carried forward through ρ . The fitted rows reach the held-out rows only through the class posterior (the second held-out row does condition on the first). That is a coherent estimand — what the latent class structure alone predicts — but it is not the forecast a persistence model is capable of, and it understates what these fits know. The model has since been corrected (*The corrected generated quantities*, below); the results below were computed offline on the same rows, with the offline machinery first validated by reproducing the fits’ own stored held-out densities to a correlation of 0.9999998. The AR-conditional forecast uses the same class posterior but carries the last fitted deviation forward, $\mu_k(a_h) + \rho_k^G(y_{\text{last}} - \mu_k(a_{\text{last}}))$, with the gap-adjusted predictive variance. The state-space holdout fits were run after the correction and store the history-conditioned estimand, which is why the two sets of stored numbers cannot be

compared directly.

Two benchmarks are estimated on the fitted rows only: an age-quadratic Gaussian, and last-observation-carried-forward (LOCF), which is a strong naive predictor for a persistent quantity like health.

Channel	Predictor	LPD/obs	RMSE	R^2	cover ₅₀	cover ₉₀
PCS	AR(1), history-conditioned	-1.005	0.671	.614	.56	.89
	AR(1), class-only	-1.043	0.706	.573	.60	.90
	LOCF	-1.118	0.740	.531	.61	.90
	age-quadratic	-1.416	0.991	.159	.52	.88
GRM phys (P-FULL)	AR(1), history-conditioned	-0.889	0.589	.672	.54	.91
	AR(1), class-only	-1.015	0.665	.582	.53	.91
	AR(1) + ME, history-conditioned	-0.834	0.561	.703	.55	.91
	AR(1) + ME, class-only	-1.040	0.660	.588	.52	.90
	LOCF	-0.993	0.653	.596	.56	.90
	age-quadratic	-1.376	0.958	.133	.50	.90
GRM phys (P-FUNC)	AR(1) + ME, history-conditioned	-0.896	0.597	.665	.54	.91
	AR(1) + ME, class-only	-1.066	0.687	.556	.53	.91
	LOCF	-1.059	0.697	.542	.55	.89
	age-quadratic	-1.396	0.976	.104	.49	.90
GRM original	AR(1) + ME, history-conditioned	-0.938	0.622	.636	.54	.91
	AR(1) + ME, class-only	-1.088	0.707	.530	.53	.91
	LOCF	-1.107	0.732	.496	.55	.90
	age-quadratic	-1.402	0.982	.093	.48	.90
GRM combined	AR(1), history-conditioned	-0.961	0.635	.627	.54	.90
	AR(1), class-only	-1.079	0.709	.535	.54	.90
	LOCF	-1.058	0.697	.551	.56	.90
	age-quadratic	-1.452	1.029	.020	.48	.89

Per held-out observation (two per person; 58–79k rows), on each channel’s standardised scale. 400 posterior draws; every model scored offline through one code path (`clustering/holdout_matched.py`). ME: AR(1) latent state plus one-period measurement error. History-conditioned forecasts carry the deviation (AR(1)) or filtered state (ME) at the end of the fitted window forward; class-only forecasts use the class posterior and the stationary distribution. cover₅₀ and cover₉₀ are the empirical coverage of central predictive intervals with nominal 0.50 and 0.90. Bold marks the best predictor in each block.

Age alone predicts almost nothing at the person level. The age-quadratic benchmark reaches $R^2 = .02$ –.16. Everything interesting in these data is between-person, which is the premise of the whole exercise and is worth having measured.

The class structure alone does not beat carrying the last observation forward. On the model’s own estimand, the mixture edges LOCF on the PCS (R^2 .573 vs .531) but *loses* to it on both GRM channels (.582 vs .596; .535 vs .551). A three-class typology plus an age polynomial is roughly as informative about a person’s next two observations as simply assuming they will be where they were last time — no better on the sharper measures. Stated plainly: the classes describe lifecycle shape well, but they are a coarse summary of an individual’s current position. The state-space class structure does no better on P-FULL (.588 against .596) and edges LOCF on P-FUNC and the original GRM — where LOCF is itself weaker, because it carries more measurement noise forward.

Persistence is what makes the model predict. Carrying the last deviation forward lifts R^2 to .61–.67, clearly ahead of LOCF on every channel, and improves the log density by .04–.13 nats

per observation. Carrying the filtered state forward does the same for the state-space fits, to R^2 .64–.70 and by .15–.21 nats over their own class-only forecasts. The gain over LOCF is largest on the physical GRM (.672 vs .596). The two ingredients are complementary: the class trajectory supplies where the person is heading, ρ supplies how much of where they are now to carry with them, and LOCF is the degenerate case that keeps the level and ignores the trajectory.

Separating measurement error improves the forecast — by about a third of what the stored numbers implied. Because the two sets of fits stored different estimands, both were rescored here on matched ones: the same people, the same held-out rows and the same 400 draws, through one code path that reproduces every fit’s own stored value to a correlation of 1.000. P-FULL is the one measure with both fits.

P-FULL, 29,901 people, 59,802 held-out rows	AR(1)	AR(1) + ME	difference
stored held-out density, per person	−1.830	−1.587	+0.242
estimand stored	class-only	history-conditioned	mismatched
joint density of both held-out rows, per person	−1.671	−1.587	+0.084
history-conditioned forecast, per observation	−0.889	−0.834	+0.054
RMSE	0.589	0.561	−0.028
R^2	.672	.703	+.031
class-only forecast, per observation	−1.015	−1.040	−0.025

On the joint density of a person’s two held-out observations — the estimand both models’ corrected generated quantities report — the state-space model gains 0.084 nats per person. Setting the stored numbers side by side gives 0.242, roughly three times too much, because it compares a forecast that uses the person’s history with one that does not. The gain is real and consistent across metrics: RMSE falls from 0.589 to 0.561, R^2 rises from .672 to .703, and 90% coverage is unchanged at .91. It comes entirely from a better estimate of where each person currently is. The class structure on its own forecasts slightly *worse* under the new specification (−1.040 against −1.015), so what the filter buys is separating a person’s persistent state from the noise on their last observation, not a better typology.

The cleaner instruments predict better, in the order of their measurement noise. Among the AR(1) fits the physical GRM predicts best (R^2 .672, against .627 for the combined bank and .614 for the PCS), the floor analysis (§7) showing up out of sample. The state-space fits extend the ranking: history-conditioned R^2 is .703 for P-FULL, .665 for P-FUNC and .636 for the original GRM. P-FUNC and the original GRM run on near-identical samples (38,923 and 38,971 people), so that pair is clean, and it follows their measurement error (.444 against .467). LOCF — which carries the last observation forward, noise and all — degrades along the same line, from R^2 .596 on P-FULL to .542 on P-FUNC to .496 on the original GRM, which is what a noisier instrument does to a naive forecast. P-FULL’s lead is on a different and smaller sample and should be read with that caveat. R^2 carries these comparisons because each measure is standardised over its own sample, so log densities for different measures are in different units.

Calibration is good at the tails, mildly conservative in the middle. 90% intervals cover 88–91%, close to nominal. 50% intervals cover 48–61%, over-covering on most fits — and the same pattern appears for the naive benchmarks, so it reflects health changes being more sharply peaked than Gaussian rather than a defect of the mixture. A heavier-tailed observation model (Student- t residuals) is the natural refinement.

The corrected generated quantities

The measurement above is a defect worth fixing rather than documenting, so `person_class_loglik` now takes a `prev_row` argument that anchors the AR(1) recursion: 0 opens a window at the stationary variance (correct for a person’s first fitted row), a row index opens it by carrying that row’s deviation forward. The generated-quantities block emits both estimands rather than replacing one with the other — `log_lik_heldout` is now AR-conditional, and `log_lik_heldout_marginal` preserves the class-only quantity these eight fits reported.

The model block is untouched: it passes `prev_row = 0`, and `log_prob` at a fixed parameter point returns bit-identical values before and after the change (-15686.1620000000 on the test payload). The existing posteriors therefore remain valid — only the reported held-out quantity changes. `tests/test_gq_heldout.py` recomputes both estimands in numpy from the same payload and draws and requires agreement to 10^{-5} , and checks that they coincide exactly when `ar_mode == 0`.

What the corrected block reports on these fits, per held-out observation (the joint density of the two-row window, so the second row conditions on the first’s realised value — which is why it exceeds the row-by-row forecast in the table above):

Channel	class-only (as run)	AR-conditional (corrected)	gain
PCS	−1.043	−0.964	+0.079
GRM physical	−1.015	−0.836	+0.180
GRM combined	−1.079	−0.908	+0.171

So the correction is worth 0.08–0.18 nats per held-out observation on the reported density, and most on the physical GRM — the channel whose persistence estimate is the strongest. Held-out numbers from the eight fits in this note are the class-only column, and should be read as class-structure diagnostics; anything refitted from here reports the corrected quantity. The state-space holdout fits do: -0.794 per observation on P-FULL, -0.856 on P-FUNC and -0.901 on the original GRM. Set against P-FULL’s -0.836 in this table, that is the 0.042 per observation — 0.084 per person — of the matched comparison above.

Caveats. The holdout samples are people with at least five observations, who are healthier and better-retained than the full roster; the holdout fits reproduce their full-sample siblings’ structure closely (θ within .02, ρ within .02 on every channel), which is reassuring but does not make them representative. And the combined-channel fits inherit §6’s unidimensionality caveat: a single class structure over two correlated dimensions.

10 Four classes under the state-space specification

The state-space fits of §9 were refitted with four classes for the two physical measures, P-FULL and P-FUNC. Everything else is unchanged — the AR(1) latent state, the one-period measurement error, the priors, the contracts — so each $K=4$ fit runs on exactly the same people as its $K=3$ counterpart and the class count is the only difference. This section repeats §9’s typology material at $K=4$, and adds what a change of K needs: where the extra class comes from, whether the model can recover such a class, and whether it earns its place. Two parts of §9 have no $K=4$ counterpart because the fits were not run: the same-sample contrast with AR(1), and holdout. The original GRM was not refitted at $K=4$.

Measure	K	Persons	θ (shares)	ρ (persistence)	σ	σ_{meas}	$\max \hat{R}$	wall
P-FULL	3	38,963	.261 / .478 / .261	.95 / .90 / .86	.229	.418	1.002	5.0 h
P-FULL	4	38,963	.226 / .425 / .278 / .071	.95 / .89 / .80 / .68	.241	.413	1.003	21.8 h
P-FUNC	3	50,001	.254 / .487 / .259	.95 / .89 / .82	.242	.442	1.004	9.9 h
P-FUNC	4	50,001	.223 / .445 / .261 / .071	.94 / .88 / .74 / .52	.257	.435	1.004	21.9 h

Class 1 is worst health. Wall times are not comparable across K: the K=4 fits ran four single-threaded chains on a shared machine, the K=3 fits three threads per chain.

Both K=4 fits converged cleanly. The two measures agree closely on what the fourth class is: 7.1% of people in both, placed above the other three.

Class trajectories and shares

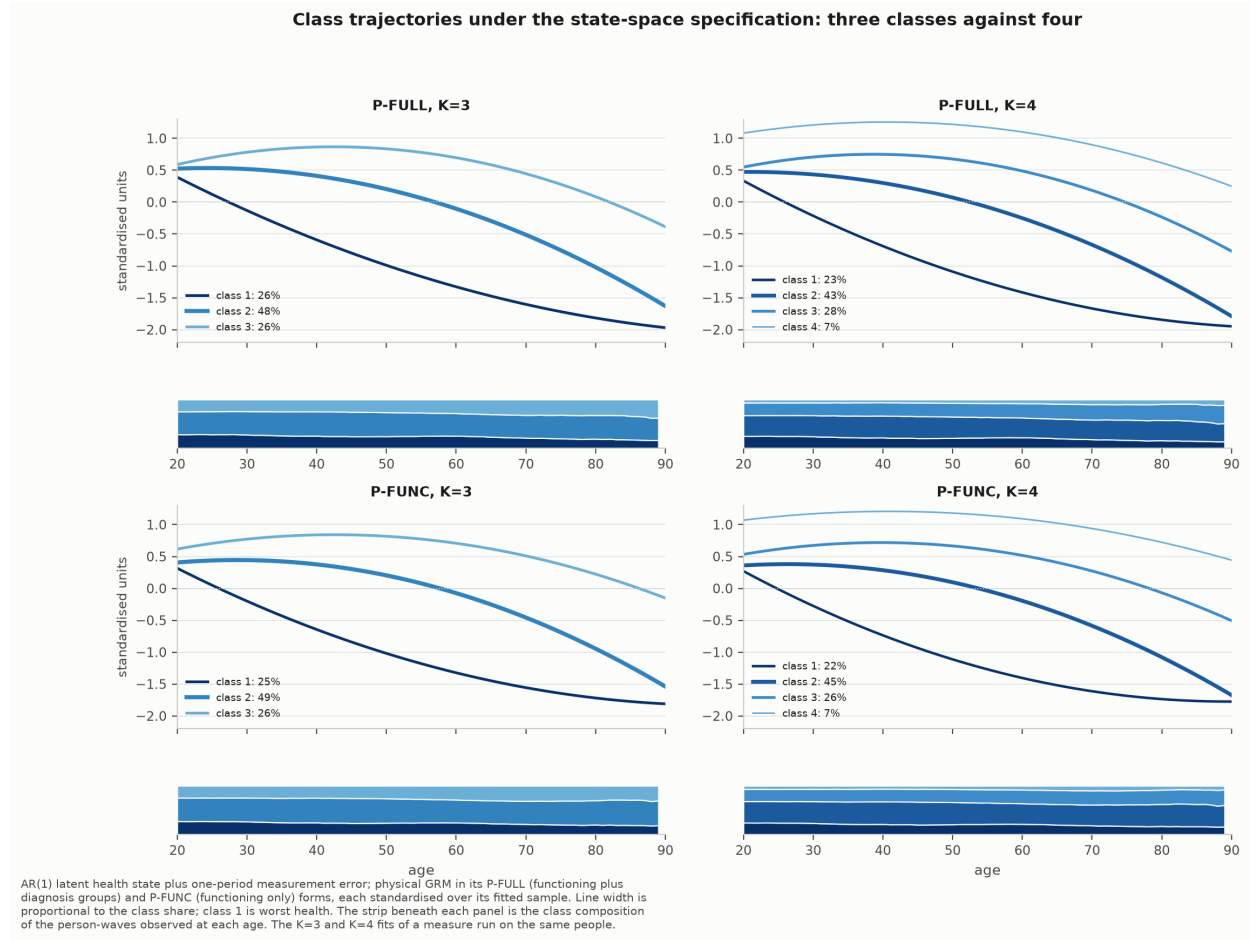


Figure 11: Fitted class mean trajectories $\alpha_k + \beta_k a + \gamma_k a^2$ (with $a = (\text{age} - 55)/10$) for the state-space specification at three and four classes. Rows: the physical GRM as P-FULL (`theta_phys_full`, functioning plus diagnosis groups) and as P-FUNC (`theta_phys_func`, functioning only), each standardised over its fitted sample. Columns: K=3 and K=4, fitted to the same people. Line width is proportional to the class share; class 1 is worst health. The strip beneath each panel is the class composition of the person-waves observed at each age.

Fit	worst-class share	level gap (best – worst) age 30	age 80	decline, age 30–80 worst	best
P-FULL, K=3	.261	0.91	1.90	–1.69	–0.70
P-FULL, K=4	.226	1.42	2.46	–1.63	–0.60
P-FUNC, K=3	.254	0.97	1.94	–1.52	–0.55
P-FUNC, K=4	.223	1.45	2.46	–1.46	–0.45

Standardised units. “Best” is class 3 at K=3 and the added class 4 at K=4, so the gaps are not like-for-like across K.

The first three classes keep their shape. Classes 1 to 3 at K=4 trace the same paths as the three classes at K=3, shifted slightly down: the worst class still starts close to the others and falls away fastest (–1.63 and –1.46 sd from 30 to 80, against –1.69 and –1.52), and it shrinks a little, from .26 to .23. The level gaps widen mainly because the “best” class is now a more extreme one; class 3 at K=4 sits at 0.71 and –0.24 at ages 30 and 80 on P-FULL, close to the K=3 top class’s 0.78 and 0.09.

The fourth class is high, flat and small. It starts at 1.21 sd on P-FULL (1.17 on P-FUNC) and declines only 0.60 (0.45) sd by age 80, the flattest path in either model. Its share of the person-waves observed at each age barely moves with age, so unlike the worst class it is not thinned by attrition.

	share	α	ρ	signal share	level, 30	level, 80
<i>P-FULL, K=4</i>						
class 1	.226	–1.26	.948	.77	–0.21	–1.85
class 2	.425	–0.08	.892	.63	0.43	–1.18
class 3	.278	0.60	.796	.48	0.71	–0.24
class 4	.071	1.17	.678	.39	1.21	0.62
<i>P-FUNC, K=4</i>						
class 1	.223	–1.27	.943	.76	–0.28	–1.74
class 2	.445	–0.04	.883	.61	0.37	–1.08
class 3	.261	0.60	.737	.43	0.67	–0.07
class 4	.071	1.14	.524	.32	1.17	0.72

α is the class level at age 55. Signal share is the persistent-state share of observation variance, $\sigma^2/(1 - \rho_k^2)$ over that plus σ_{meas}^2 .

Where the fourth class comes from

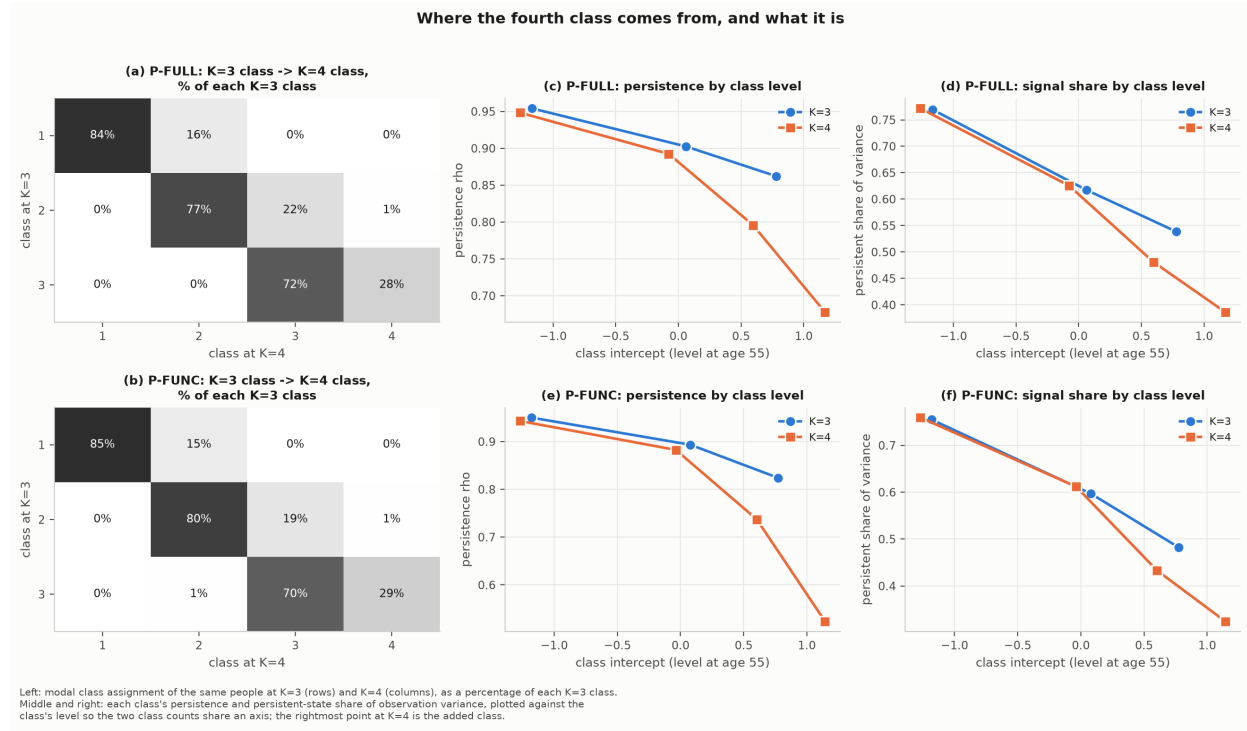


Figure 12: Left, (a) and (b): the modal class of each person at K=3 (rows) against their modal class at K=4 (columns), as a percentage of each K=3 class; the K=3 and K=4 fits run on the same people, so this is a person-by-person cross-tabulation. Modal assignment overstates how cleanly people map — the posterior-weighted version is more diffuse, see text. Middle and right, (c) to (f): each class's persistence ρ and signal share, plotted against the class's level at age 55 so the two class counts share one axis; the rightmost point at K=4 is the added class.

It is carved from the top, and every boundary moves up a step. Of the people in the healthiest K=3 class, 72% stay in the corresponding K=4 class and 28% move to the new class on P-FULL (70% and 29% on P-FUNC). Almost nobody reaches it from lower down: 1% of the middle K=3 class. The lower classes are not simply left alone, though. Each gives up its healthier members to the class above it — 16% of the worst K=3 class moves to K=4 class 2, and 22% of the middle class to K=4 class 3 — so adding a class at the top shifts the whole ladder rather than splitting one rung.

Modal assignment makes that look cleaner than it is. Weighting each person by their posterior class probabilities instead of their single most likely class, only 57% of the worst K=3 class's membership lands in K=4 class 1 and 35% in class 2; the top K=3 class splits 49/27/20 across K=4 classes 3, 2 and 4. The typical person is shared between neighbouring classes at either K, which is the overlap the fit statistics below measure.

Persistence and signal share extend the same gradient. At both K, ρ and the signal share fall monotonically as class level rises (panels c to f). The added class carries the lowest of both: $\rho = 0.68$ and a signal share of 0.39 on P-FULL, 0.52 and 0.32 on P-FUNC. Refitting also moves a little variation out of the noise term: σ_{meas} falls from .418 to .413 (P-FULL) and from .442 to .435

(P-FUNC), while the state innovation σ rises from .229 to .241 and from .242 to .257.

The instrument ranking holds. P-FULL remains the less noisy measure at $K=4$, $\sigma_{\text{meas}} = .413$ against .435 — the same ordering as at $K=3$, where §9’s same-roster control showed the gap is a property of the instrument rather than the sample. The $K=4$ P-FUNC fit is on the larger roster, so this particular comparison carries the cross-roster caveat of §9, which that control resolved in P-FULL’s favour.

Can the model recover a fourth class?

The real-data fits cannot say whether the model would find a fourth class of this shape if one truly existed. So a panel of 5,000 people was simulated from the fitted P-FULL $K=4$ posterior — class-specific persistence included, no ceiling imposed — and refitted with the same model at $K=4$ (four chains, 1,000 warmup and 600 sampling iterations).

Class	share		level α		persistence ρ		
	truth	estimate	truth	estimate	truth	estimate	90% interval
1	.226	.200	−1.02	−1.14	.948	.939	.927–.949
2	.425	.409	0.08	−0.00	.893	.887	.856–.914
3	.278	.299	0.70	0.66	.796	.789	.736–.836
4	.071	.093	1.23	1.14	.678	.765	.706–.814

On the simulated panel’s standardised scale. σ is recovered as .223 against .223, σ_{meas} as .383 against .383. Bold marks the two parameters, of 22, whose true value falls outside the 90% posterior interval.

It does. The fit converges cleanly — maximum $\hat{R}$ 1.008, no divergences, and the four chains agree on every persistence parameter to within .009, so there is no mode split. The true value lies inside the 90% interval for 20 of the 22 parameters, about the rate expected. The fourth class comes back at the right size and in the right place: a share of .093 against .071, inside its interval. The fitted fourth class is therefore not an artefact of the estimation: were such a class real, this model would find it.

Its persistence is overstated, in a telling direction. Both misses are in the fourth class. Its level is understated (1.14 against 1.23) and its persistence overstated (.765 against .678, with the truth below the interval). A third of the people truly in the fourth class (31.6%) are assigned to class 3, which is more persistent, and the estimate is pulled towards its neighbour. The real fit has roughly eight times as many people, so the pull there should be weaker; but its direction means the real fourth class’s persistence of .68 is, if anything, itself an overstatement. The class is not more persistent than it appears.

A true fourth class is no easier to assign. At the *true* parameters the average person’s most likely class has probability .689 (relative entropy .504), and only 69% of people are assigned to the class they were simulated in; at the estimates, .711 and 69%. That is the overlap this configuration produces when four classes are genuinely present: a persistent state blurred by one-period noise leaves neighbouring classes indistinguishable for many people. Members of the true fourth class are found about as often as anyone else (68%, against 59% for the worst class).

Does the fourth class earn its place?

Measure	K	log likelihood	parameters	BIC	classification	
					mean max. posterior	relative entropy
P-FULL	3	−295,489.9	16	591,149	.696	.403
P-FULL	4	−295,203.4	21	590,629	.668	.478
P-FUNC	3	−404,460.7	16	809,095	.704	.421
P-FUNC	4	−403,963.2	21	808,154	.679	.495

In-sample marginal log likelihood at the posterior-mean parameters, class states integrated out by the Kalman filter. BIC uses the number of people. Mean max. posterior is the average probability of each person’s most likely class; relative entropy is $1 - \text{entropy}/(N \ln K)$.

On fit, yes — but at this sample size that settles little. Four classes raise the log likelihood by 287 (P-FULL) and 498 (P-FUNC) for five extra parameters, and BIC prefers K=4 by 520 and 941. But a penalty of five parameters costs only about 27 log-likelihood units at this sample size ($5 \ln N/2$), against gains of 287 and 498, so BIC is weighing little here and is not the test that matters. Larger K has not been fitted.

On separation, the statistics cannot decide. People are assigned slightly less confidently at K=4: the average probability of each person’s most likely class falls from .696 to .668 on P-FULL and from .704 to .679 on P-FUNC. Relative entropy moves the other way, from .40 to .48, but only because it divides by $\ln K$. Neither is high at either K — relative entropies of .40–.50 fall well short of the values near .8 usually read as clean separation — but the recovery check shows that a genuine four-class population of this shape yields .69 at its true parameters. Certainty in this range is what a real fourth class would produce as well, so it is not evidence against one.

And the added class sits at the measurement ceiling. A sizeable share of the physical GRM is piled at its maximum, and that mass is concentrated in exactly the people the fourth class describes:

Person-mean level (sd)	P-FULL		P-FUNC	
	people	waves at maximum	people	waves at maximum
≥ 1.0	3,862	45.6%	4,322	58.8%
0.5 to 1.0	8,733	8.9%	11,517	13.8%
0 to 0.5	9,621	2.0%	12,566	3.5%
below 0	16,747	0.3%	21,596	0.5%
all person-waves		6.8%		9.3%

Share of each person’s observations recorded at the measure’s exact maximum, by the person’s mean standardised level, in the fitted lifecycle sample; hence 6.8% and 9.3% overall rather than §7’s whole-panel 7.2% and 9.5%.

People at the fourth class’s level have close to half of their observations recorded at the exact maximum. The model treats observations as Gaussian and has no notion of a ceiling, so a score that alternates between the maximum and just below it can only be read as transient noise around a weakly persistent state — which is precisely the fourth class’s signature: the lowest ρ and the lowest signal share of any class. The two measures line up the same way. P-FUNC, which piles more people at its ceiling (58.8% against 45.6%), gives its fourth class markedly lower persistence (0.52 against 0.68) and signal share (0.32 against 0.39). That is what a ceiling artefact predicts, and it echoes §7, where the diagnosis items were found to relieve the ceiling on P-FULL.

Reading. The fourth class is not an estimation artefact. It is recovered at the same size and position by two measures, it converges cleanly, it improves the in-sample fit, and simulation shows the model finds a class of this shape when one is truly there. What tells against reading it as a distinct trajectory is the ceiling, and the ceiling alone: its defining dynamics are the ones a measurement ceiling produces, and about half of its members’ observations sit at the exact maximum. The classification statistics bear on the question neither way. It is best read as the people at the top of the physical GRM’s range, whose recorded health cannot rise further, rather than as a distinct lifecycle trajectory. That is informative in its own right — it locates where measurement rather than health drives the estimated dynamics — but it is not a reason to prefer four classes to three for describing health trajectories, and the rest of this note keeps $K=3$.

What is not established. The recovery check shows that the model can find a fourth class, not that the real one is genuine: it was run without a ceiling, so it tests the model rather than the ceiling reading. It also used 5,000 people, an eighth of the fitted sample, so the size of the persistence bias it found is not the size in the real fit. There is no AR(1) fit at $K=4$, so §9’s same-sample attenuation contrast cannot be repeated, and no holdout, so whether $K=4$ predicts better out of sample is unknown. The ceiling reading therefore remains an interpretation of the pattern rather than a test of it. The direct tests would be to repeat this simulation from *three* true classes with scores censored at the measure’s maximum and see whether a small, weakly persistent top class appears, or to fit an observation model that treats scores at the maximum as censored.

11 Towards a multidimensional model

The predecessor’s five-channel mixed model used PCS and MCS (Gaussian), self-rated health (ordered logit), a chronic count (negative binomial) and an ADL count (hurdle negative binomial). Everything in this note argues for a different set, and the binding constraint is coverage.

Candidate channel	person-waves	persons	limiting gate
θ physical, P-FUNC	497,423	84,679	none (every wave)
θ mental, without DEPR	489,278	81,729	none (every wave)
θ mental, as fitted (with DEPR)	379,462	67,017	diagnosis inventory
θ physical, P-FULL	387,599	69,958	diagnosis inventory
chronic count	76% of person-waves		diagnosis inventory
ADL count	4 waves ($\sim 8\%$)		scheduled module

One weak item is costing a quarter of the sample. The mental bank as fitted includes ever-diagnosed depression, which inherits the BHPS exclusion of §3.4. Dropping that single item — one hurdle, $a = 0.93$, the weakest item in the bank bar the rare diagnoses — recovers 109,816 person-waves and 14,712 people, a 22.4% increase. Jointly observing a physical and a mental channel then covers 484,832 person-waves on 81,458 people (49,545 of them with three or more observations), against 376,019 / 66,841 / 38,478 if either channel carries a diagnosis item. That is a 29% larger estimation sample for the loss of almost no information.

Recommended specification. Three channels, and the ragged-channel machinery already in `runner/mixed.py` handles them as they stand:

1. θ **physical (P-FUNC)**, Gaussian. Replaces PCS and, because self-rated health enters as the GH item and functioning as FUNC, it also absorbs the old model’s separate self-rated-health

channel and its ADL channel — the latter traded for `disdif`, which measures the same construct in every wave rather than four.

2. **θ mental without DEPR**, Gaussian. Replaces MCS, without the orthogonality artifact, and with GHQ-12 doing most of the work.
3. **Chronic condition count**, negative binomial, on the 76% of person-waves where the inventory exists. Kept as a channel rather than folded into a measurement model, which is where §5 concluded it belongs: “ever” is the correct state variable for a diagnosis, its accumulation is a process to be modelled rather than an artefact to be corrected, and as a channel its missingness is explicit instead of silently deleting people.

Two properties of this design matter. First, the class structure is identified off channels 1 and 2, which are observed for everyone in every wave, so the BHPS members contribute to the typology even though channel 3 is missing for them — the opposite of the current arrangement, where they are dropped outright. Second, the missingness in channel 3 is a survey-design gate, not a health-related process: the excluded people are indistinguishable on every health measure (§3.4), so treating it as ignorable conditional on the observed channels is defensible in a way that ADL’s scheduled-module missingness also is, but that attrition would not be.

What to leave out, and why. Utilisation (§8) is an outcome for validating the measures, not a health dimension to be clustered on. SF-6D is a mixed physical–mental utility (§6) and would duplicate both channels at once. The combined GRM is physically dominated (Figure 5) and its residuals say the data want two dimensions, not one. SWEMWBS, loneliness and the ADL module all fail the every-wave rule.

An extension worth considering later. Channels 1 and 2 currently compress each domain to a point estimate and then treat it as data, discarding the posterior sd the GRM already computes (§7: 0.27 near the population mean, rising to 0.65 at the extremes). Passing that through as a known measurement-error variance per observation is a small change to the Gaussian channels and would stop the model reading heteroskedastic measurement noise as heterogeneity in health — which, given the ceiling and floor results, is concentrated exactly in the classes the model cares most about.

12 The multidimensional model

The specification of §11 was fitted in four variants — baseline, holdout, AR(1), AR(1)+holdout — on one common sample of 38,268 people and 388,774 person-waves, so the four differ by specification alone. Three channels: the physical GRM (P-FUNC) and the mental GRM without DEPR, both Gaussian, and the chronic count as a negative binomial, masked on the 24% of rows with no diagnosis inventory. Mortality was toggled off for these four fits; it is switched on in the later fits reported under *Measurement error and mortality* below. All four converged.

Variant	$\max \hat{R}$	θ (shares)	ρ	chronic at 55	wall
baseline	1.0032	.189 / .377 / .434	—	2.6 / 1.4 / 0.13	8.0 h
holdout	1.0049	.187 / .371 / .442	—	2.6 / 1.4 / 0.14	6.8 h
AR(1)	1.0014	.190 / .398 / .413	.73 / .61 / .60	2.9 / 1.1 / 0.01	19.0 h
AR(1) + holdout	1.0023	.195 / .389 / .416	.72 / .58 / .58	2.9 / 1.1 / 0.01	18.2 h

“chronic at 55” is the implied mean condition count at age 55 for each class.

Extra channels stabilise the classes against the AR term. In the univariate fits of §9 adding persistence roughly doubled the worst class ($.16 \rightarrow .33$ on both GRM channels). Here the shares are flat across all four variants at about $.19 / .38 / .42$. The mechanism in §9 was that a single channel cannot separate a small extreme class from the autocorrelation of repeated low observations, so once ρ can absorb the latter the boundary moves; with three channels the class is identified from agreement *across* channels, and persistence no longer competes for the same variation.

Persistence is much less class-graded than the univariate fits implied. $\rho = .73/.61/.60$ here, against $.78/.27/.39$ for the physical GRM alone. The worst class remains the most persistent, but the other two are far more so once a second correlated channel is observed: a shock that looks transient in one measure reads as persistent when two move together. That is a warning about reading ρ off a single channel.

The classes absorb the overdispersion in the chronic count. The implied means at 55 — $2.6/1.4/0.13$ conditions — weight up to 1.08 against an observed mean of 1.045, so the mixture reproduces the marginal. And `phi_chronic` settles at 656 (baseline) to 920 (AR), which for a negative binomial is effectively Poisson: after conditioning on class and age there is almost no overdispersion left to model.

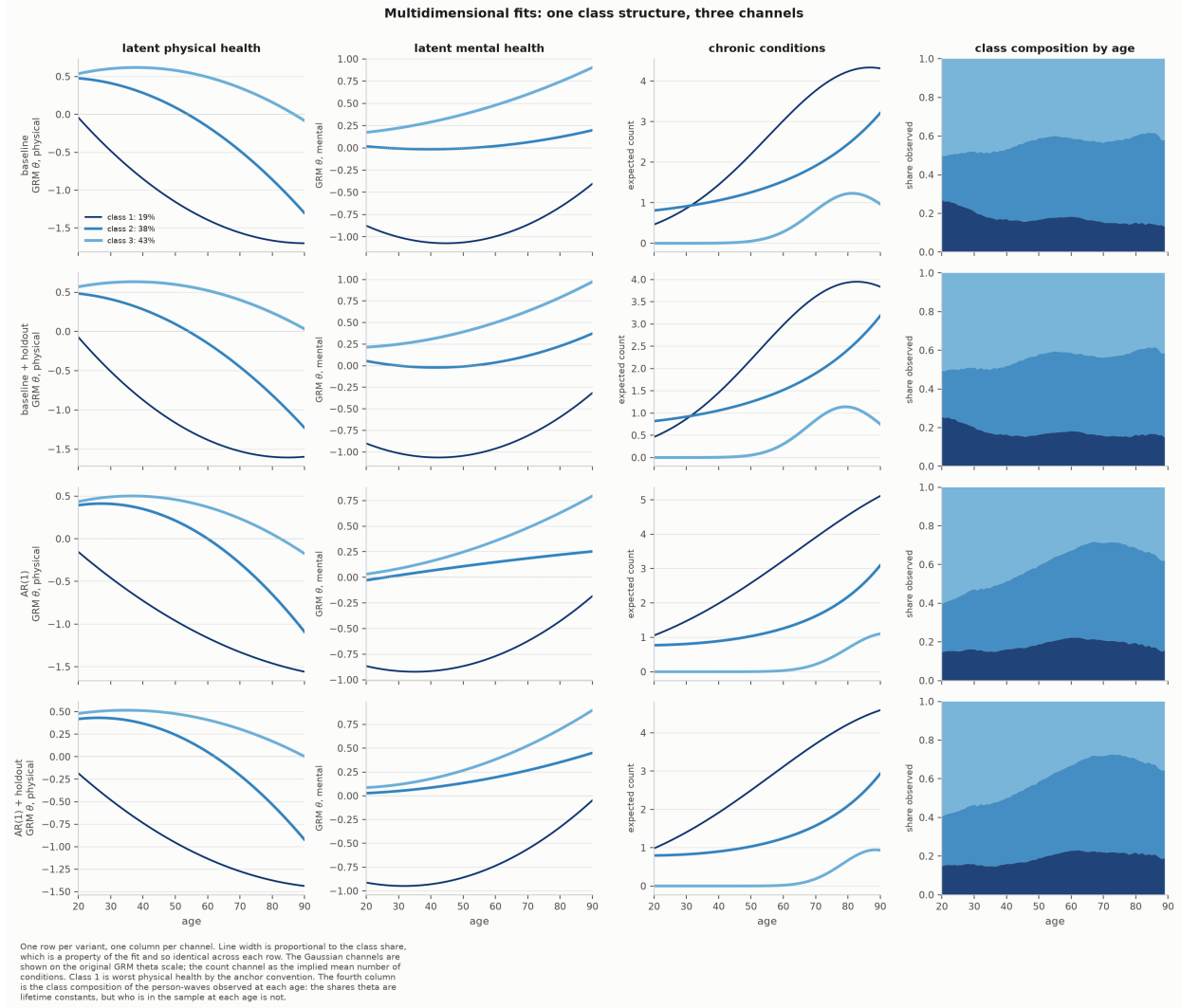


Figure 13: Class trajectories and shares for the four multidimensional fits, one row per variant and one column per channel. A fit has a single class structure, so the shares are identical across a row and the line width (proportional to the share) repeats; what changes across the row is what each class *does* on each channel. The Gaussian channels are returned to the original GRM θ scale using each fit's standardising moments; the count channel shows the implied mean number of conditions. Class 1 is worst physical health by the anchor convention. Together with Figure 10, which covers the eight univariate fits, every Bayesian model in this note has its trajectories and shares drawn.

Figure 13 makes the classes legible as types rather than as parameter vectors. The physical channel behaves as the univariate fits led us to expect: three ordered, non-crossing, accelerating declines, with class 1 falling from about -0.1 at age 20 to -1.7 at 90 while class 3 stays near $+0.5$ until the sixties. The mental channel is where the multidimensional model earns its keep. Its trajectories *rise* with age for every class, recovering the late-life wellbeing pattern of Figure 2 inside a model that is simultaneously fitting a steep physical decline — and they rise fastest for class 3, so the healthiest physical type is also pulling away mentally. Class 1 is worst on both channels at every age, which is worth stating because nothing in the model forces it: only the physical channel carries the ordering constraint, so the mental ordering is an estimate. The count channel then separates

the classes most sharply of the three, from about 4.5 conditions at age 80 in class 1 to 0.7 in class 3. The composition strips answer a question the shares alone cannot. In a mixture θ is a lifetime constant, so any movement in a strip is selection into the observed sample, not people changing class. Across the twelve fits that movement is modest but consistent in sign for the physical measures: the worst class rises from 12.1% of person-waves at age 30 to 18.8% at 80 on the PCS, and from 29.8% to 34.5% on P-FULL under AR(1) — the healthiest are observed at every age while the least healthy accumulate in the older cells. The multidimensional baselines tilt the other way (21.0% at 30 falling to 15.3% at 80), which is mortality and attrition removing the sickest, and a reminder that these strips describe who is present rather than how anyone’s health evolved.

Held-out prediction, channel by channel

Chronic is nearly free to predict — carrying the last observed count forward is exactly right on 93.4% of the 58,600 held-out chronic rows — so a single held-out number would be flattered by it. The density is therefore decomposed by channel offline, validated by reproducing the fits’ own stored per-person values (correlation 0.999999, maximum deviation 0.04).

Channel	AR(1) + holdout		holdout (no AR)
	AR-conditional	class-only	
θ physical	−0.936	−1.059	−1.121
θ mental	−1.138	−1.245	−1.321
chronic count	−0.981	−0.981	−1.086
all three	−1.021	−1.105	−1.184

Log predictive density per held-out row. 76,536 rows on each Gaussian channel, 58,623 on the count.

Three readings. First, the two Gaussian channels gain 0.12 and 0.11 nats per row from carrying the last value forward, while **the chronic channel gains exactly zero** — to three decimal places, because the model carries no persistence term on that channel at all. It predicts the count from class and age, never from the person’s own previous count.

Second, that omission is expensive. The distribution of the change from a person’s last observed count is 93.4% zero, 5.8% +1, 0.6% +2 or more, with an entropy of 0.274 nats. A channel that used the previous count could therefore reach about −0.27 per row where the fitted model achieves −0.98: roughly **0.7 nats per row left on the table**, far more than the 0.12 the AR term buys on the Gaussian side. This is the count-channel analogue of the generated-quantities defect of §9, and the obvious next fix: the chronic channel should condition on the last observed count, either as an offset or through a monotone accumulation process.

Third, the physical channel predicts better than the mental one on every variant (−0.94 against −1.14), the same ordering the univariate fits found, and the non-AR holdout has AR-conditional and class-only densities identical to four decimal places — correct by construction when there is no persistence to carry, and a useful check that the toggle behaves.

Measurement error and mortality

Three further multidimensional fits were run on the same 38,268 people and 388,774 person-waves: the state-space specification of §9 on both Gaussian channels, with and without the mortality

hazard, and the mortality hazard on the conditionally independent baseline. Measurement error enters the two GRM channels only. A condition count is a report of an accumulated stock rather than a noisy reading of a continuous state, and mortality is administrative, so neither has a measurement-error analogue. Mortality is a per-wave logit hazard with a quadratic in age, observed on every person-wave, with 1,996 deaths among the 38,268 people.

Both state-space fits split into two modes. The ordering constraint sits on the physical-GRM intercept, but under this specification classes 2 and 3 differ mainly on the chronic channel and lie close on the physical one. A chain that starts with those two labels the wrong way round cannot exchange them without crossing the constraint, so it settles against the boundary with the two intercepts pressed together — a gap of 0.000, against 0.32 in the other chains. That mode sits 480 log-posterior units below the other in both fits, so it is decisively worse rather than a competing account of the data, but pooling chains across the two modes produced split $\hat{R}$ values up to 54. Relabelling does not repair it, because a trapped chain sits at different parameter values rather than at a permutation of the same ones. The state-space results below therefore use only the chains in the better mode — three of four without mortality, two of four with — on which split $\hat{R}$ is 1.003 and 1.004. The baseline with mortality converged without a split on all four chains ($\hat{R}$ 1.001). The durable fix is to put the ordering constraint on the channel that actually separates the classes; until that is done the two state-space fits should be treated as provisional.

Variant	chains	max $\hat{R}$	θ (shares)	ρ	σ_{meas}	wall
baseline	4	1.003	.189 / .377 / .434	—	—	8.0 h
baseline + mortality	4	1.001	.189 / .377 / .434	—	—	13.2 h
AR(1)	4	1.001	.190 / .398 / .413	.73 / .61 / .60	—	19.0 h
AR(1)+ME	3 of 4	1.003	.171 / .413 / .416	.96 / .94 / .93	.445 / .549	35.1 h
AR(1)+ME + mortality	2 of 4	1.004	.171 / .413 / .416	.96 / .94 / .93	.445 / .549	37.3 h

ME: AR(1) latent state plus one-period measurement error on the two Gaussian channels; σ_{meas} is physical / mental. “chains” is the number used; for the state-space fits the rest were in the inferior mode. Implied condition counts at 55 are 2.6/1.4/0.13 for both baselines and 3.1/1.1/0.01 for both state-space fits.

The univariate finding carries into the joint model. Separating measurement error lifts persistence from .73/.61/.60 to .96/.94/.93. The innovation scale falls from 0.60 and 0.71 to 0.23 and 0.25 on the physical and mental channels, and measurement error takes up 0.445 and 0.549 respectively. The mental GRM is the noisier instrument, in line with its weaker held-out prediction above. And persistence becomes almost flat across the classes, as the univariate state-space fits of §9 found on a single channel.

Mortality is strongly graded by class, and changes nothing else.

Class	baseline + mortality			AR(1) + ME + mortality		
	age 55	age 70	age 85	age 55	age 70	age 85
1	0.53%	2.18%	8.58%	0.52%	2.01%	8.62%
2	0.17%	0.88%	5.79%	0.14%	0.78%	5.66%
3	0.08%	0.46%	3.53%	0.08%	0.48%	3.74%

Probability of dying in a wave, from the class-specific logit hazard at posterior-mean parameters. Class 1 is worst health.



Figure 14: Per-wave mortality hazard by latent class in the multidimensional model (physical GRM as P-FUNC, mental GRM without DEPR, chronic count), on a log scale from age 40, below which deaths are too few to identify the age quadratic. Panel (a): conditionally independent baseline, all four chains. Panel (b): AR(1) latent state plus measurement error, the two chains in the better mode.

At age 55 a person in the worst class is 7.0 times as likely (on the odds scale) to die in a given wave as one in the healthiest, and 6.5 times under measurement error. The hazards converge with age: by 85 the gap is 8.6% against 3.5%, or 2.4 times. The two specifications agree closely on all of it. Adding the channel leaves the typology exactly as it was: shares, intercepts and chronic profiles agree to three decimal places with and without mortality, on both specifications. With 1,996 deaths set against 777,548 Gaussian and 295,247 count observations, the channel cannot move a class boundary. It describes survival by class; it does not help define the classes.

Caveats. Mortality enters only the fits of the subsection above, where it describes survival by class without shaping the classes. The GRM thetas enter as point estimates: the posterior standard deviation the measurement model already computes (0.27 near the population mean, 0.65 at the extremes, §7) is discarded, so the mixture reads heteroskedastic measurement noise as heterogeneity in health, and does so most where the extreme classes live. Passing that through as a known per-observation variance is a small change to the Gaussian channels and the natural next extension. Finally, the univariate AR(1) comparison in §9 used P-FULL while this model uses P-FUNC, so those two are not a like-for-like pair. The univariate P-FUNC state-space fit of §9 is on the same measure, though not the same sample, and its measurement error of 0.442 sits almost exactly on the joint model's 0.445 for the physical channel; a univariate P-FUNC AR(1) fit would close the remaining gap.

13 Predicting outcomes the models never saw

Everything so far scores a model against the health variables it was fitted to. This section asks a different question: given a person's class posterior, how well can we predict outcomes that appear in *no* channel of *any* fitted model — death, and healthcare use? None of the fits scored here carries

a mortality channel — the later fits that do (§12) are deliberately left out, since for them death would no longer be out of sample — and utilisation was never a channel, so this is criterion validity, and it is the comparison that decides whether the typology is useful beyond describing its own inputs.

Three outcomes: death during the panel (person-level), any in-patient stay in the last twelve months, and the GP visit band 0–4 (h12gp; both person-wave, waves 7–15). Three predictor sets, each also carrying a quadratic in age so that what is measured is what each adds to age alone: age only; age plus the two free class posteriors; and age plus the model’s own continuous health measure, which is the fair competitor because it is what the classes summarise. People are split 70/30, the outcome model fitted on the 70 and scored on the 30.

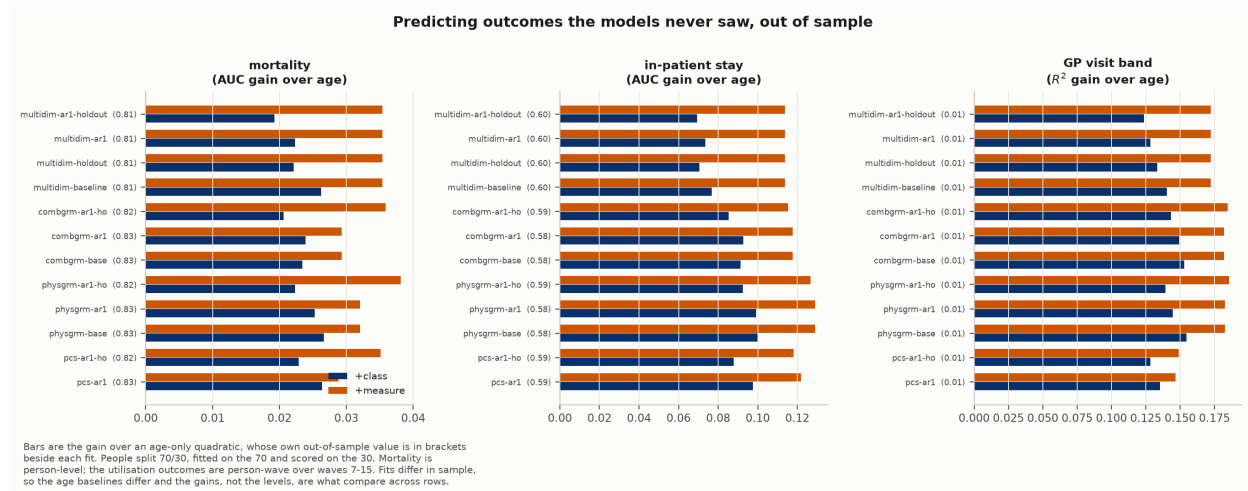


Figure 15: Out-of-sample gain over an age-only quadratic, whose own value is in brackets beside each fit. Fits differ in sample, so the age baselines differ and it is the gains, not the levels, that compare across rows.

	mortality AUC, in / out	in-patient AUC, in / out	GP visits R^2 , in / out
<i>age only</i>			
PCS sample	.836 / .831	.576 / .588	.011 / .012
multidim sample	.828 / .815	.586 / .596	.010 / .012
<i>age + class posterior</i>			
PCS, AR(1)	.856 / .857	.678 / .686	.143 / .148
P-FULL, baseline	.853 / .861	.674 / .681	.155 / .166
P-FULL, AR(1)	.852 / .860	.675 / .680	.148 / .157
combined, baseline	.850 / .856	.664 / .672	.157 / .165
multidim, baseline	.843 / .841	.667 / .673	.152 / .153
multidim, AR(1)	.841 / .837	.663 / .670	.144 / .141
<i>age + the continuous measure</i>			
P-FULL	.860 / .866	.705 / .710	.187 / .194
multidim (P-FUNC)	.855 / .850	.705 / .710	.182 / .185

The continuous measure beats the classes on every outcome, in every fit, in and out of sample. For GP visits the gap is large: R^2 of .194 against .166, so discretising into three classes

throws away about a sixth of what the measure knows. For in-patient stays it is .710 against .681 in AUC. This is the sharpest statement of a theme running through the note — the classes are a summary, and summarising costs. If prediction is the goal, use θ ; the typology earns its place by being interpretable and by describing trajectories, not by predicting better than the thing it summarises.

Age already predicts mortality well, and health adds little. The age-only AUC is .83, and the best class specification reaches .861 — a gain of about .027. Health measured in mid-life is a weak marginal predictor of death once age is known, at least at this horizon. Utilisation is the opposite: age alone gives AUC .59 for in-patient stays and $R^2 = .01$ for GP visit bands, and health does the overwhelming majority of the work. Prevention targeted on these measures should expect to move utilisation far more readily than mortality.

The multidimensional classes are not better than the univariate ones at this, and are slightly worse. On gain over age they are comparable for mortality (+.026 against +.027 for P-FULL baseline) but behind on both utilisation outcomes (+.077 against +.100 AUC for in-patient; +.141 against +.154 R^2 for GP visits). Adding channels bought stability and interpretation (§12), not external predictive power. Note the samples differ, so only the gains are comparable, and the multidim measure column is P-FUNC where the univariate is P-FULL.

The AR variants predict external outcomes slightly worse than their baselines, consistently across all three outcomes and both families. That is coherent with §9: the AR term absorbs within-person variation that would otherwise separate the classes, so the classes it leaves behind carry marginally less information about anything else.

Caveat. The class posteriors come from mixtures fitted on all people, so the 70/30 split tests the class-to-outcome relationship out of sample but not the whole pipeline; a fully clean version would refit each mixture on the training people. In-sample and out-of-sample figures are nonetheless close throughout — often the out-of-sample number is the higher of the two — so there is little sign of the outcome models overfitting.

14 A cost proxy, and whether health matters more with age

§8 could not put pounds on the vertical axis, and said so. A staged plan for a cost proxy built from UKHLS alone is set out in `cost_proxy_plan.pdf`; stages 1–3 are built and validated there. This section uses it for the two questions the plan was for.

The index prices three tiers of self-reported contact — in-patient spells and excess bed days from `hosp/hospd`, out-patient attendances from `h12hop`, GP visits from `h12gp` — at indicative national average unit costs, maternity separable. It averages £539 per person-year against NHS spend per head near £3,000, because it carries no prescribing, A&E, community or social care. It reproduces the published concentration of spending (top decile 65.6% against a published benchmark above half) and understates the age profile roughly threefold, for reasons diagnosed in the plan: unit cost per contact is held fixed. It is a bed-day-weighted index in pounds, not a cost, and is named that way throughout.

14.1 Convexity when the vertical axis is money

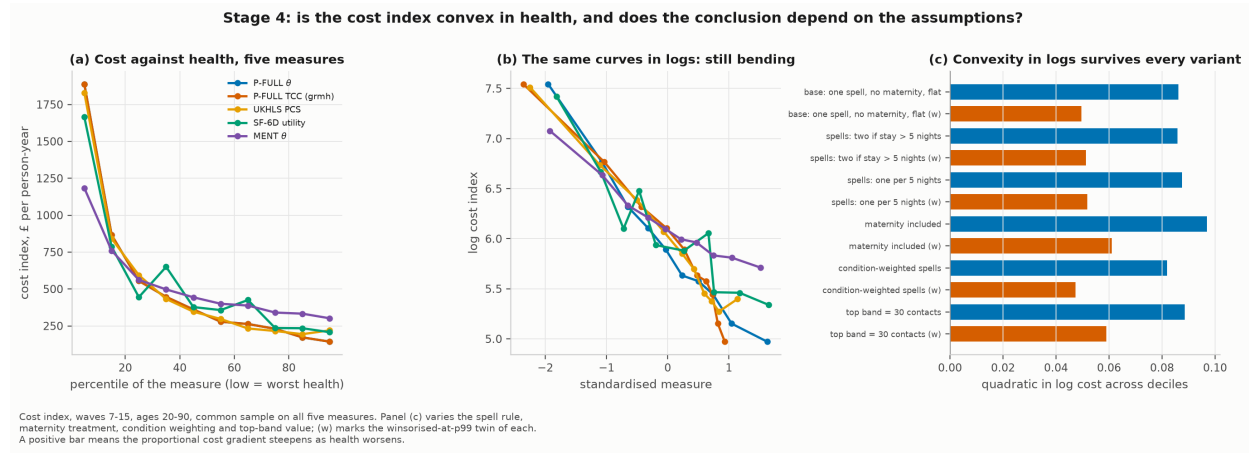


Figure 16: Cost index against five health measures, in levels and in logs, and the sensitivity of the log-scale convexity to every costing assumption the index rests on. Measures oriented so higher means better health. Panel (c) varies the spell rule, maternity treatment, condition weighting and top-band value, each with a winsorised-at-the-99th-percentile twin.

Measure	quad, levels (t)	quad, logs	bottom decile	top decile	ratio
P-FULL θ	+205 (67)	+0.086	£1,888	£143	13.2×
P-FULL TCC (grmh)	+125 (38)	-.111	£1,888	£143	13.2×
UKHLS PCS	+129 (40)	+0.002	£1,830	£220	8.3×
SF-6D utility	+207 (57)	+0.116	£1,665	£207	8.0×
MENT θ	+89 (30)	+0.079	£1,184	£302	3.9×

In pounds, every measure is convex — $t = 30$ to 67 — and the bottom decile costs 8 to 13 times the top on the physical measures.

But the scale problem of §8 does not go away. The worry there was that a binary outcome has no natural vertical scale. Pounds supply one, and the horizontal cardinalisation still decides the sign: on the log scale P-FULL θ gives +.086 and its own TCC gives -.111 on identical people with an identical ranking, while the PCS gives +.002, which is nothing. Fixing the vertical axis was necessary and is not sufficient. A claim that healthcare costs are convex in health remains a claim about the health ruler, and the honest statement is the conditional one: convex in θ , flat in PCS units, concave in grmh.

The costing assumptions, by contrast, do not matter. All twelve variants in panel (c) — one spell, two spells, one per five nights; maternity in or out; flat or condition-weighted; top band at 15 or 30 contacts; each with and without winsorising — are convex in levels and in logs. Winsorising at the 99th percentile roughly halves the magnitude (+.086 to +.050) and changes nothing else, and the health ratio stays between 8.3× and 14.6×. The spell ambiguity, which looked like the most damaging gap in the data, is not binding for this conclusion.

14.2 Does the health–cost relationship change with age?

The framework’s question is what a unit of health improvement is worth and whether that depends on when in life it happens. This is only askable because the GRM is fitted multigroup by single year of age against a pooled $N(0, 1)$: a θ of -1 means the same latent health at 30 as at 80. An age-standardised measure would build the answer in.

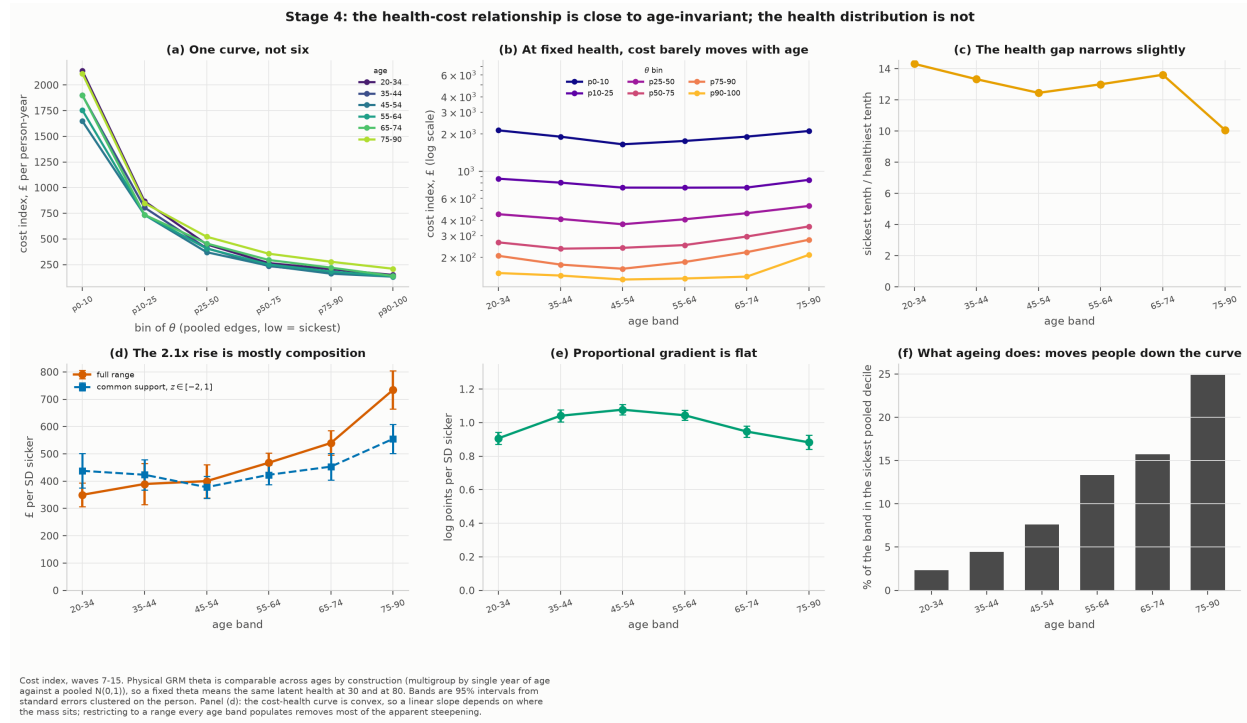


Figure 17: The cost index against physical health θ within age bands. Panel (a) plots cost across pooled θ bins, one line per age band; (b) the same table read down the other way; (c) the ratio of the sickest to the healthiest tenth; (d) the linear gradient over the full range and over the common support every band populates; (e) the same gradient in logs; (f) the share of each band falling in the sickest pooled tenth.

θ bin	20–34	35–44	45–54	55–64	65–74	75–90
p0–10	2138	1899	1648	1754	1902	2112
p10–25	868	806	735	734	736	849
p25–50	447	410	371	407	456	522
p50–75	265	235	239	252	295	356
p75–90	206	175	162	183	220	278
p90–100	149	142	132	135	140	210
sickest/healthiest	14.3×	13.3×	12.5×	13.0×	13.6×	10.1×

The answer is that it barely changes, and the raw table is the clearest way to see it. Read along the top row: a person in the sickest tenth of the health distribution costs £2,138 at 20–34 and £2,112 at 75–90, with a shallow minimum of £1,648 in midlife. A very sick 25-year-old and a very sick 85-year-old cost the index about the same. The sickest-to-healthiest ratio is 10–14× at every age and if anything narrows.

The absolute gradient does rise, and it is mostly composition. The linear slope goes from £350 per SD at 20–34 to £734 at 75–90, a $2.1\times$ rise. But the cost–health curve is convex, so a linear slope depends on where the mass sits, and the mass moves: 2.3% of the 20–34 band falls in the sickest pooled tenth against 24.9% of the 75–90 band, an elevenfold shift. Restricting to the common support $z \in [-2, 1]$, which every band populates, the slope runs £438 to £555 — a $1.27\times$ rise, non-monotone, with a midlife minimum. The proportional gradient is flatter still, 0.907 log points per SD at 20–34 against 0.882 at 75–90, hump-shaped with a peak of 1.077 at 45–54. Formally, with age in decades centred at 50 and standard errors clustered on the person, the $\theta \times$ age interaction is +9.48 ($t = 1.2$) in pounds — indistinguishable from zero — and +0.0196 ($t = 4.0$) in logs, statistically present and economically about two percent per decade.

What this means for the framework. The return to preventing a unit of ill-health is close to age-invariant in proportional terms. Older cohorts cost more because more of them are ill, not because illness is more expensive when you are old. That is a substantive claim and it cuts against the intuition that late-life prevention has mechanically larger returns: on this index it does not, once you condition on health rather than age. It also raises the value of preventing ill-health early, since a young person in poor health is already carrying close to the full cost.

Two cautions. The index understates the age profile threefold (`cost_proxy_plan.pdf` §Stage 3), and although that shortfall is diagnosed as flat unit costs rather than anything about health, a measure that recovered the true age profile could tilt these gradients. And UKHLS attrits and does not follow people into care homes, so the sickest old are under-represented in a way the sickest young are not; both cautions push in the direction of understating late-life cost.

A third caution: this is a statement about θ . The same test on `grmh`, the 0–1 expected-score scale, gives a proportional gradient that falls by about a third across the life course: on P-FULL, 1.12 log points per SD at 20–34 against 0.73 at 75–90, with an interaction of +0.048 ($t = 10.3$) against +0.020 ($t = 4.0$) on θ . Nothing about health differs between the two. h is a fixed monotone transform of θ and the ranking is identical; what differs is the unit. The test characteristic curve is steepest where the limitation and condition items do their work, low down, so the 0–1 scale is stretched exactly where the old sit: at the mean θ of the 75–90 band its slope is 0.28 against 0.10 at 20–34 on P-LIM3. One standard deviation of h there is a smaller move in latent health, and so a smaller cost difference. A gradient per standard deviation is not invariant to a nonlinear re-cardinalisation — the same point the costing sensitivity made when θ and its own TCC gave log-scale quadratics of opposite sign (`cost_proxy_plan.pdf` §Stage 4) — so §15 and §16 report both scales, and the workbook carries the gradient at 20–34 and at 75–90 for every measure.

15 Physical SF-12 plus limitations: six banks, two models

The baseline measure for the prevention paper is narrowing to the physical SF-12 items plus the impairment questions. This section builds six such banks, fits each under the graded-response model and under the generalised partial credit model, and takes all twelve scores through the baseline criteria: floor, cost convexity, age profile, identification and clusters. The same criteria for these and the 26 earlier candidates are in `redo_concepts/baseline_measure_comparison.xlsx`, where the new rows are shaded blue.

15.1 The banks

Every bank has the four validated SF-12 testlets, GH, PF, RP and BP, and adds counts over the impairment areas that follow the long-standing illness question (§3.4): whether health problems cause substantial difficulty with 1 mobility, 2 lifting, carrying or moving objects, 3 manual dexterity, 4 continence, 5 hearing, 6 sight, 10 physical coordination, 11 personal care, or 12 another health problem or disability. The cognitive areas, 7 to 9, stay out for a possible mental-health version.

Bank	Limitation testlets added to GH, PF, RP, BP
P-FUNC	physical function (1, 2, 3, 10, 11), capped at 3+
P-LIM1	physical function, uncapped 0–5
P-LIM	physical function 0–5; sensory and continence (4, 5, 6)
P-LIM4	mobility and lifting (1, 2); dexterity, coordination and care (3, 10, 11); continence (4); hearing and sight (5, 6)
P-LIM3	functional limitations (1, 2, 3, 10); self-care (4, 11); hearing and sight (5, 6)
P-LIM3+O	P-LIM3 plus other health problem (12)

Two coding rules hold throughout. Anyone without a long-standing illness counts as having no limitation in every wave: the routing of waves 1 to 7, applied to all waves. From wave 8 those people are asked too, so in waves 8 to 15 the rule discards 14% of all physical-function mentions and 19.5% of sensory and continence mentions (23% for hearing). And a top count category holding less than 0.5% of person-waves merges into the one below, decided before any fitting. Only one category merges: three sensory or continence difficulties, at 0.18%. All six banks share P-FUNC’s 497,423 person-waves from 84,679 people.

Testlet	Label	Areas	% with 0 / 1 / 2 / ... difficulties	Banks
Lim. 3+	Physical function, capped	1, 2, 3, 10, 11	82.82 / 6.57 / 4.86 / 5.75 (top is 3+)	P-FUNC
Lim. 0–5	Physical function	1, 2, 3, 10, 11	82.82 / 6.57 / 4.86 / 2.98 / 1.79 / 0.98	P-LIM1, P-LIM
Sens.+cont.	Sensory and continence	4, 5, 6	92.94 / 5.75 / 1.31 (top is 2+)	P-LIM
Mob.+lift.	Mobility and lifting	1, 2	84.05 / 7.37 / 8.58	P-LIM4
Dex.+care	Dexterity, coordination, personal care	3, 10, 11	91.51 / 5.24 / 2.21 / 1.04	P-LIM4
Cont.	Continence	4	96.41 / 3.59	P-LIM4
Sensory	Hearing and sight	5, 6	95.60 / 3.83 / 0.56	P-LIM4, P-LIM3, P-LIM3+O
Func. lim.	Functional limitations	1, 2, 3, 10	83.01 / 6.67 / 5.33 / 3.27 / 1.72	P-LIM3, P-LIM3+O
Self-care	Self-care	4, 11	94.21 / 4.70 / 1.09	P-LIM3, P-LIM3+O
Other	Other health problem	12	93.13 / 6.87	P-LIM3+O

15.2 Two measurement models

The graded-response model describes the chance of answering at or above each category. The generalised partial credit model (Muraki 1992) describes the chance of each category against the one below. Its likelihood depends on θ only through the weighted sum $T = \sum_j a_j(y_j - 1)$, so under one prior the score is exactly a fixed monotone curve applied to T , with one weight per testlet for each category step. The estimator, `measures/gpcm.py`, mirrors the graded-response code line for

line and recovers known parameters in simulation. On every bank the fitted score never falls as T rises.

Bank	Testlets	Patterns	$\log L$, GRM	$\Delta \log L$, GPCM	$\max Q_3$, GRM	$\max Q_3$, GPCM	$r(\theta)$
P-FUNC	5	3,833	-2,474,821	-13,475	-0.048	-0.048	0.995
P-LIM1	5	4,756	-2,502,311	-14,920	-0.045	-0.052	0.995
P-LIM	6	9,191	-2,615,869	-15,190	+0.083	+0.056	0.995
P-LIM4	8	15,008	-2,684,779	-14,258	+0.074	+0.066	0.995
P-LIM3	7	12,459	-2,653,722	-14,170	+0.061	+0.052	0.995
P-LIM3+O	8	17,201	-2,759,791	-14,164	+0.074	+0.076	0.995

The graded-response model fits every bank better, by 13,500 to 15,200 log-likelihood points for the same number of parameters. The shortfall comes from PF and RP. Each sums two items, and answers that agree across the pair are more common than the totals in between, so the partial credit step parameters come out of order (RP's eight steps alternate), which the cumulative model absorbs. The two models' θ still correlate at 0.995. No bank shows local dependence. The largest Q_3 is +0.083, between the two counts of P-LIM, which share a question and the illness gate. Functional limitations and self-care reach only +0.051, so moving personal care and continence into their own testlet created no dependence. The non-specific item's largest residual is with general health, +0.074.

15.3 Criteria

Floor is the share of 60 to 84 year olds at the worst score, and Distinct the number of distinct values in the bottom decile at 85 to 90. Cost t is the person-clustered t -statistic on the quadratic term with costs capped at their 99th percentile. Var. is the variance at 85 to 89 over the variance at 60 to 64, with 0.90 to 1.10 counted as flattening. C3 and C2 mark the sign path, positive $\text{Corr}(H, d)$ at 25 to 40 and negative at 75 to 90, under Case 3 on balanced moments and Case 2 on pairwise moments.

Bank	Model	Floor	Distinct	h scale				θ scale			
		60-84	85-90	Cost t	Var.	C3	C2	Cost t	Var.	C3	C2
GRM original		1.43%	21	20.3	1.01	✓	✓	38.2	0.77	✓	✓
	phys	1.42%	21	20.0	0.99	✓	✓				
P-FUNC	GRM	1.11%	42	15.0	1.07	✓	✓	37.5	0.77	✓	✓
	GPCM	1.11%	44	13.1	1.10	✓	×	38.1	0.74	✓	✓
P-LIM1	GRM	0.37%	98	13.2	1.08	✓	×	36.9	0.77	✓	✓
	GPCM	0.37%	101	11.0	1.11	✓	×	37.4	0.74	✓	✓
P-LIM	GRM	0.10%	227	11.9	1.11	✓	×	36.9	0.78	✓	✓
	GPCM	0.10%	232	9.6	1.15	✓	×	37.9	0.75	✓	✓
P-LIM4	GRM	0.02%	344	10.6	1.13	✓	×	36.6	0.78	✓	✓
	GPCM	0.02%	351	8.4	1.16	✓	×	37.5	0.75	✓	✓
P-LIM3	GRM	0.02%	348	11.4	1.12	✓	×	36.8	0.78	✓	✓
	GPCM	0.02%	353	9.2	1.15	✓	×	37.8	0.75	✓	✓
P-LIM3+O	GRM	0.01%	428	11.3	1.11	✓	×	37.4	0.77	✓	✓
	GPCM	0.01%	431	9.1	1.15	✓	×	38.4	0.75	✓	✓

- **The floor goes.** The share of 60 to 84 year olds at the worst score falls from 1.11% for P-FUNC to 0.37% once the count is uncapped, 0.10% with sensory and continence, 0.02% for

P-LIM3 and P-LIM4, and 0.01% with the other item. Distinct values in the bottom decile at 85 to 90 rise from 42 for P-FUNC to 98 for P-LIM1, 227 for P-LIM, and between 344 and 431 for the three finer banks.

- **Cost convexity survives but weakens.** On h the capped t falls from 15.0 to between 10.6 and 13.2 under the graded-response model and to between 8.4 and 13.1 under partial credit, all well above 2. Spreading out the sickest straightens the cost curve, as it did for the earlier composites. On θ it stays near 37 for every score.
- **Variance flattens on h and falls on θ .** Growth from 60 to 85 is 1.07 for P-FUNC and 1.08 to 1.13 for the new banks, so most sit just above the flattening band; partial credit adds about 0.03. On θ the variance falls, to about three quarters.
- **Identification splits on h .** Case 3 is admissible in every band with the sign path for all twelve scores on both scales. Case 2 keeps the sign path on θ everywhere, but on h only for P-FUNC under the graded-response model: at 75 to 90 its correlation of -0.02 becomes $+0.01$ to $+0.05$ in the new banks and $+0.01$ to $+0.10$ under partial credit. Section 15.4 shows that the moments, not the noise specification, make the difference.
- **Mortality barely changes.** The log-odds slope per standard deviation of h is -0.86 to -0.89 , against -0.91 for P-FUNC.

15.4 Moments or noise? The estimators swapped, and Case 4

The two scorecard estimators differ twice over: Case 3 frees the noise variance at each age and uses balanced moments, from people observed at all four ages; Case 2 has one noise variance and uses pairwise moments, where each cell uses everyone observed at both of its ages, so the variances are cross-sectional. Running each case on both kinds of moments separates the two. Case 4 adds a one-period error to Case 2's noise, $\sigma_{\text{slow}}^2 \rho^{|s-t|} + \sigma_{\text{spike}}^2 \mathbf{1}[s = t]$, with ρ profiled over 0.30 to 0.97 as in `redo_concepts/testing_metrics`, because below 0.30 the slow term becomes a second spike. On four ages two years apart it has five coefficients and ρ against ten moments. Admissible now also requires both noise variances to be non-negative.

		Balanced moments						Pairwise moments					
		Case 2		Case 3		Case 4		Case 2		Case 3		Case 4	
Bank	Model	path	75–90	path	75–90	path	75–90	path	75–90	path	75–90	path	75–90
<i>h scale</i>													
P-FUNC	GRM	✓	−0.05	✓	−0.22	✓ ^o	−0.02	✓	−0.02	×	−0.34	×	+0.49
	GPCM	✓	−0.05	✓	−0.22	✓ ^o	−0.02	×	+0.01	×	−0.34	×	+0.21
P-LIM1	GRM	✓	−0.04	✓	−0.20	✓ ^o	−0.02	×	+0.01	×	−0.34	×	—
	GPCM	✓	−0.04	✓	−0.19	✓ ^o	−0.02	×	+0.06	×	−0.33	×	+0.52
P-LIM	GRM	✓	−0.04	✓	−0.19	✓ ^o	−0.01	×	+0.04	×	−0.33	×	—
	GPCM	✓	−0.03	✓	−0.19	✓ ^o	−0.01	×	+0.09	×	−0.32	×	+1.18
P-LIM4	GRM	✓	−0.03	✓	−0.18	✓ ^o	−0.01	×	+0.05	×	−0.33	×	—
	GPCM	✓	−0.02	✓	−0.19	✓ ^o	−0.00	×	+0.09	×	−0.32	×	+0.53
P-LIM3	GRM	✓	−0.04	✓	−0.17	✓ ^o	−0.01	×	+0.05	×	−0.33	×	—
	GPCM	✓	−0.03	✓	−0.19	✓ ^o	−0.00	×	+0.10	×	−0.32	×	+0.88
P-LIM3+O	GRM	✓	−0.04	✓	−0.17	✓ ^o	−0.02	×	+0.04	×	−0.33	×	—
	GPCM	✓	−0.03	✓	−0.17	✓ ^o	−0.01	×	+0.09	×	−0.33	×	+0.77
<i>θ scale</i>													
P-FUNC	GRM	✓ ^o	−0.39	✓	−0.38	✓ ^o	−0.51	✓	−0.37	×	−0.39	✓ ^o	−0.87
	GPCM	✓ ^o	−0.53	✓	−0.41	✓ ^o	−0.89	✓	−0.46	×	−0.40	✓ ^o	−1.12
P-LIM1	GRM	✓ ^o	−0.39	✓	−0.38	✓ ^o	−0.50	✓	−0.37	×	−0.38	✓ ^o	−0.77
	GPCM	✓ ^o	−0.53	✓	−0.40	✓ ^o	−1.02	✓	−0.46	×	−0.39	✓ ^o	−0.91
P-LIM	GRM	✓ ^o	−0.38	✓	−0.37	✓ ^o	−0.46	✓	−0.35	×	−0.38	✓ ^o	−0.72
	GPCM	✓ ^o	−0.52	✓	−0.40	✓ ^o	−0.87	✓	−0.44	×	−0.39	✓ ^o	−0.94
P-LIM4	GRM	✓ ^o	−0.39	✓	−0.37	✓ ^o	−0.47	✓	−0.38	×	−0.38	✓ ^o	−0.77
	GPCM	✓	−0.49	✓	−0.40	✓ ^o	−0.80	✓	−0.43	×	−0.38	✓ ^o	−1.06
P-LIM3	GRM	✓ ^o	−0.37	✓	−0.37	✓ ^o	−0.45	✓	−0.35	×	−0.38	✓ ^o	−0.74
	GPCM	✓ ^o	−0.50	✓	−0.40	✓ ^o	−0.83	✓	−0.44	×	−0.39	✓ ^o	−1.04
P-LIM3+O	GRM	✓ ^o	−0.38	✓	−0.37	✓ ^o	−0.45	✓	−0.35	×	−0.38	✓ ^o	−0.74
	GPCM	✓ ^o	−0.50	✓	−0.40	✓ ^o	−0.77	✓	−0.44	×	−0.39	✓ ^o	−0.99

✓: positive $\text{Corr}(H, d)$ at 25 to 40 and negative at 75 to 90, admissible in all four bands at the best-fitting ρ . ✓^o: the same signs, but not admissible in every band. ×: no sign path. 75–90 is the oldest band’s correlation at the best fit; values beyond ± 1 are inadmissible, and a dash marks a correlation that is undefined because $V[H]$ and $V[d]$ have opposite signs.

- **The moments decide.** On balanced moments, Case 2 keeps the sign path on h for every bank and both models, admissible throughout, with the oldest band at -0.02 to -0.05 . On pairwise moments Case 3 loses it too, at the youngest band rather than the oldest: its correlation there is -0.13 to -0.16 . So the new banks’ failure on pairwise moments is not an artefact of Case 2’s single noise variance; the noise specification only decides which band fails.
- **Case 3 on pairwise moments never gives the sign path.** Its youngest band is negative for all 48 measures in the workbook, the largest being -0.09 , while it stays admissible.
- **Case 4 follows Case 2.** It is signed on balanced moments for every bank and unsigned on pairwise moments on h . As in `testing_metrics`, its best fit is never admissible in all four bands: in the youngest band $\hat{\rho}$ runs to 0.91 – 0.97 and the persistent term absorbs the level, leaving $V[H]$ near zero and a correlation outside ± 1 . At the best-fitting admissible ρ in each band it is admissible and signed on balanced moments for all twelve h scores.
- **On θ the pattern shifts.** Case 3 on balanced and Case 2 on pairwise moments keep the sign path. Case 2 on balanced moments and both versions of Case 4 are signed but inadmissible at the best fit, and Case 3 on pairwise moments fails the youngest band as it does on h .

15.5 Implied weights

Weights are shares of total weight on standardised testlets. On the latent scale, each discrimination becomes a loading on the underlying response, $\lambda = a/\sqrt{a^2 + 1.702^2}$, weighted by λ/ψ with $\psi = 1 - \lambda^2$; the polychoric one-factor model is its twin. On the observed scale, each model's h is regressed on the standardised codes, the partial credit model's exact weights are a times the testlet's standard deviation, and the Pearson one-factor model and first principal component are the linear comparisons. The tables below give every weight.

- **Each twin pair agrees.** Graded-response and polychoric loadings correlate at 0.97 to 0.99 within each bank. On the observed scale, the graded-response projection (R^2 0.993), the partial credit exact weights, its projection (R^2 0.999) and the Pearson factor weights agree to within about two points in every bank, and four for the uncapped physical-function count. The graded-response score is close to a weighted sum with Pearson factor-score weights; partial credit makes the weighted sum exact, with nearly the same weights.
- **Counts lose weight between the scales.** Most people sit at zero, so a count moves the observed score less than its latent loading suggests. In P-LIM3, functional limitations get 21% of latent weight and 16% of observed weight, and self-care 12% and 6%. The polychoric weights push the counts the other way, to 30% for functional limitations and 35% for mobility and lifting in P-LIM4. That is why polychoric weights on the codes over-weight them.
- **The added testlets carry little weight.** On the observed scale, self-care has 6%, hearing and sight 3%, and the other item 3%. They break ties among the sickest more than they move anyone's score. The first principal component, which does not discount weak items, gives them 12%, 8% and 7%.

P-FUNC	GH	PF	RP	BP	Lim. 3+	
GRM discrimination a	1.98	3.68	3.36	2.40	3.31	
GPCM slope a	1.29	2.30	1.29	1.46	2.25	
GRM, latent: λ/ψ	10	29	24	14	24	
Polychoric one-factor	10	29	21	13	28	
GRM h on codes (R^2 .993)	14	27	28	16	17	
GPCM, exact: $a \times sd$	13	27	27	15	17	
GPCM h on codes (R^2 .999)	12	27	28	15	18	
Pearson one-factor	12	29	28	16	16	
Pearson first component	18	21	21	20	20	
P-LIM1	GH	PF	RP	BP	Lim. 0–5	
GRM discrimination a	1.98	3.67	3.36	2.40	3.26	
GPCM slope a	1.29	2.27	1.28	1.45	2.08	
GRM, latent: λ/ψ	10	29	24	14	23	
Polychoric one-factor	10	29	21	13	27	
GRM h on codes (R^2 .993)	13	27	27	15	17	
GPCM, exact: $a \times sd$	13	26	27	15	19	
GPCM h on codes (R^2 .999)	11	27	27	15	19	
Pearson one-factor	12	29	29	16	15	
Pearson first component	18	21	21	20	20	
P-LIM	GH	PF	RP	BP	Lim. 0–5	Sens.+cont.
GRM discrimination a	1.99	3.63	3.31	2.39	3.38	1.61
GPCM slope a	1.30	2.24	1.25	1.43	2.20	1.44
GRM, latent: λ/ψ	9	26	22	13	23	7
Polychoric one-factor	9	25	18	11	30	6
GRM h on codes (R^2 .993)	13	25	26	15	17	4
GPCM, exact: $a \times sd$	13	25	25	14	19	4
GPCM h on codes (R^2 .999)	11	25	26	14	19	4
Pearson one-factor	11	27	27	15	15	5
Pearson first component	16	19	19	17	18	11

P-LIM4	GH	PF	RP	BP	Mob.+lift.	Dex.+care	Cont.	Sensory
GRM discrimination a	1.99	3.57	3.20	2.38	3.83	3.05	1.80	1.37
GPCM slope a	1.30	2.19	1.19	1.43	3.18	2.59	1.95	1.26
GRM, latent: λ/ψ	7	20	16	10	22	15	6	4
Polychoric one-factor	7	14	11	7	35	17	5	3
GRM h on codes (R^2 .993)	12	22	23	14	15	8	3	2
GPCM, exact: $a \times sd$	12	22	22	13	16	10	3	3
GPCM h on codes (R^2 .999)	11	23	23	13	16	9	3	2
Pearson one-factor	10	24	22	13	16	8	4	3
Pearson first component	13	15	15	14	15	13	8	7
P-LIM3	GH	PF	RP	BP	Func. lim.	Self-care	Sensory	
GRM discrimination a	1.99	3.60	3.28	2.38	3.38	2.43	1.36	
GPCM slope a	1.30	2.21	1.24	1.43	2.32	2.36	1.25	
GRM, latent: λ/ψ	8	23	20	11	21	12	5	
Polychoric one-factor	9	20	15	10	30	12	4	
GRM h on codes (R^2 .993)	13	24	25	14	16	6	3	
GPCM, exact: $a \times sd$	12	24	24	14	17	6	3	
GPCM h on codes (R^2 .999)	11	25	25	14	18	6	3	
Pearson one-factor	11	26	24	14	16	6	3	
Pearson first component	15	17	17	16	16	12	8	
P-LIM3+O	GH	PF	RP	BP	Func. lim.	Self-care	Sensory	Other
GRM discrimination a	2.01	3.57	3.27	2.38	3.37	2.45	1.37	1.19
GPCM slope a	1.32	2.19	1.23	1.43	2.31	2.39	1.26	1.26
GRM, latent: λ/ψ	8	22	19	11	20	11	5	4
Polychoric one-factor	9	20	15	10	28	12	4	3
GRM h on codes (R^2 .993)	13	23	24	14	15	6	3	3
GPCM, exact: $a \times sd$	12	23	24	14	17	6	3	3
GPCM h on codes (R^2 .999)	11	24	24	13	17	6	3	3
Pearson one-factor	11	25	24	14	15	6	3	3
Pearson first component	14	16	15	14	15	11	7	7

15.6 Figures by bank

Each bank has four figures, all on h with the graded-response and partial credit scores paired. Exhibit 1 and the barcharts use the identification sample, person-ages flagged `s4`, and the four-ages-two-years-apart estimators of `redo_concepts/testing_metrics`. Exhibit 1's panels (c) and (d) show Cases 2, 3 and 4 on balanced and on pairwise moments. The first barchart has the scorecard's two estimators, Case 2 on pairwise moments and Case 3 on balanced moments; the second has Case 4 on balanced moments, with its noise split into the persistent part and the one-period error. In each, the graded-response model is above partial credit. The clusters use the 50,055 people with at least three observed ages from 20 to 90, 50 seeded starts, and the settings of Figure 3. In every bank about 10%, 26 to 28% and 61 to 64% of people fall in the three clusters, and about 97% sit in the same cluster under both models.

15.6.1 P-FUNC

The reference bank. Case 2's oldest-band correlation is -0.02 under the graded-response model and $+0.01$ under partial credit.

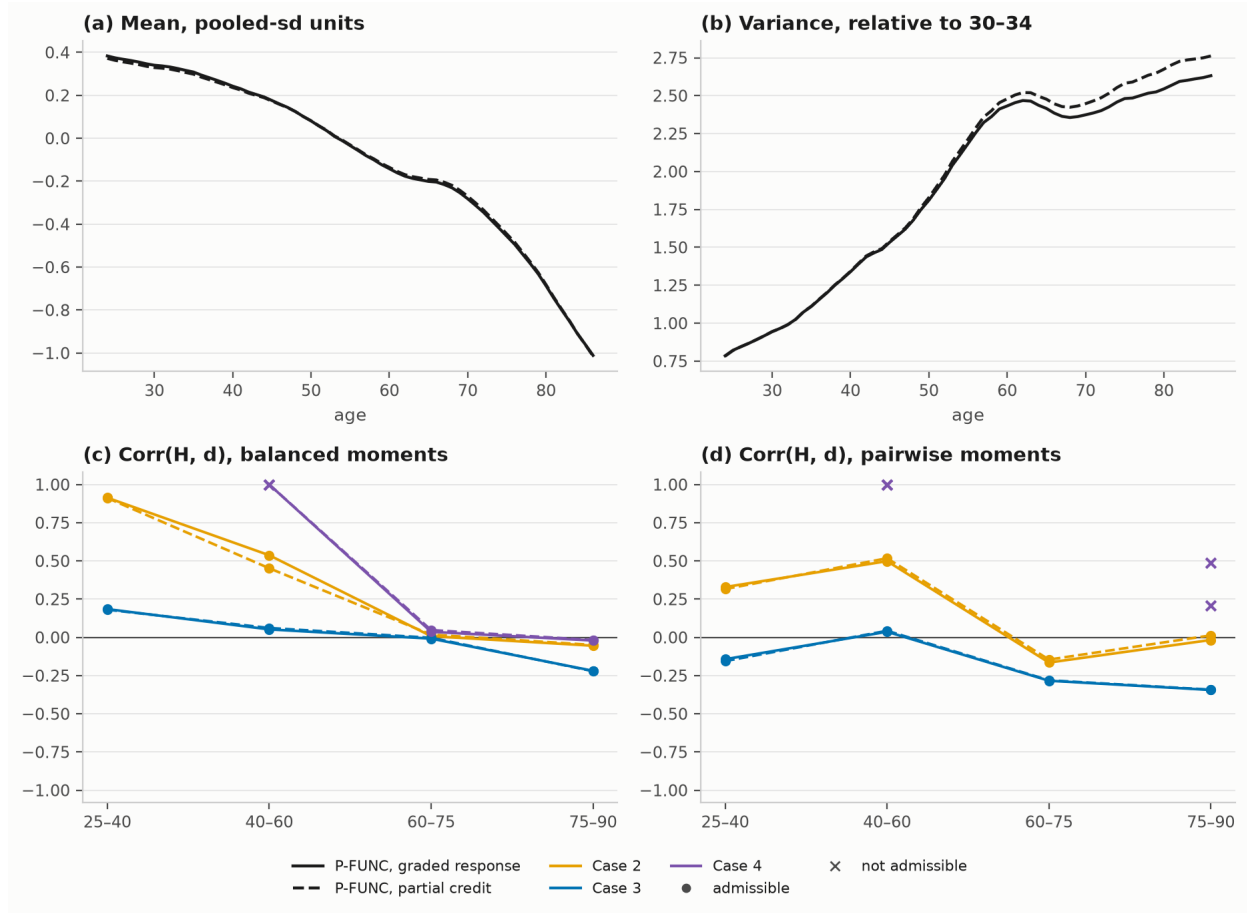


Figure 18: Exhibit 1 for P-FUNC on h . (a) Mean in pooled-sd units and (b) variance relative to ages 30–34, five-year centred means; graded response solid, partial credit dashed. (c) and (d) $\text{Corr}(H, d)$ by band under Cases 2, 3 and 4 on balanced and on pairwise moments, at the best-fitting ρ . Crosses mark bands where the fit is not admissible, correlations beyond ± 1 are drawn at the edge, and undefined correlations are left out.

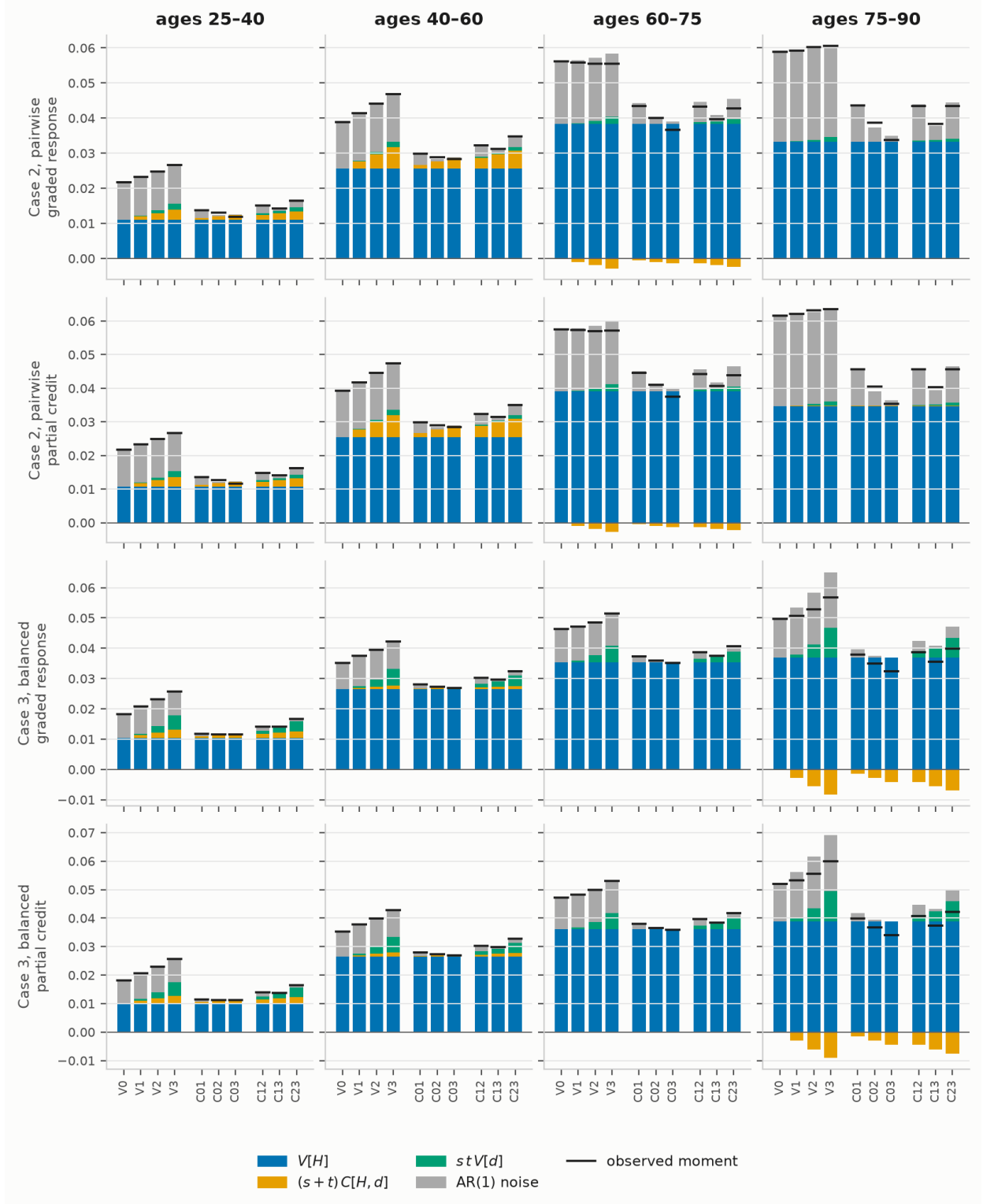


Figure 19: Identification barcharts for P-FUNC on h : observed moments (black rules) and fitted layers for the ten cells of four ages two years apart, pooled over base ages within each band. A panel would be marked if its fit were not admissible; none is, here or for any other bank.

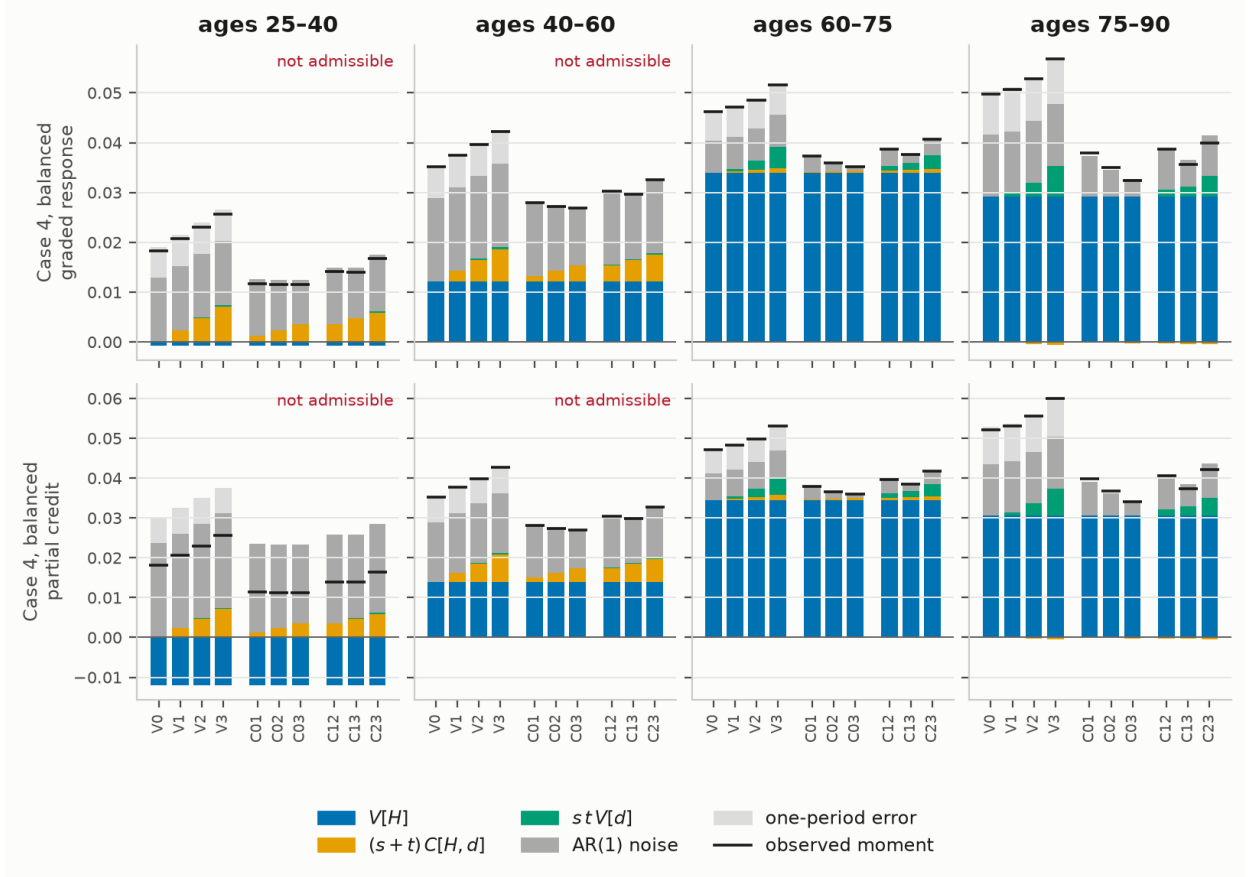


Figure 20: Case 4 barcharts for P-FUNC on h : balanced moments, noise split into the AR(1) part and the one-period error, at the best-fitting ρ in each band; as Figure 19 otherwise.

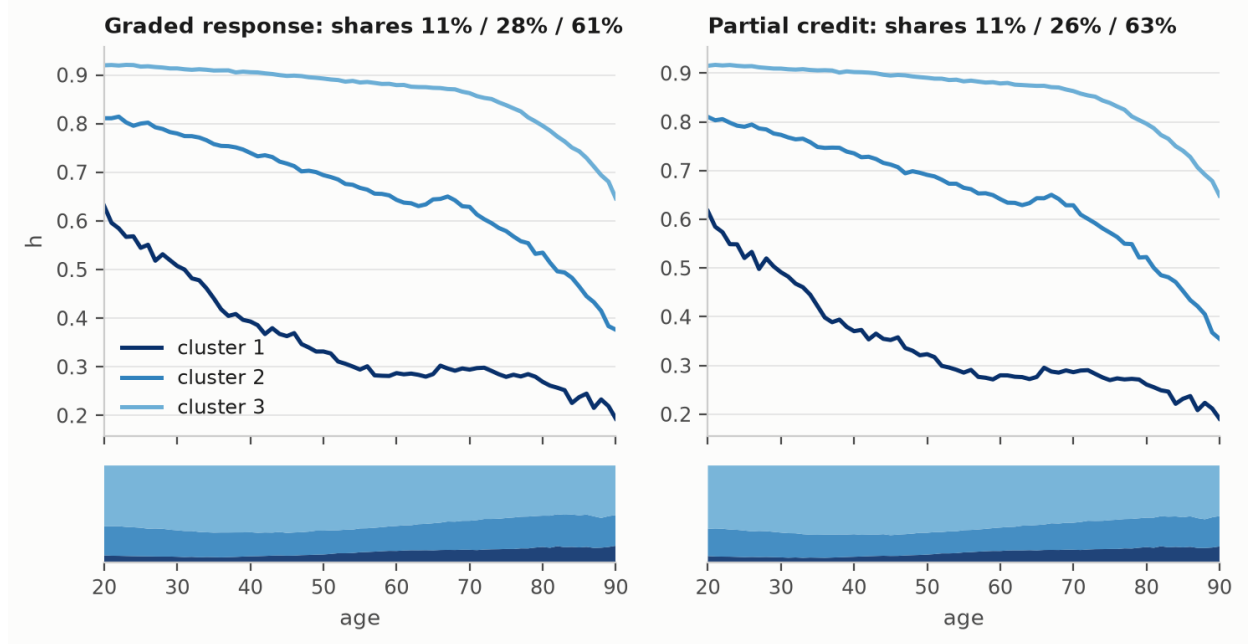


Figure 21: Partial K-means, $K = 3$, for P-FUNC on h : cluster means by age where at least 25 members are observed, and cluster shares of the observed sample by age beneath. Graded response left, partial credit right.

15.6.2 P-LIM1

Uncapping the count alone takes the floor from 1.11% to 0.37%. Case 2's oldest-band correlation turns positive, +0.01, and the clusters barely move: the adjusted Rand index against P-FUNC's clusters is 0.98.

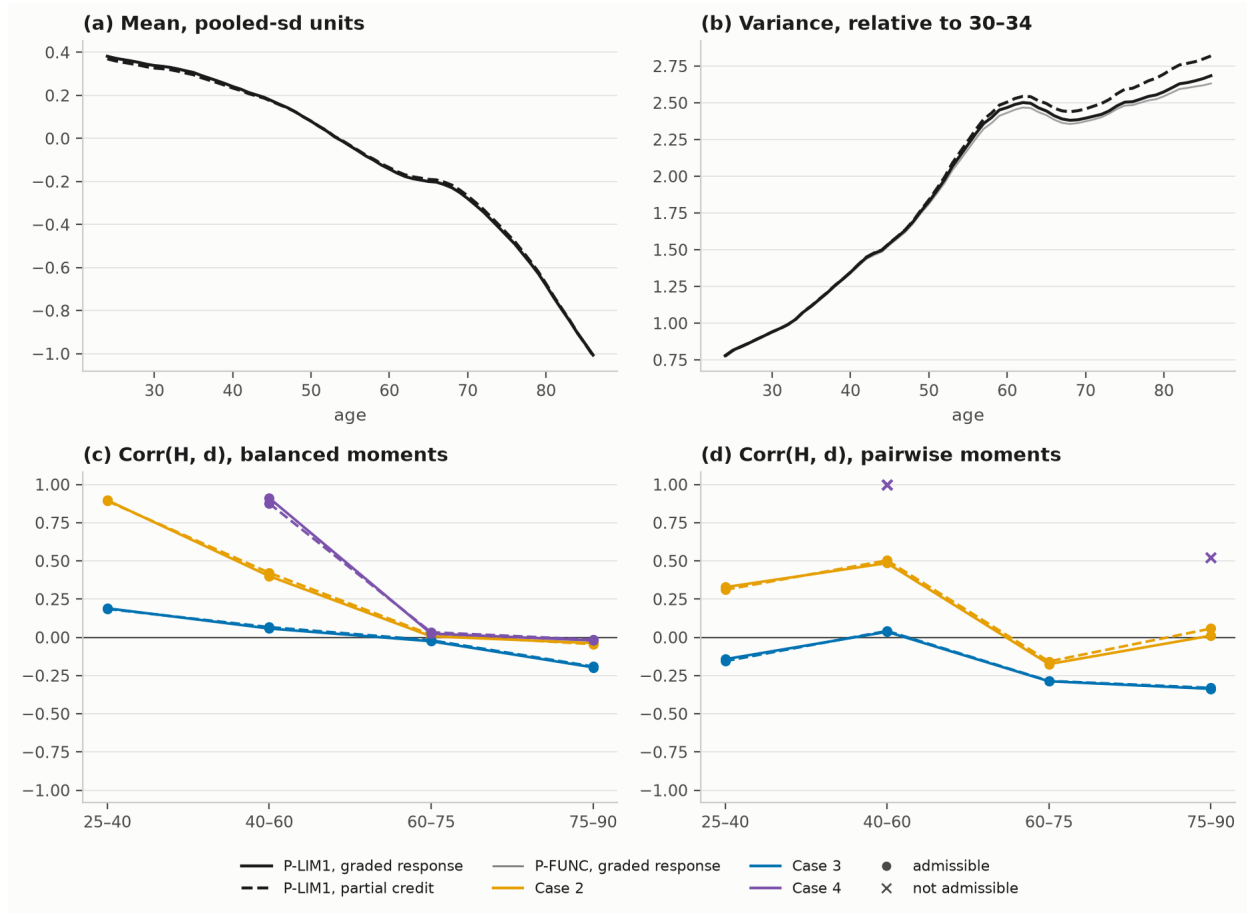


Figure 22: Exhibit 1 for P-LIM1 on h , as Figure 18; the grey lines in (a) and (b) are P-FUNC under the graded-response model.

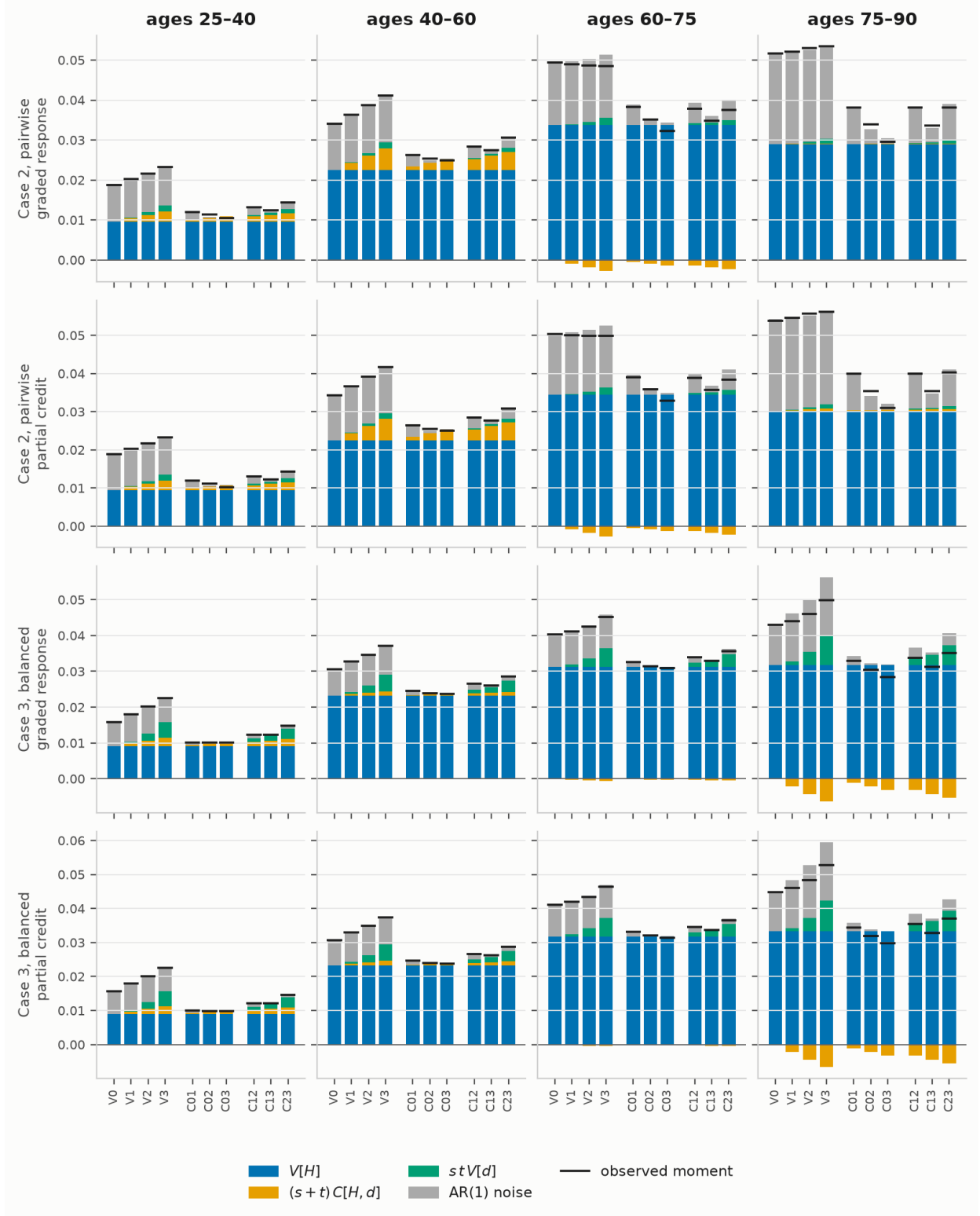


Figure 23: Identification barcharts for P-LIM1 on h , as Figure 19.

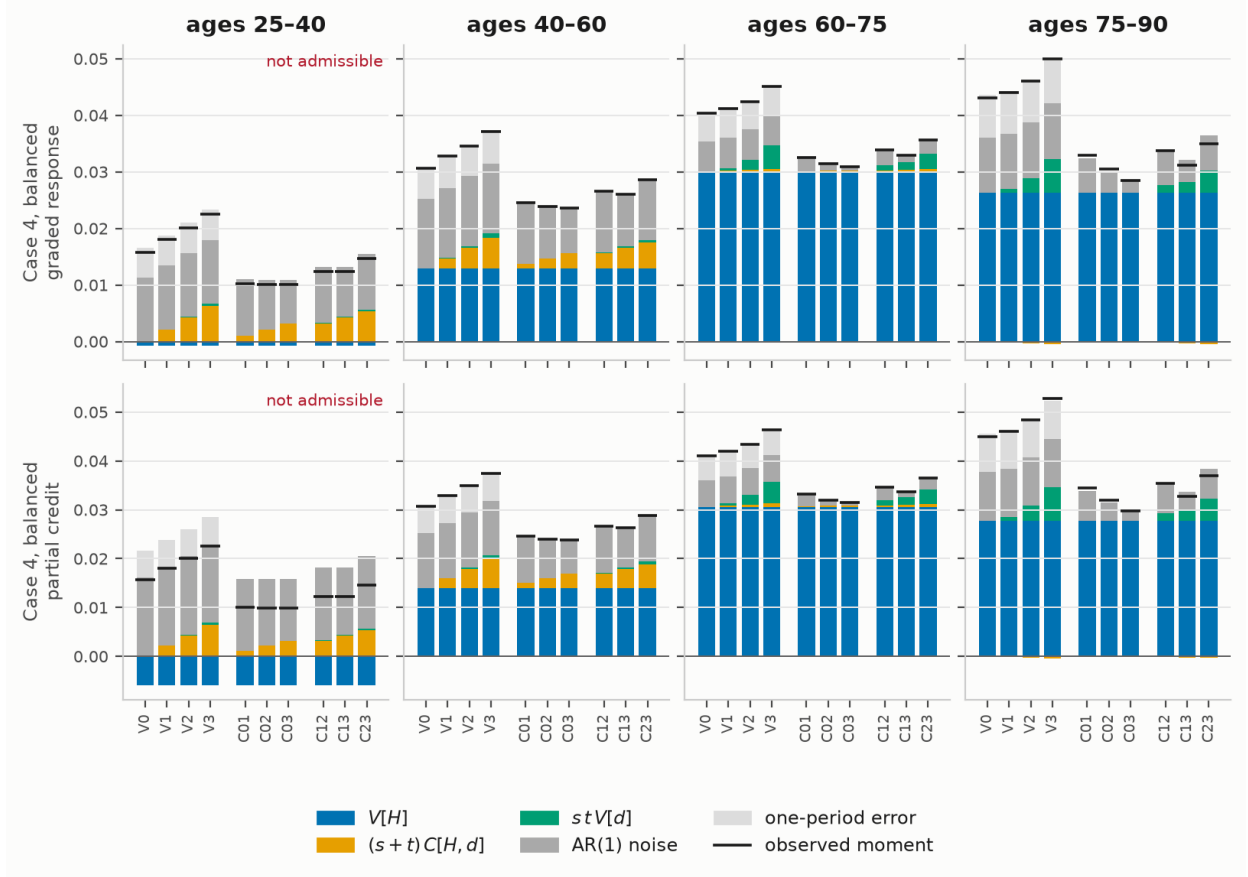


Figure 24: Case 4 barcharts for P-LIM1 on h : balanced moments, noise split into the AR(1) part and the one-period error.

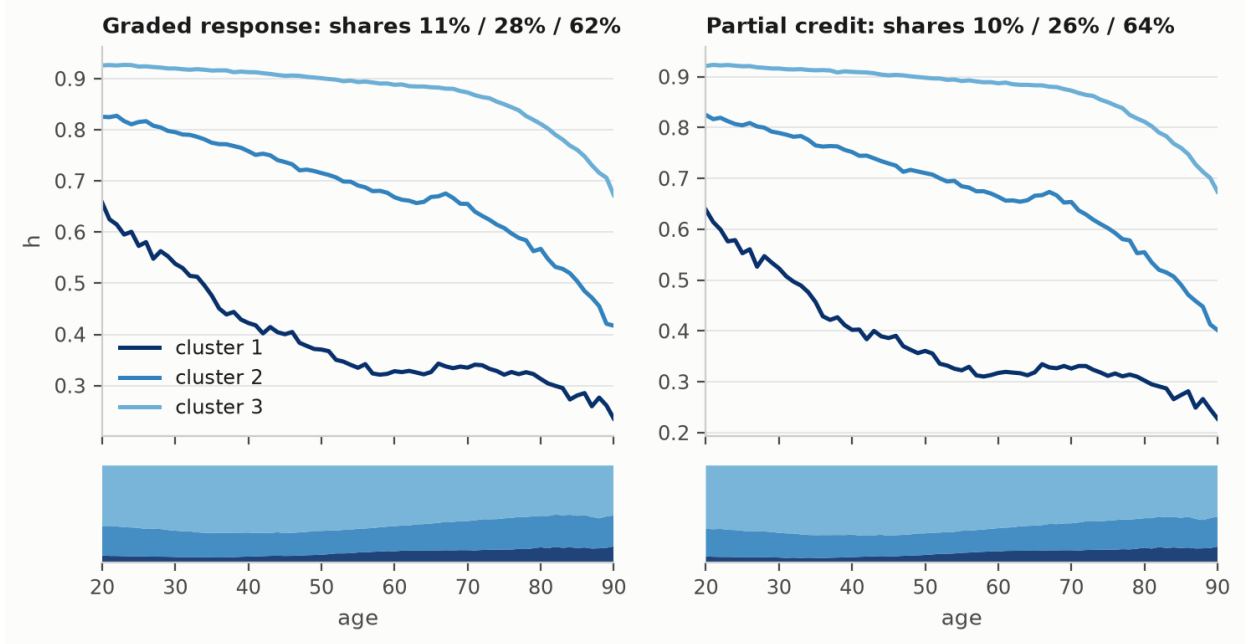


Figure 25: Partial K-means, $K = 3$, for P-LIM1 on h , as Figure 21.

15.6.3 P-LIM

Sensory and continence take the floor to 0.10%, with 4% of observed weight. The two counts carry the largest residual dependence of any bank, +0.083. Case 2's oldest-band correlation is +0.04, or +0.09 under partial credit.

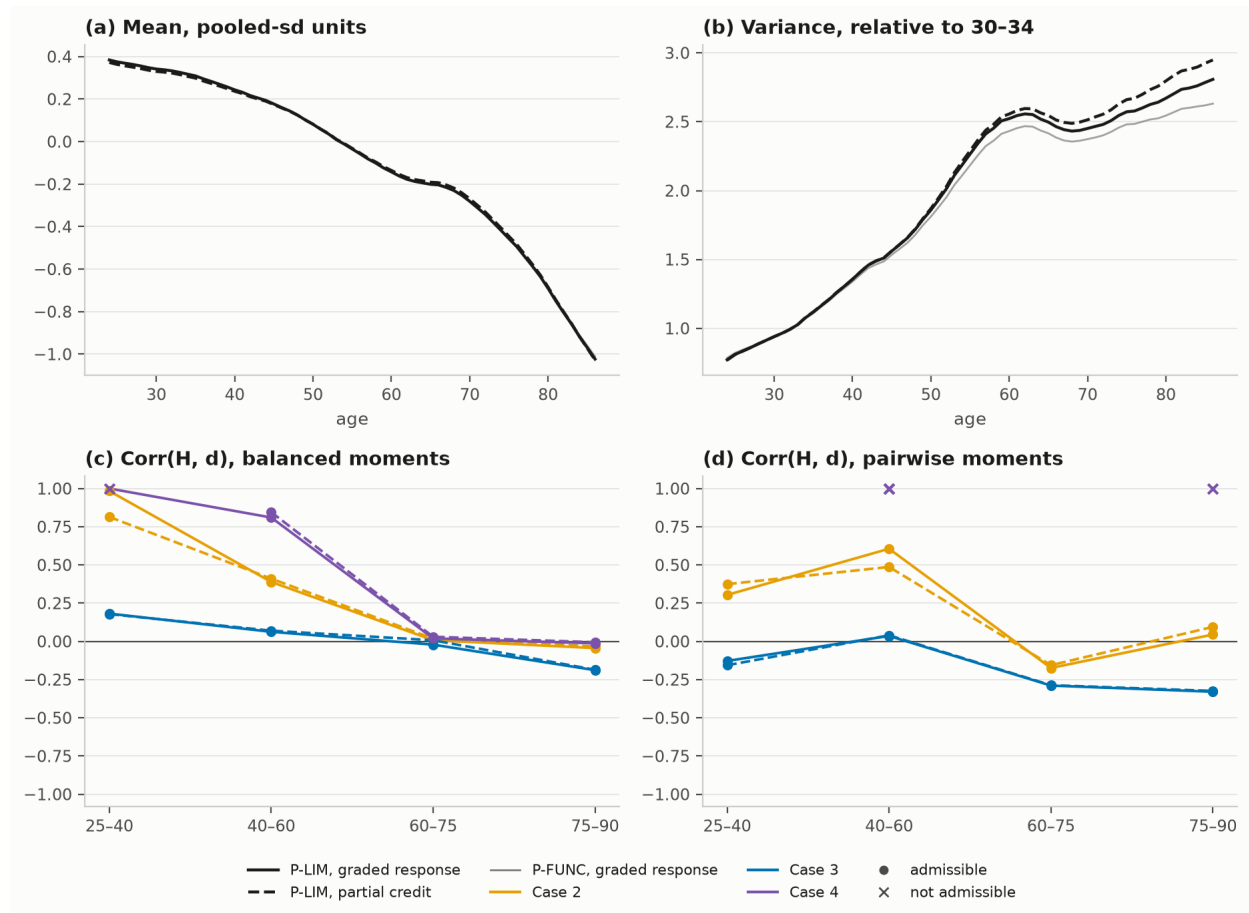


Figure 26: Exhibit 1 for P-LIM on h , as Figure 22.

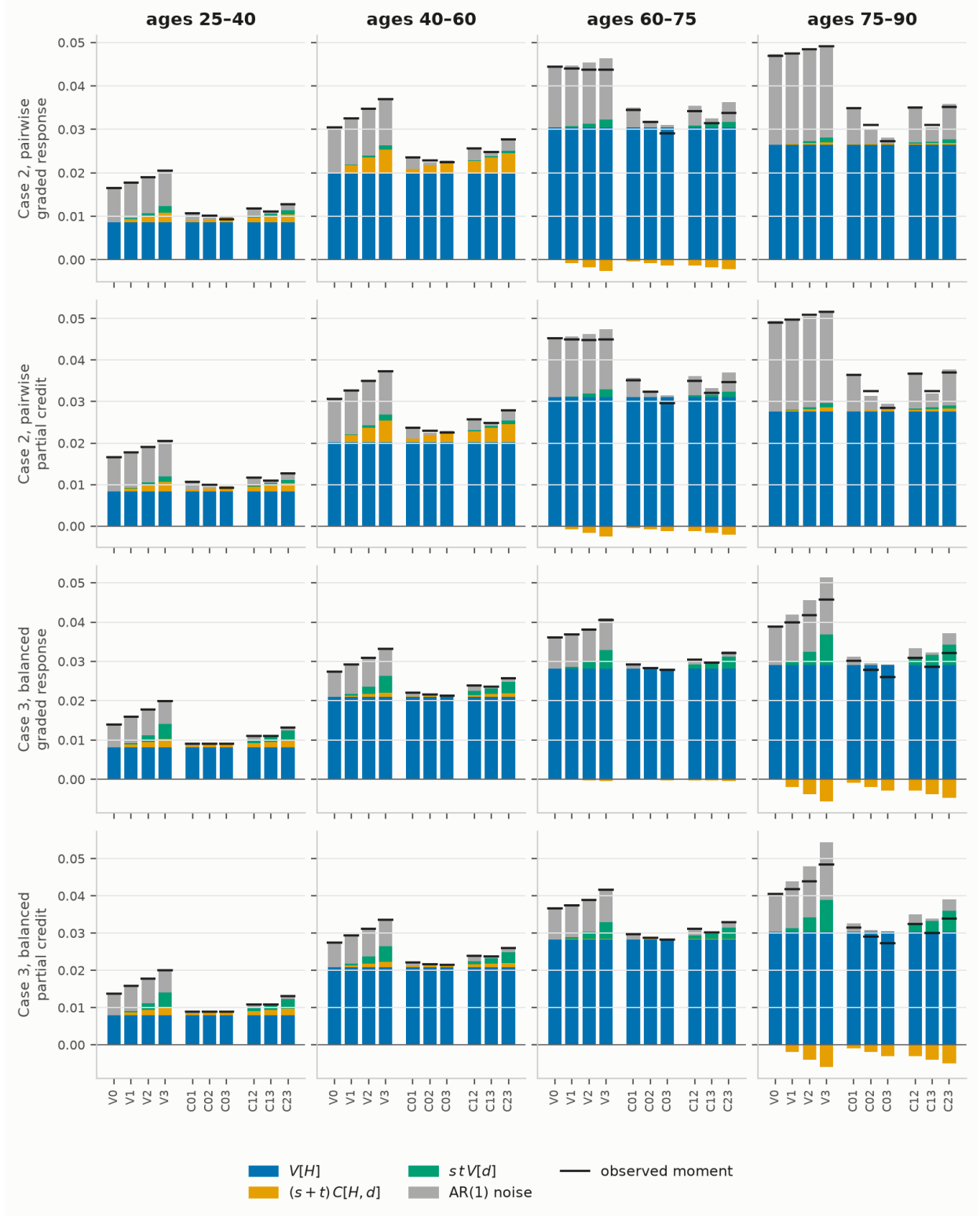


Figure 27: Identification barcharts for P-LIM on h , as Figure 19.

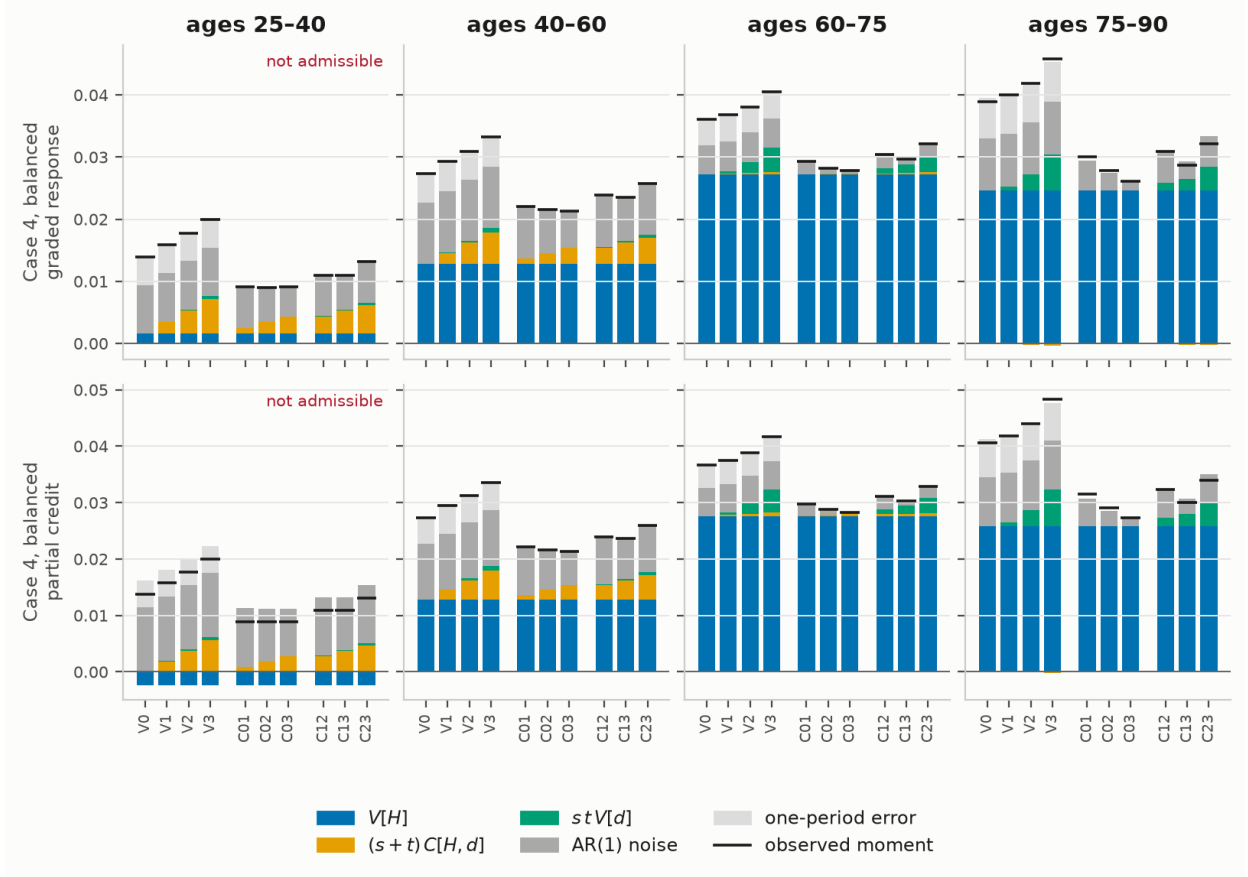


Figure 28: Case 4 barcharts for P-LIM on h : balanced moments, noise split into the AR(1) part and the one-period error.

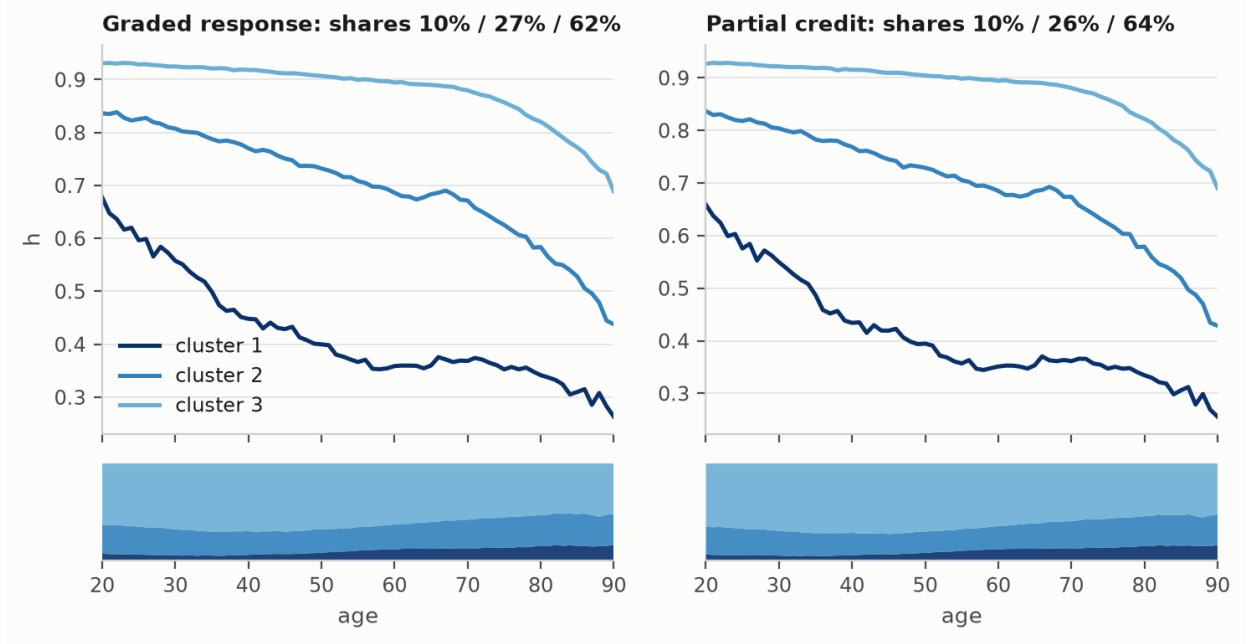


Figure 29: Partial K-means, $K = 3$, for P-LIM on h , as Figure 21.

15.6.4 P-LIM4

The finest split. Mobility and lifting get 22% of latent weight but 15% of observed weight, and continence and hearing and sight get 3% or less of observed weight. This bank has the weakest capped cost t , 10.6 under the graded-response model and 8.4 under partial credit, and the most variance growth, 1.13.

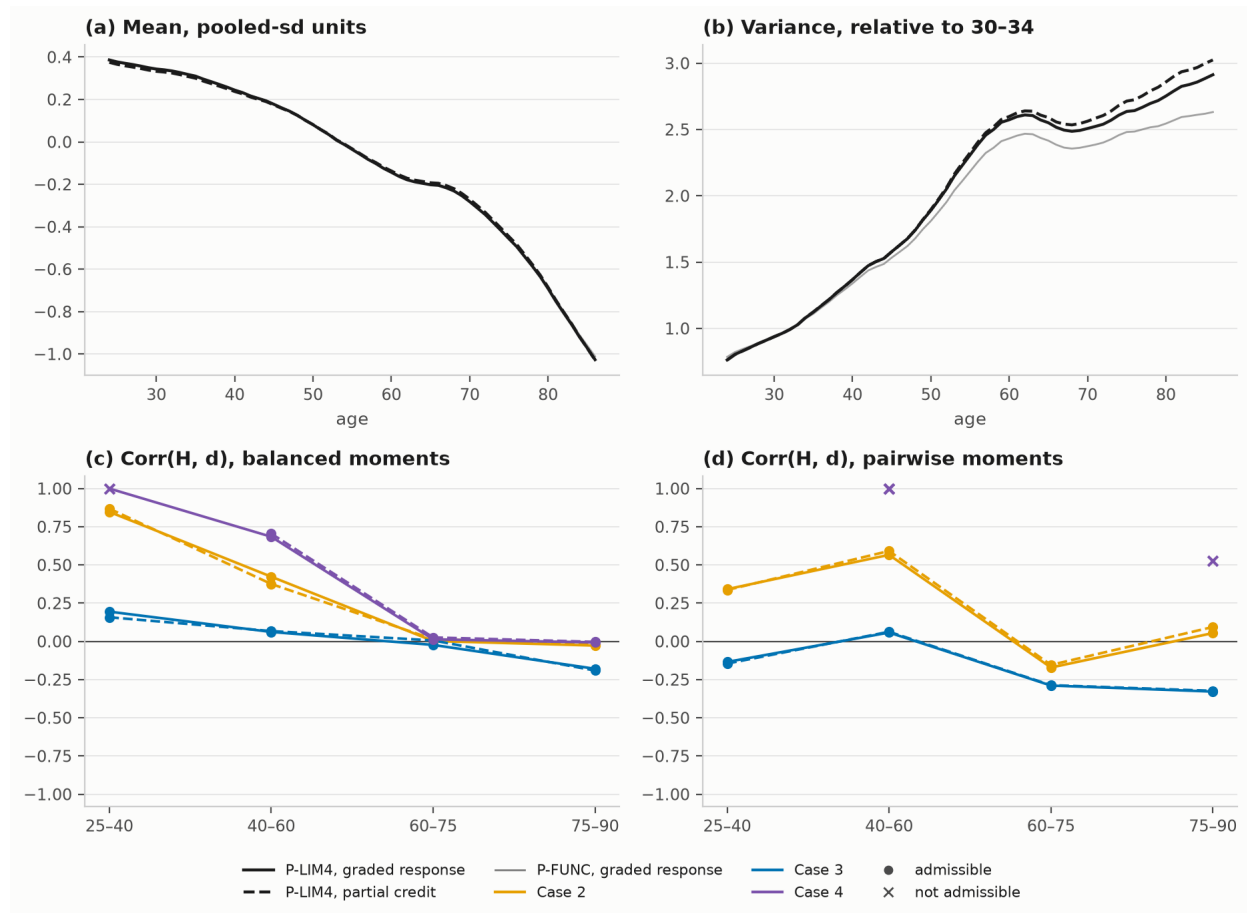


Figure 30: Exhibit 1 for P-LIM4 on h , as Figure 22.

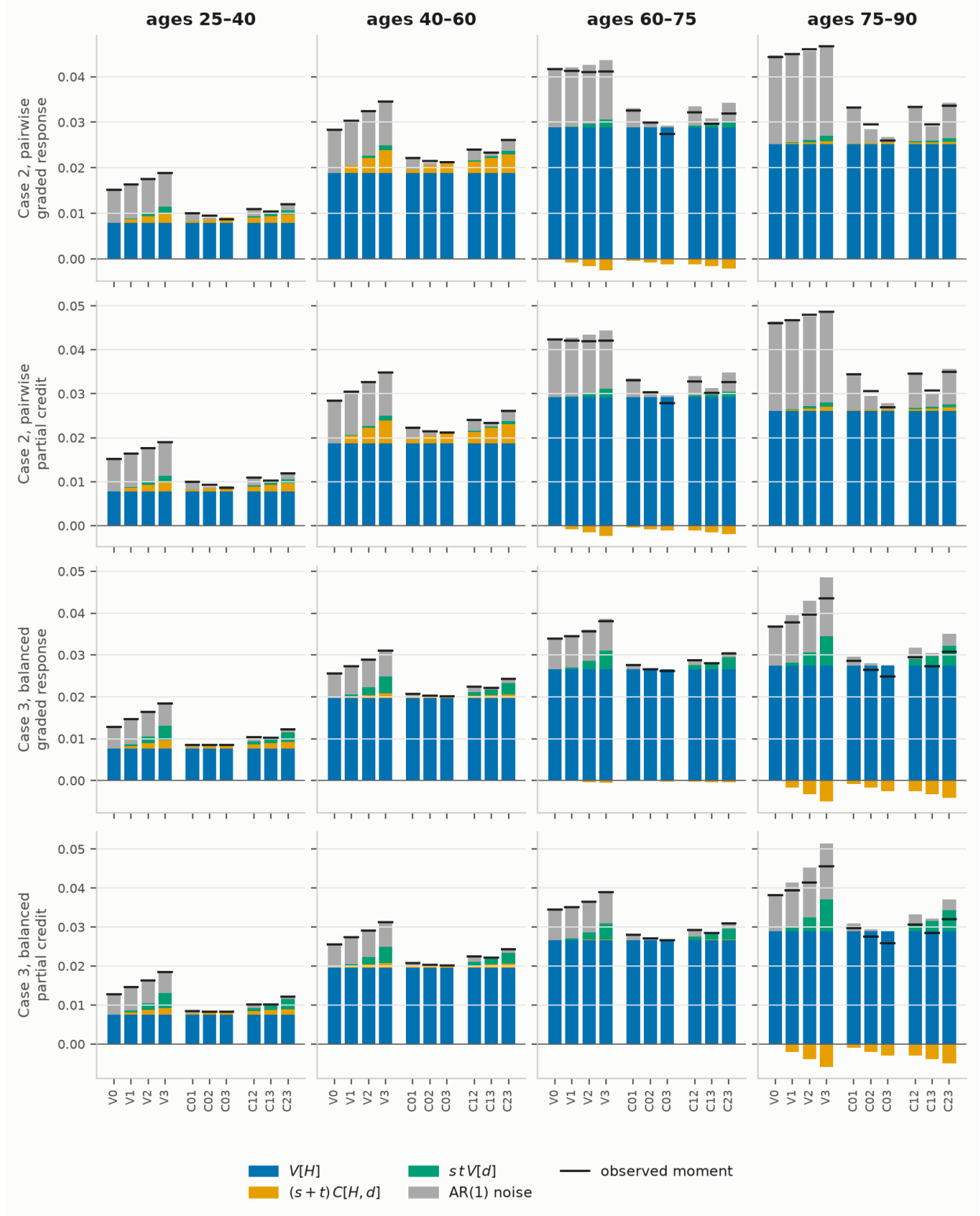


Figure 31: Identification barcharts for P-LIM4 on h , as Figure 19.

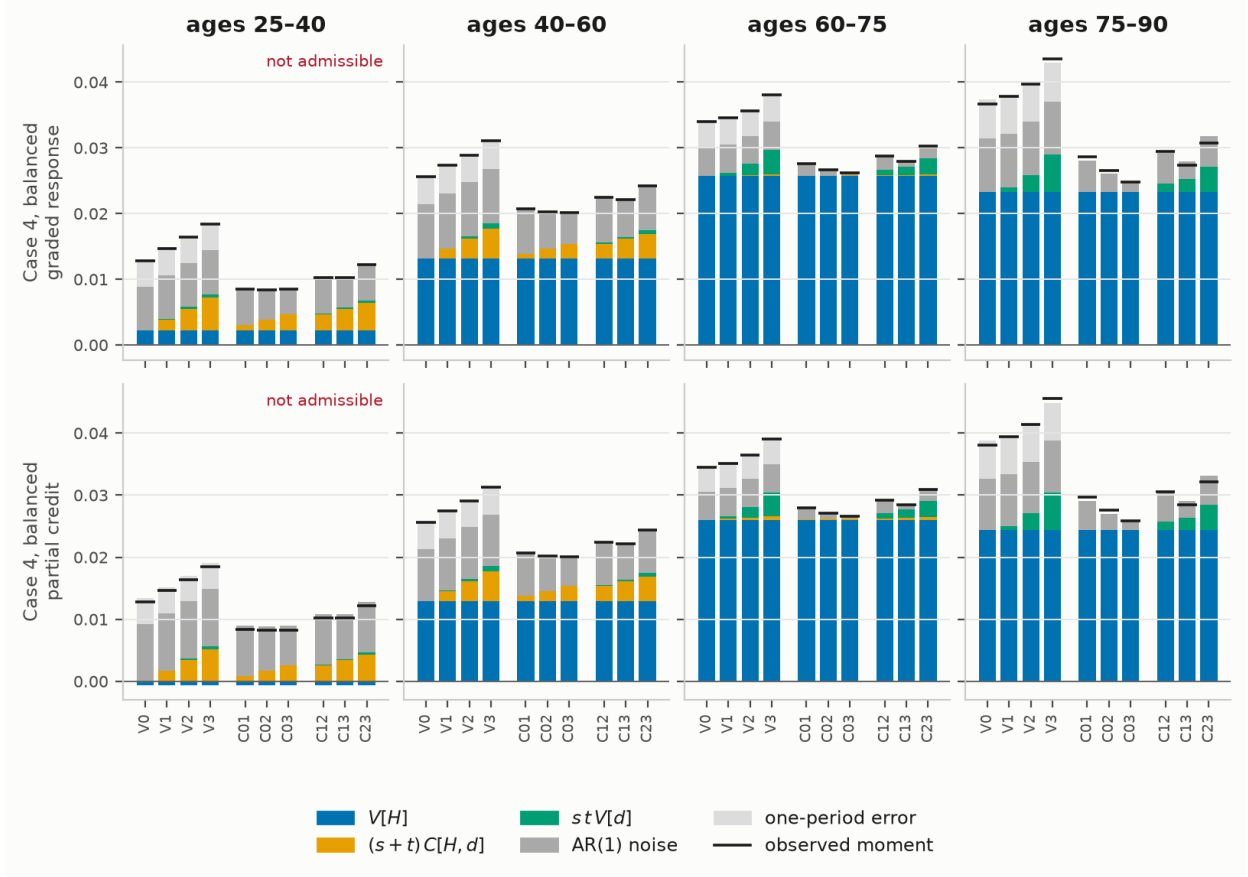


Figure 32: Case 4 barcharts for P-LIM4 on h : balanced moments, noise split into the AR(1) part and the one-period error.

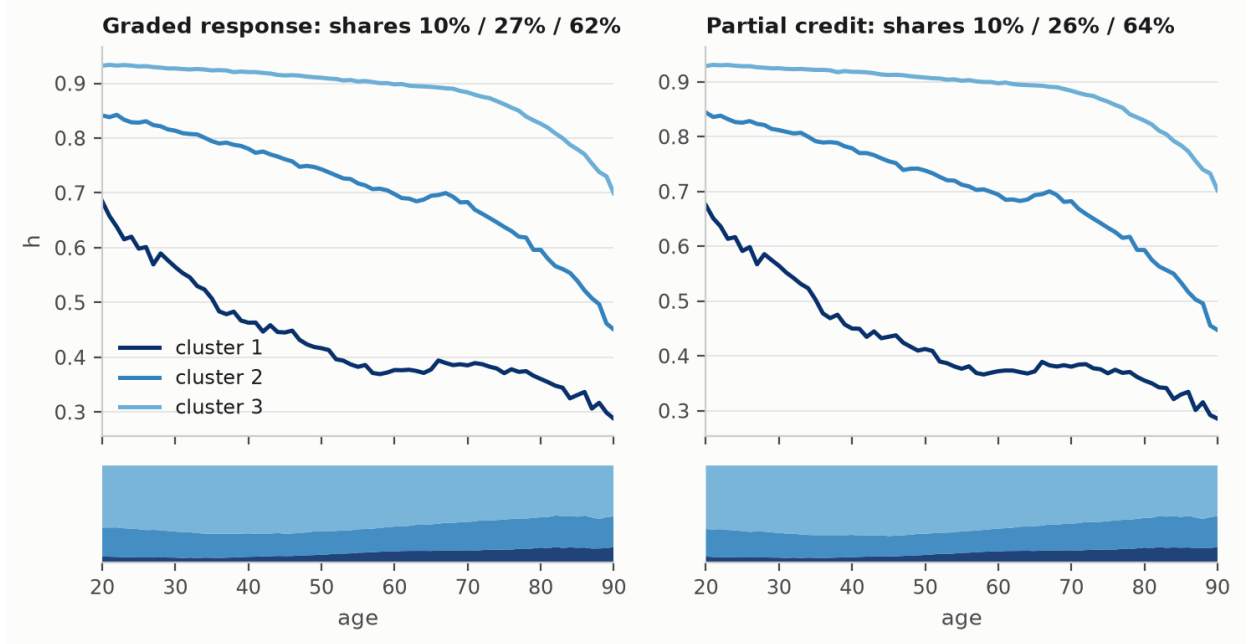


Figure 33: Partial K-means, $K = 3$, for P-LIM4 on h , as Figure 21.

15.6.5 P-LIM3

Each testlet has a name in the disablement literature: functional limitations (Verbrugge and Jette 1994), basic self-care as in the Katz activities of daily living (Katz et al. 1963), and the senses. The floor matches P-LIM4 at 0.02%, with a stronger cost t of 11.4 and no local dependence above +0.061. Case 3's oldest-band correlation is -0.17 ; Case 2's is $+0.05$.

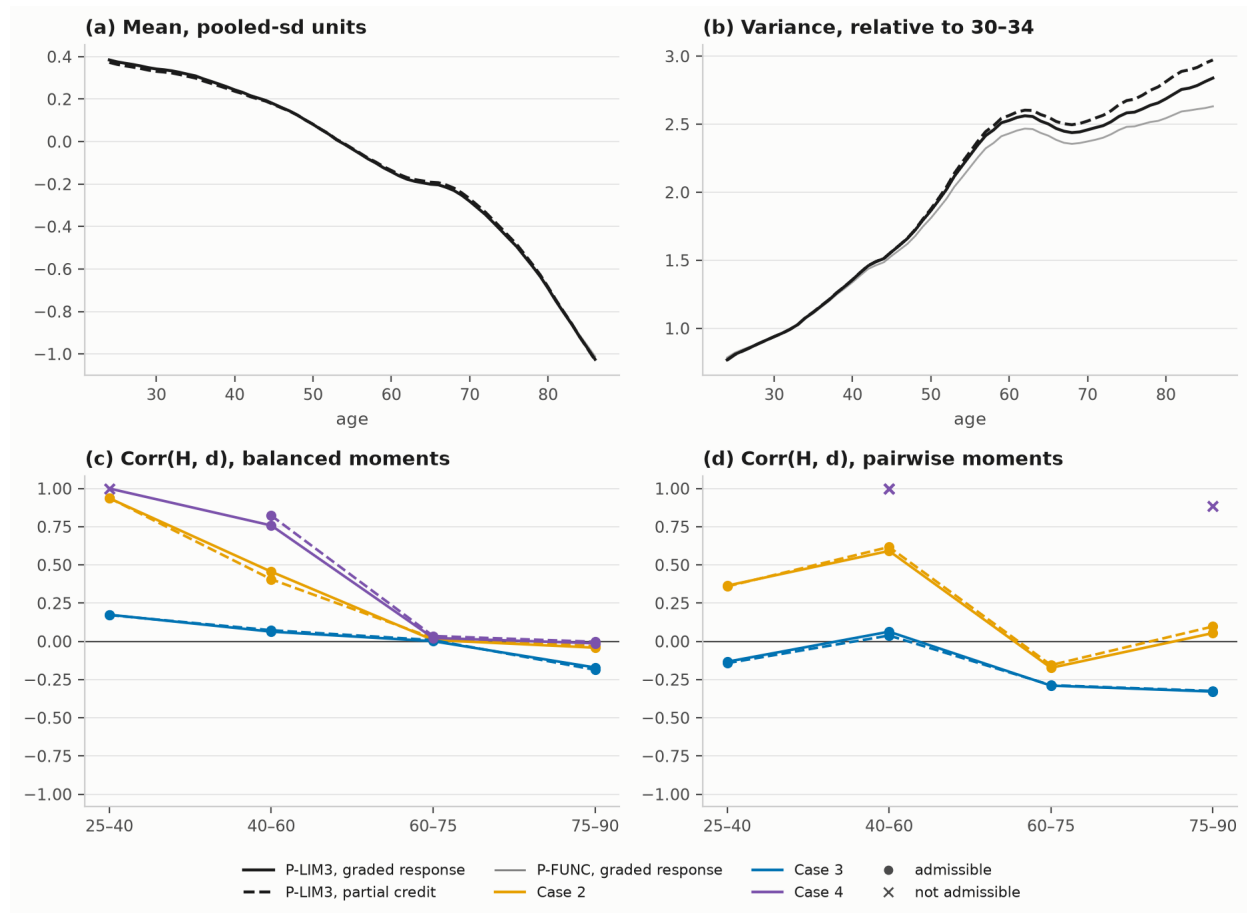


Figure 34: Exhibit 1 for P-LIM3 on h , as Figure 22.

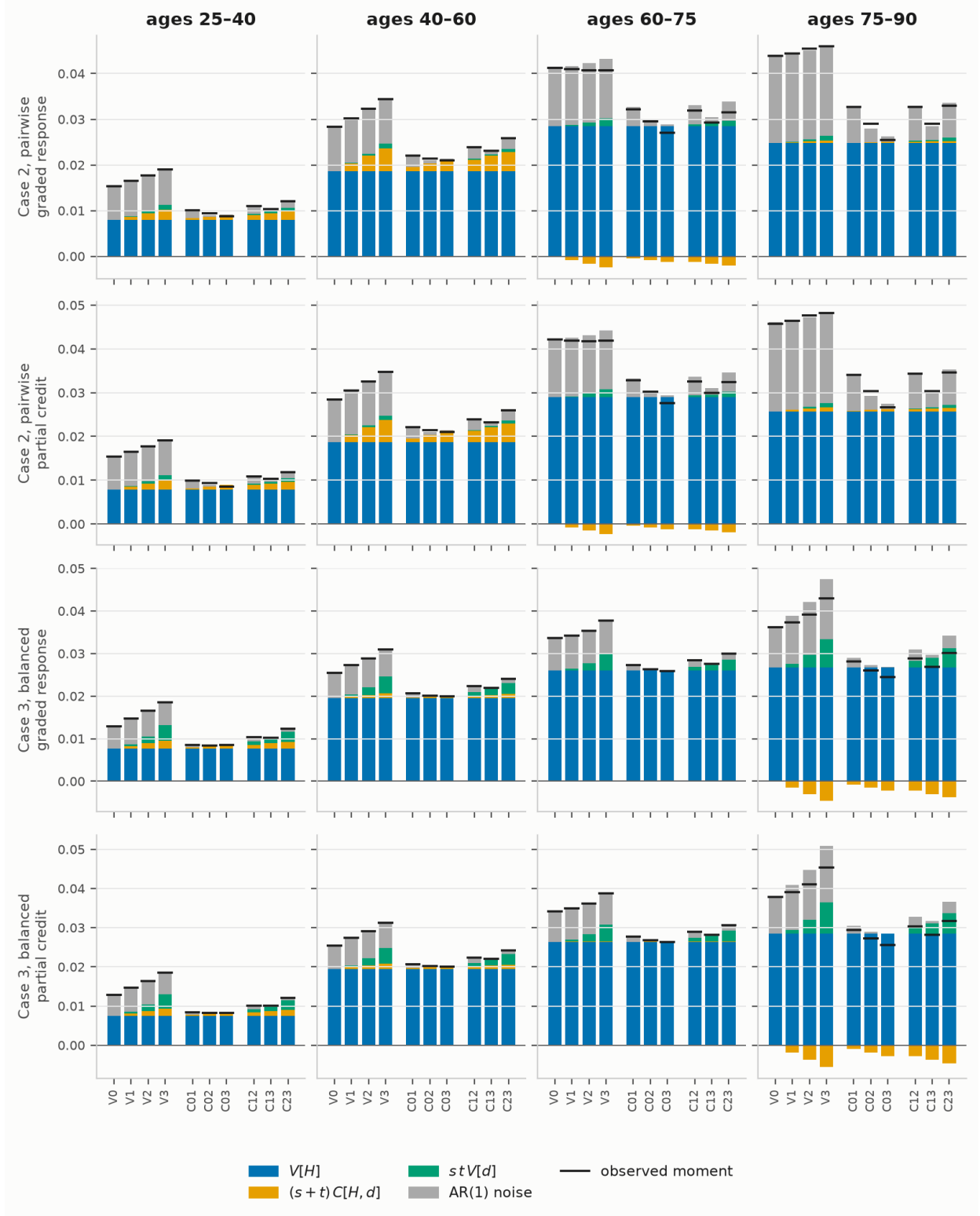


Figure 35: Identification barcharts for P-LIM3 on h , as Figure 19.

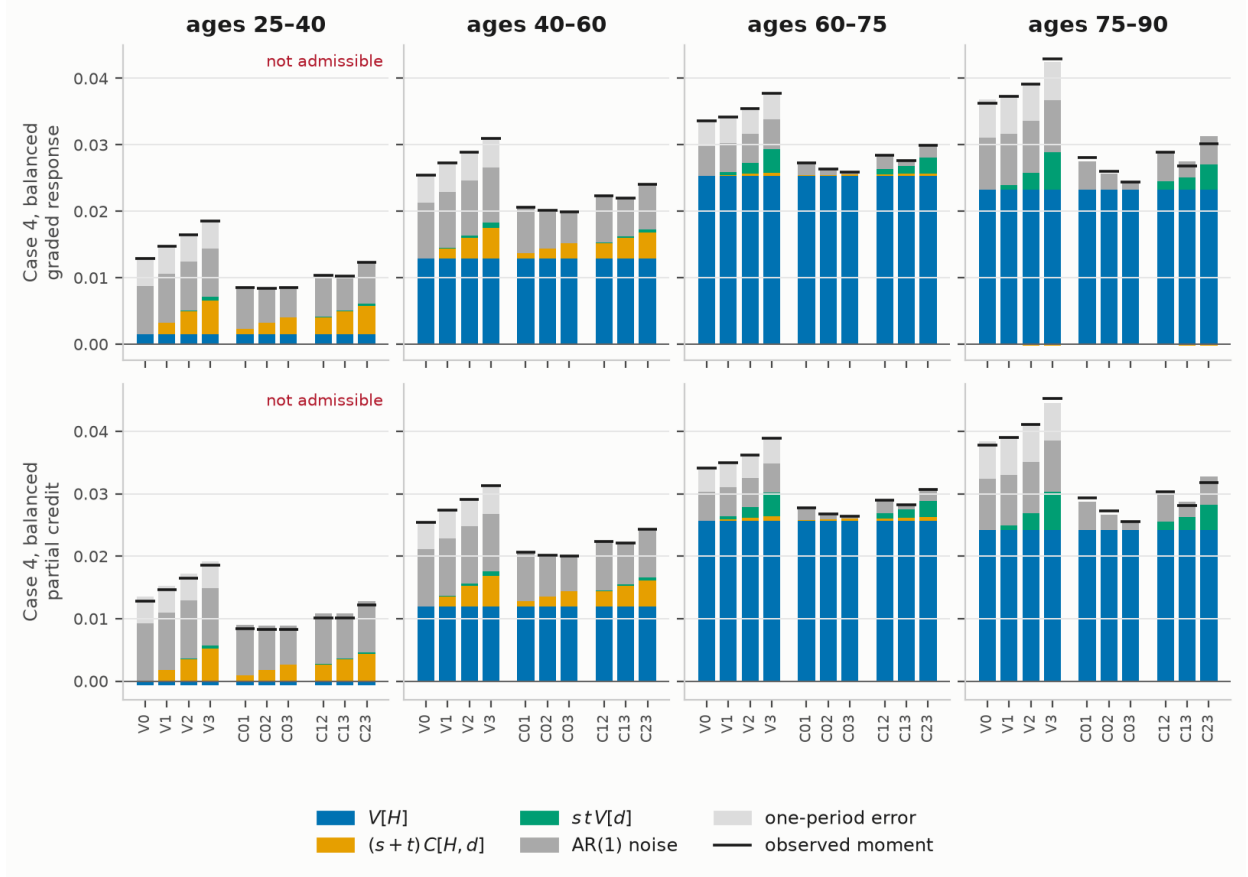


Figure 36: Case 4 barcharts for P-LIM3 on h : balanced moments, noise split into the AR(1) part and the one-period error.

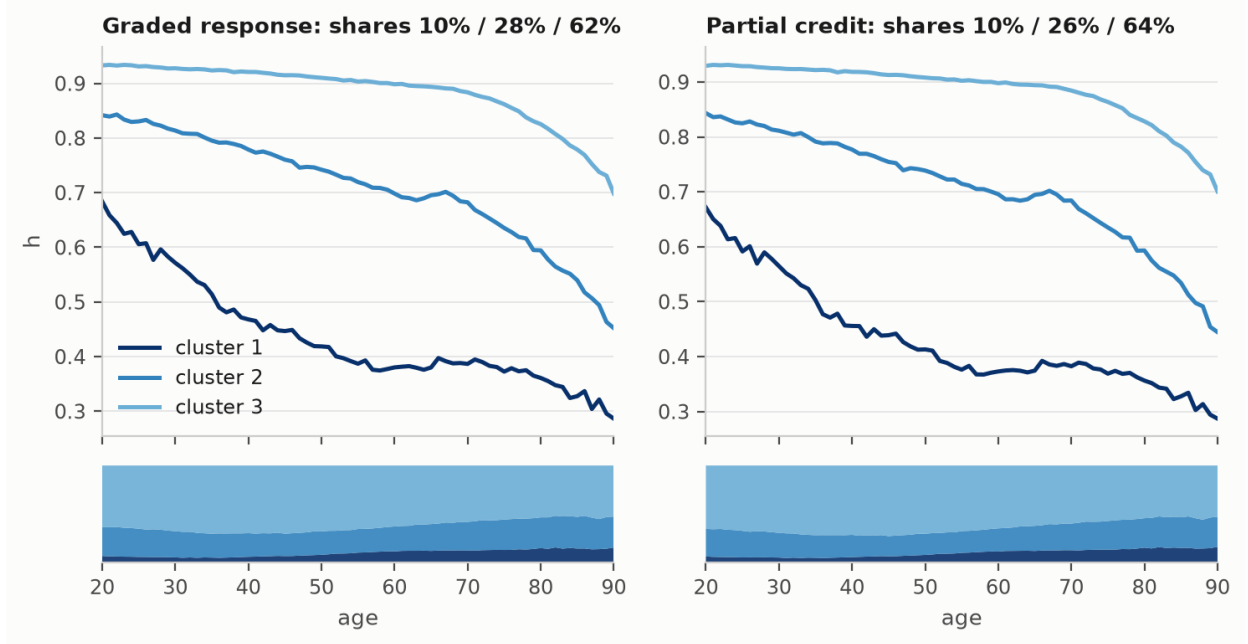


Figure 37: Partial K-means, $K = 3$, for P-LIM3 on h , as Figure 21.

15.6.6 P-LIM3+O

The non-specific item takes the floor from 0.02% to 0.01% with 3% of observed weight. Its small residual dependence with general health, +0.074, fits it picking up more than physical function. Every other criterion is close to P-LIM3's.

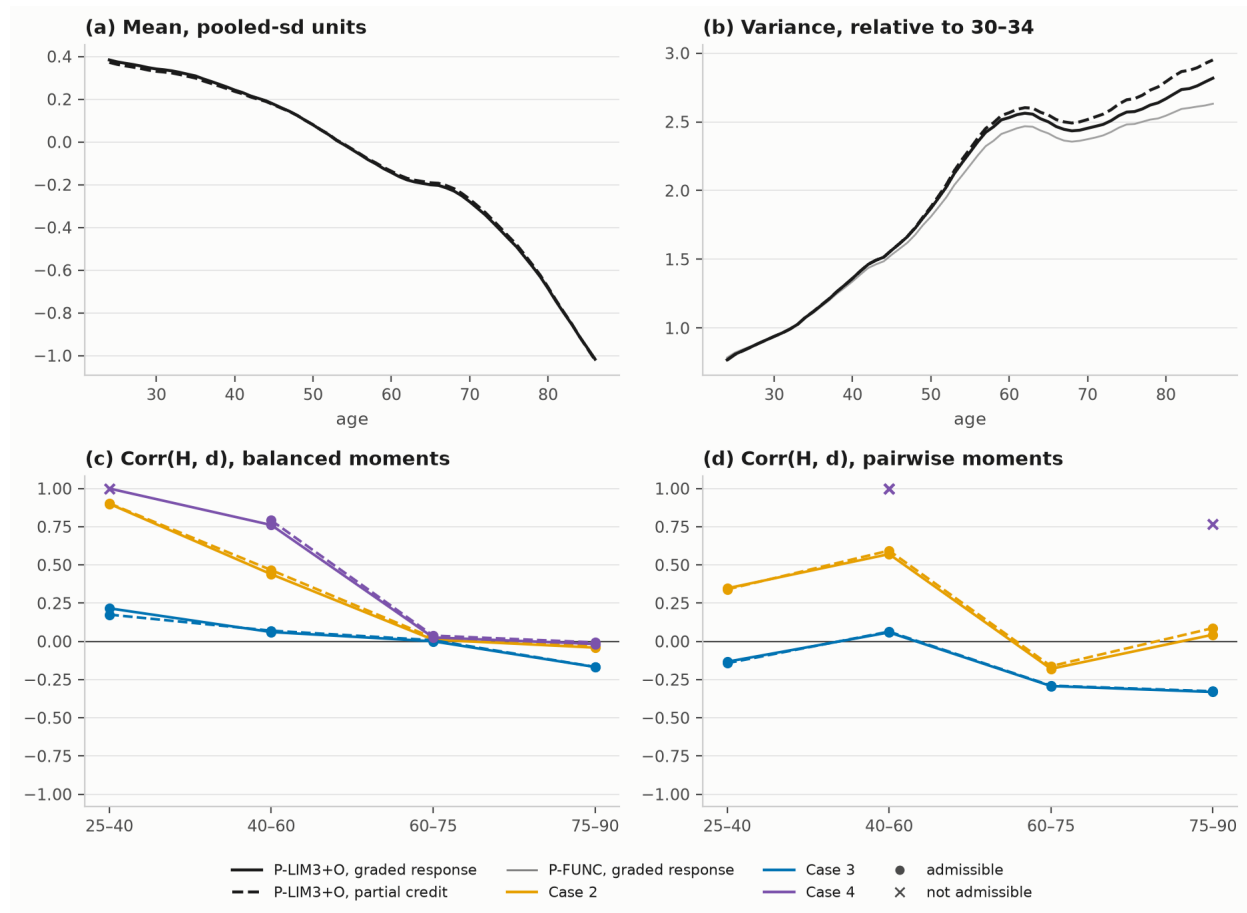


Figure 38: Exhibit 1 for P-LIM3+O on h , as Figure 22.

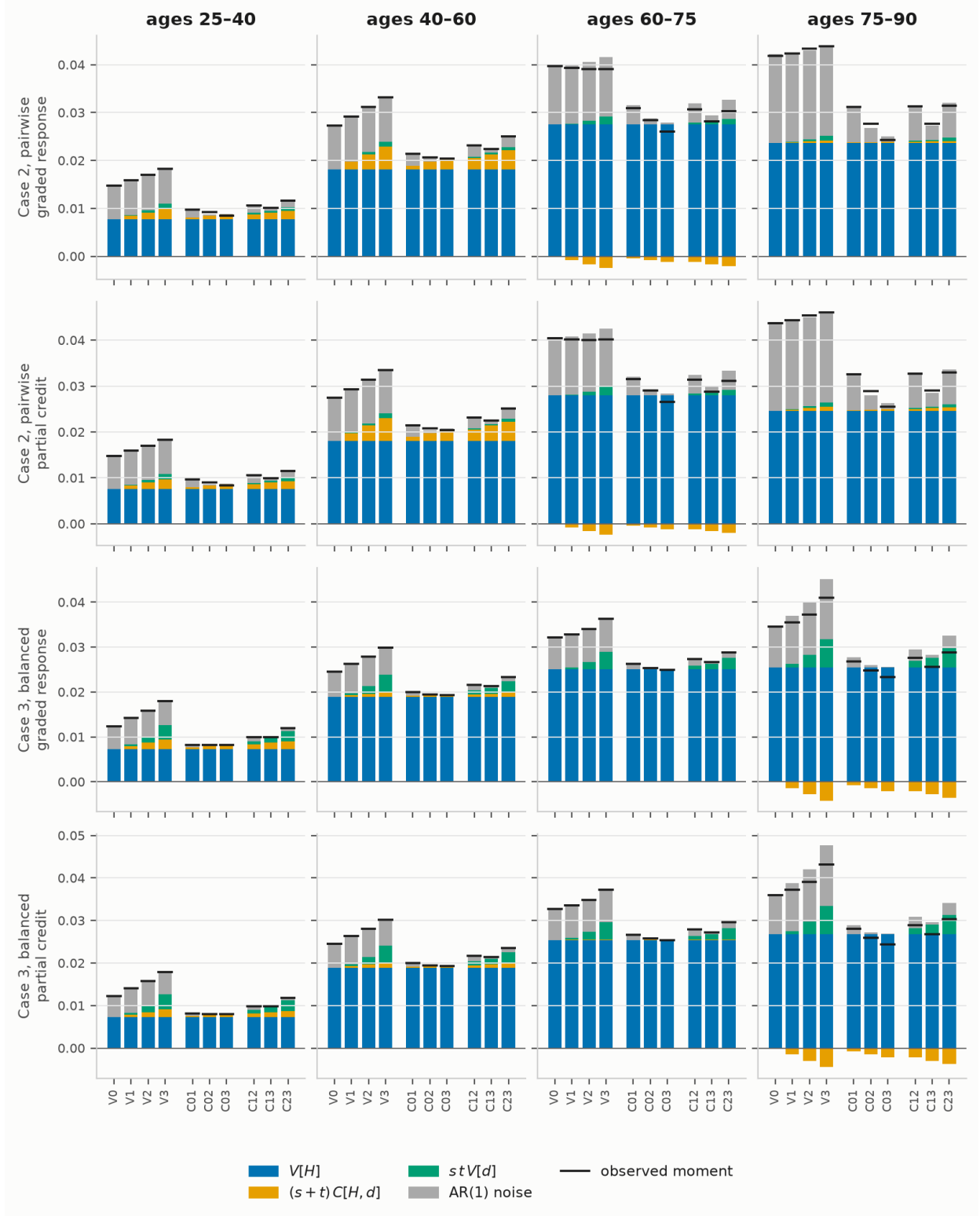


Figure 39: Identification barcharts for P-LIM3+O on h , as Figure 19.

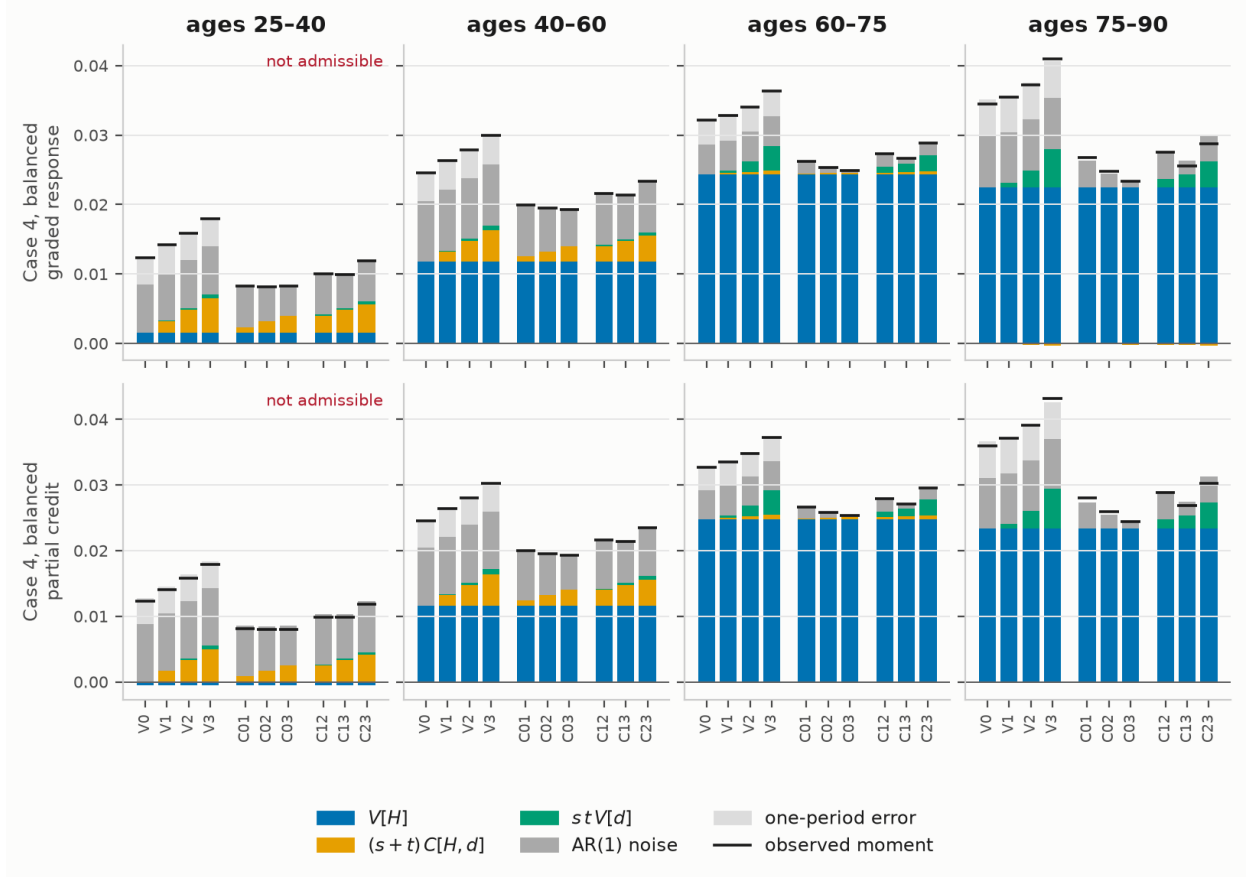


Figure 40: Case 4 barcharts for P-LIM3+O on h : balanced moments, noise split into the AR(1) part and the one-period error.

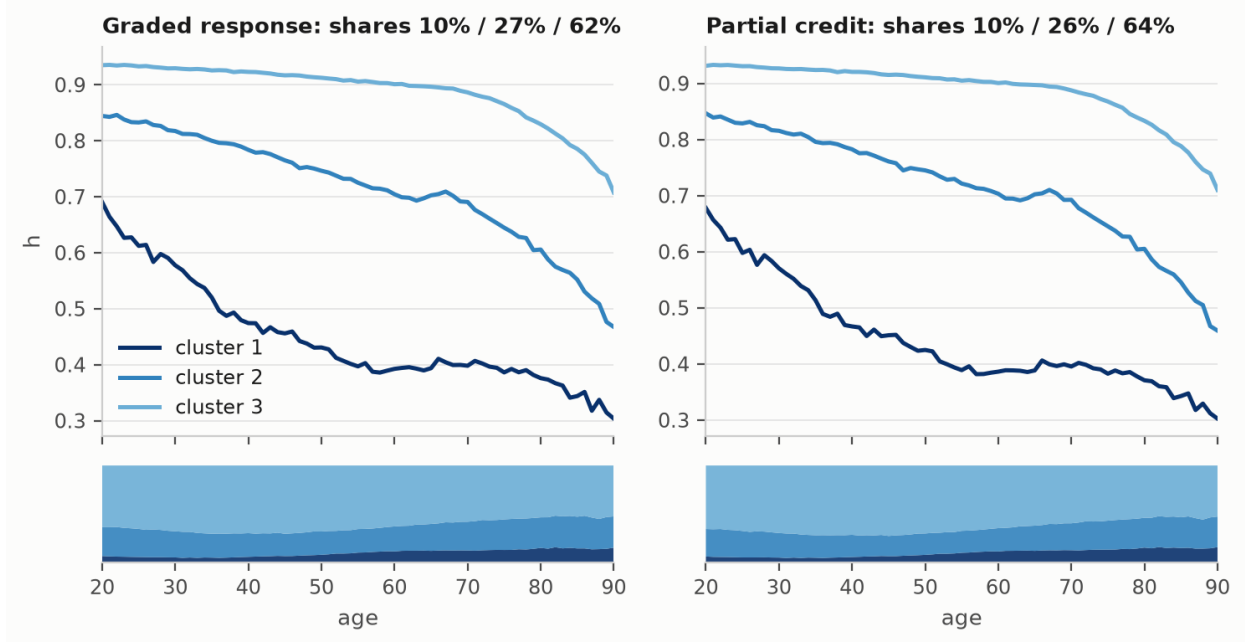


Figure 41: Partial K-means, $K = 3$, for P-LIM3+O on h , as Figure 21.

15.7 What this suggests

- Uncapping the physical-function count does most of the work on the floor. The further testlets finish the job with little weight, and none of them creates local dependence.
- P-LIM3 matches the finest split on the floor, with a clearer description and stronger cost convexity. The non-specific item adds almost nothing measurable and has no stated content.
- The graded-response model fits better than partial credit in every bank. Partial credit's exact weighted sum costs fit, weakens convexity on h , and pushes Case 2's oldest band further positive. Its weights are nearly those of the graded-response projection, so the transparency argument can be made with that projection (R^2 0.993) instead.
- On h every bank keeps the sign path on balanced moments under Cases 2, 3 and 4; on pairwise moments only P-FUNC's Case 2 under the graded-response model keeps it. So the question for Exhibit 2 is less the scale than the moments: pairwise moments give each cell its own set of people, and that is enough to turn the sign at one end of the life course or the other. On θ the pairwise Case 2 path survives, but Case 2 and Case 4 on balanced moments stop being admissible.

16 Adding chronic conditions to the limitation banks

Each limitation bank of Section 15 except P-FUNC gets the chronic conditions in two ways, both under the graded-response model. +CC adds one testlet, the count of the sixteen ever-diagnosed physical conditions (depression excluded), from 0 to 5 and 6 or more: seven or more hold 0.39% of person-waves and merge into six under the same rule as the limitation counts. +CG adds P-FULL's

six group items instead. P-FUNC with the six groups is P-FULL itself, and refitting it reproduces P-FULL’s scores exactly.

Conditions exist only from a person’s first condition inventory, so all ten banks use P-FULL’s 387,599 person-waves from 69,958 people, and people first interviewed at wave 2, mostly BHPS continuers, keep only 5% of their person-years. To separate the conditions from that change of sample, each limitation bank without conditions is also scored on exactly those person-waves (“same people” below).

Testlet	Conditions	Codes	% with 0 / 1 / 2 / ...
Chronic count	all sixteen below	0–5, 6+	48.28 / 28.88 / 12.90 / 5.59 / 2.44 / 1.05 / 0.86
Heart disease, stroke	heart failure, coronary heart disease, angina, heart attack, stroke	0, 1, 2+	92.38 / 5.17 / 2.45
Diabetes, blood pressure	diabetes, high blood pressure	0, 1, 2	73.91 / 22.09 / 4.00
Asthma, bronchitis, emphysema	asthma, emphysema, chronic bronchitis	0, 1, 2+	82.92 / 15.66 / 1.42
Arthritis	arthritis	0, 1	82.84 / 17.16
Cancer	cancer	0, 1	94.51 / 5.49
Thyroid, liver, epilepsy	hyper- and hypothyroidism, liver condition, epilepsy	0, 1+	91.30 / 8.70

16.1 Fit and local dependence

Bank	Testlets	Patterns	log L	Largest Q_3
P-LIM1+CC	6	16,203	–2,418,571	+0.025, general health with the condition count
P-LIM1+CG	11	47,469	–2,836,723	+0.145, heart disease with diabetes and blood pressure
P-LIM+CC	7	24,370	–2,506,444	+0.073, limitations with sensory and continence
P-LIM+CG	12	56,951	–2,924,370	+0.144, heart disease with diabetes and blood pressure
P-LIM4+CC	9	31,411	–2,559,843	+0.070, mobility and lifting with dexterity and care
P-LIM4+CG	14	62,497	–2,977,545	+0.145, heart disease with diabetes and blood pressure
P-LIM3+CC	8	28,360	–2,535,876	+0.052, functional limitations with hearing and sight
P-LIM3+CG	13	59,922	–2,953,756	+0.145, heart disease with diabetes and blood pressure
P-LIM3+O+CC	9	35,403	–2,619,024	+0.068, general health with other problem
P-LIM3+O+CG	14	67,345	–3,036,992	+0.144, heart disease with diabetes and blood pressure

The count’s discrimination is 1.17 to 1.19 in every bank, below any SF-12 testlet. The six groups keep their P-FULL values, from 0.41 for respiratory conditions to 1.30 for arthritis. No bank shows local dependence: the count banks’ largest Q_3 is +0.073, between two limitation testlets, and the group banks’ is +0.145, heart disease with diabetes and blood pressure, as in P-FULL.

16.2 Criteria, with the people held fixed

Columns are as in the limitation-bank criteria table, plus the share of wave-2 entrants’ person-years kept. Cost criteria use one common sample for every measure in the workbook, which already requires a condition inventory, so the “same people” rows need no separate cost figure. Their floor, variance and identification entries are computed on the condition sample.

Bank	Wave-2	Floor	Distinct	h scale				θ scale			
	kept	60–84	85–90	Cost t	Var.	C3	C2	Cost t	Var.	C3	C2
P-LIM1	100%	0.37%	98	13.2	1.08	✓	×	36.9	0.77	✓	✓
same people	5%	0.38%	95		1.06	✓	✓				
+CC	5%	0.08%	271	13.1	1.04	✓	✓	37.0	0.76	✓	✓
+CG	5%	0.00%	479	12.5	1.05	✓	✓	38.3	0.74	✓	✓
P-LIM	100%	0.10%	227	11.9	1.11	✓	×	36.9	0.78	✓	✓
same people	5%	0.11%	204		1.09	✓	×				
+CC	5%	0.03%	404	11.9	1.07	✓	✓	37.1	0.77	✓	✓
+CG	5%	0.00%	514	11.3	1.08	✓	×	38.4	0.75	✓	✓
P-LIM4	100%	0.02%	344	10.6	1.13	✓	×	36.6	0.78	✓	✓
same people	5%	0.02%	298		1.10	✓	×				
+CC	5%	0.01%	451	10.5	1.08	✓	✓	36.7	0.77	✓	✓
+CG	5%	0.00%	524	10.0	1.09	✓	×	38.0	0.75	✓	✓
P-LIM3	100%	0.02%	348	11.4	1.12	✓	×	36.8	0.78	✓	✓
same people	5%	0.02%	298		1.10	✓	×				
+CC	5%	0.01%	458	11.3	1.08	✓	✓	37.0	0.77	✓	✓
+CG	5%	0.00%	525	10.8	1.09	✓	×	38.3	0.75	✓	✓
P-LIM3+O	100%	0.01%	428	11.3	1.11	✓	×	37.4	0.77	✓	✓
same people	5%	0.01%	358		1.09	✓	×				
+CC	5%	0.00%	483	11.2	1.07	✓	✓	37.6	0.76	✓	✓
+CG	5%	0.00%	531	10.7	1.08	✓	×	38.9	0.75	✓	✓

- **The count restores Case 2 on pairwise moments.** On the same people, without conditions, only P-LIM1 has the pairwise sign path. With the count every bank has it; with the six groups only P-LIM1 does. Dropping the BHPS continuers already moves the oldest band’s correlation from +0.04 to +0.05 to between +0.00 and +0.01 in the four banks that start positive, and the count takes it to −0.02 to −0.05 in all five.
- **Variance flattens.** Growth from 60 to 85 is 1.04 to 1.08 with the count, inside the flattening band for every bank, against 1.06 to 1.10 for the same people without conditions and 1.05 to 1.09 with the groups.
- **Convexity barely moves with the count.** Capped cost t is 10.5 to 13.1 with the count, within 0.1 of each bank without conditions; the groups take 0.6 to 0.7 off.
- **The floor goes completely.** With the groups no score has more than 0.003% of 60 to 84 year olds at its worst value; with the count the most is 0.08%, for P-LIM1+CC.

16.3 Identification under every estimator

Symbols as in Section 15.4. On h the table shows each limitation bank on the same people, then with the count and with the groups.

Bank	Balanced moments						Pairwise moments					
	Case 2		Case 3		Case 4		Case 2		Case 3		Case 4	
	path	75-90	path	75-90	path	75-90	path	75-90	path	75-90	path	75-90
<i>h scale</i>												
P-LIM1, same people	✓	-0.12	✓	-0.19	✓ ^o	-0.11	✓	-0.02	×	-0.32	×	+0.16
+CC	✓	-0.10	✓	-0.18	✓ ^o	-0.10	✓	-0.05	×	-0.31	×	+0.11
+CG	✓	-0.09	✓	-0.15	✓ ^o	-0.08	✓	-0.02	×	-0.30	×	+0.15
P-LIM, same people	✓	-0.11	✓	-0.16	✓ ^o	-0.11	×	+0.00	×	-0.30	×	+0.17
+CC	✓	-0.09	✓	-0.16	✓ ^o	-0.10	✓	-0.03	×	-0.30	×	+0.14
+CG	✓	-0.08	✓	-0.14	✓ ^o	-0.08	×	+0.01	×	-0.29	×	+0.21
P-LIM4, same people	✓	-0.10	✓	-0.17	✓ ^o	-0.10	×	+0.01	×	-0.30	×	+0.22
+CC	✓	-0.09	✓	-0.16	✓ ^o	-0.09	✓	-0.02	×	-0.29	×	+0.17
+CG	✓	-0.08	✓	-0.15	✓ ^o	-0.08	×	+0.03	×	-0.29	×	+0.23
P-LIM3, same people	✓	-0.11	✓	-0.16	✓ ^o	-0.11	×	+0.01	×	-0.30	×	+0.20
+CC	✓	-0.10	✓	-0.15	✓ ^o	-0.10	✓	-0.02	×	-0.29	×	+0.14
+CG	✓	-0.08	✓	-0.14	✓ ^o	-0.09	×	+0.02	×	-0.29	×	+0.18
P-LIM3+O, same people	✓	-0.11	✓	-0.15	✓ ^o	-0.12	×	+0.00	×	-0.30	×	+0.15
+CC	✓	-0.10	✓	-0.15	✓ ^o	-0.11	✓	-0.03	×	-0.29	×	+0.12
+CG	✓	-0.09	✓	-0.14	✓ ^o	-0.09	×	+0.01	×	-0.29	×	+0.16
<i>θ scale</i>												
P-LIM1+CC	✓	-0.35	✓	-0.34	✓ ^o	-0.35	✓	-0.32	×	-0.35	✓ ^o	-0.37
P-LIM1+CG	✓ ^o	-0.37	✓	-0.35	✓ ^o	-0.38	✓	-0.33	×	-0.35	✓ ^o	-0.40
P-LIM+CC	✓	-0.35	✓	-0.33	✓ ^o	-0.35	✓	-0.30	×	-0.35	✓ ^o	-0.35
P-LIM+CG	✓	-0.37	✓	-0.35	✓ ^o	-0.37	✓	-0.31	×	-0.34	✓ ^o	-0.38
P-LIM4+CC	✓	-0.35	✓	-0.34	✓ ^o	-0.35	✓	-0.30	×	-0.34	✓ ^o	-0.34
P-LIM4+CG	✓	-0.37	✓	-0.35	✓ ^o	-0.37	✓	-0.31	×	-0.34	✓ ^o	-0.37
P-LIM3+CC	✓	-0.34	✓	-0.33	✓ ^o	-0.35	✓	-0.30	×	-0.34	✓ ^o	-0.34
P-LIM3+CG	✓	-0.37	✓	-0.35	✓ ^o	-0.37	✓	-0.31	×	-0.34	✓ ^o	-0.38
P-LIM3+O+CC	✓	-0.35	✓	-0.33	✓ ^o	-0.35	✓	-0.30	×	-0.34	✓ ^o	-0.35
P-LIM3+O+CG	✓	-0.37	✓	-0.35	✓ ^o	-0.37	✓	-0.32	×	-0.34	✓ ^o	-0.38

- On h and balanced moments nothing changes: Cases 2 and 3 are signed and admissible for every version, and Case 4 is signed but, as before, not admissible at its best fit in every band.
- On pairwise moments only Case 2 responds to the conditions, as above; Case 3 still fails the youngest band and Case 4 the oldest.
- On θ both codings keep the sign path under Case 3 on balanced and Case 2 on pairwise moments, as the banks without conditions did.

16.4 How much weight the conditions get

Shares of weight on the condition testlets, defined as in Section 15. R^2 is that of the graded-response score projected on the codes. The last column is the condition count's discrimination, or the range across the six groups.

Bank	Share of weight on the condition testlets, %					R^2	Condition a
	GRM, latent	Polychoric	GRM h on codes	Pearson	First component		
P-LIM1+CC	5	5	7	7	14	0.992	1.17
P-LIM1+CG	17	16	15	15	34	0.993	0.41–1.29
P-LIM+CC	4	5	6	7	12	0.992	1.18
P-LIM+CG	16	15	15	15	32	0.993	0.41–1.30
P-LIM4+CC	3	3	6	6	10	0.993	1.19
P-LIM4+CG	12	11	14	14	27	0.993	0.41–1.31
P-LIM3+CC	4	4	6	6	11	0.993	1.18
P-LIM3+CG	14	14	14	14	29	0.993	0.41–1.30
P-LIM3+O+CC	4	4	6	6	10	0.993	1.19
P-LIM3+O+CG	14	14	14	14	28	0.993	0.41–1.30

- **The count is a light item.** It gets 3 to 5% of latent weight and 6 to 7% of observed weight. The six groups together get 11 to 17% on either scale, about three times the count’s latent weight and twice its observed weight, matching the 15% the Pearson factor score gives conditions in P-FULL.
- **The first principal component doubles both,** to 10 to 14% for the count and 27 to 34% for the groups, because it does not discount weak items.

16.5 Figures by bank

Each bank has four figures laid out as in Section 15, with the count solid and the groups dashed. The grey lines in Exhibit 1 are the bank without conditions on the same people. The clusters use the 39,004 people with at least three observed ages in the condition sample. In every bank the count and the groups put 98% of them in the same cluster, with an adjusted Rand index of 0.95, against 0.90 with the bank’s clusters without conditions.

16.5.1 P-LIM1 with conditions

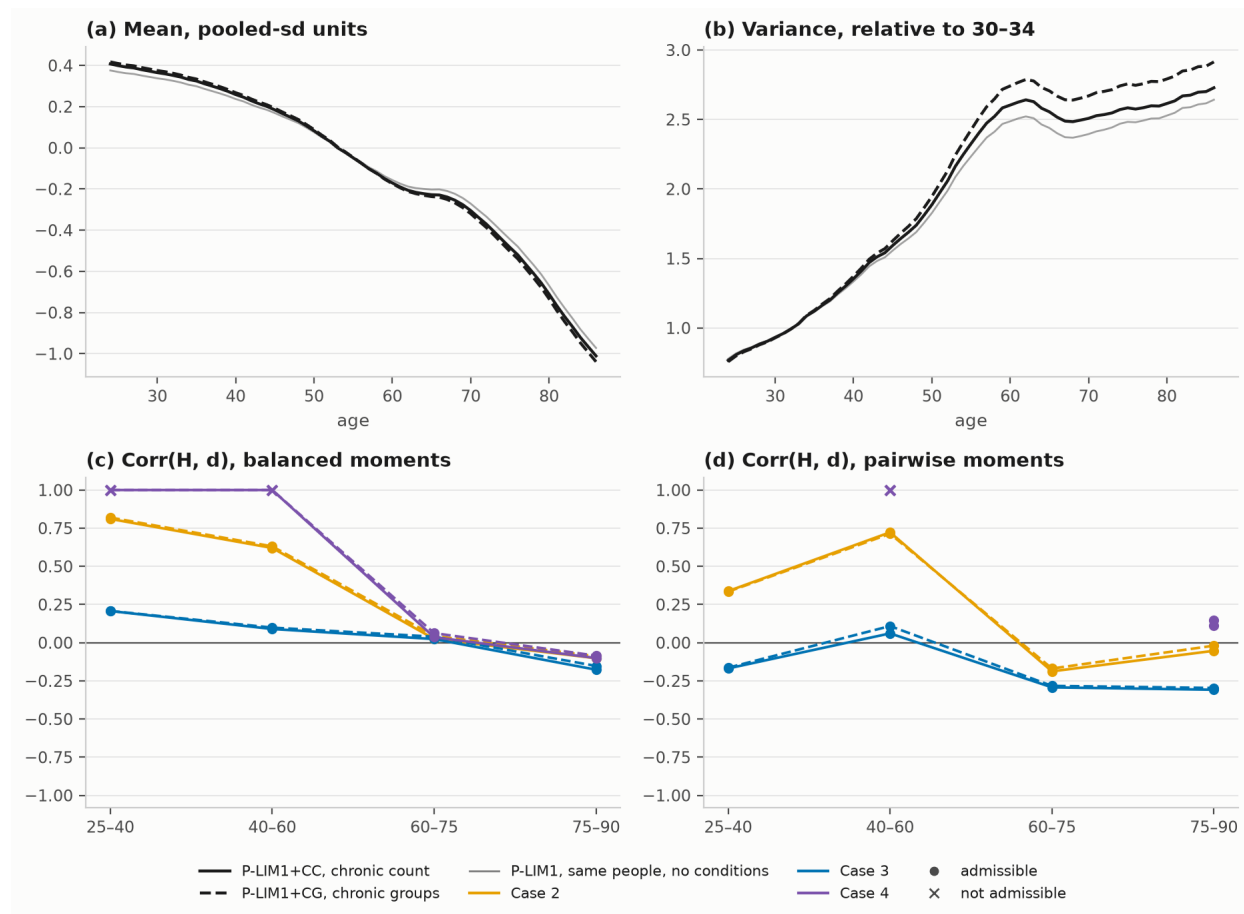


Figure 42: Exhibit 1 for P-LIM1+CC (solid) and P-LIM1+CG (dashed) on h , as Figure 18; grey lines are P-LIM1 without conditions on the same people.

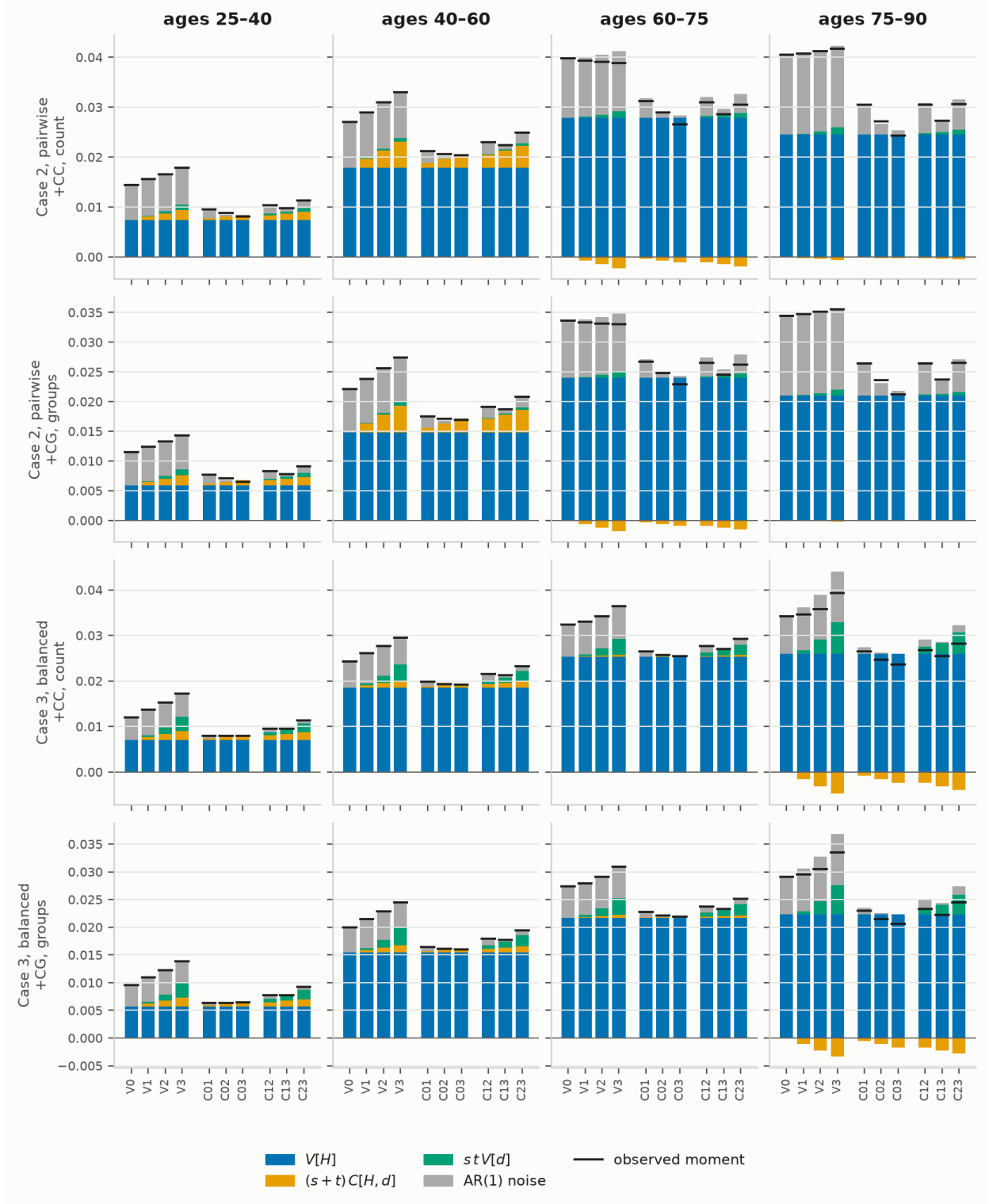


Figure 43: Identification barcharts for P-LIM1+CC and P-LIM1+CG on h , as Figure 19.

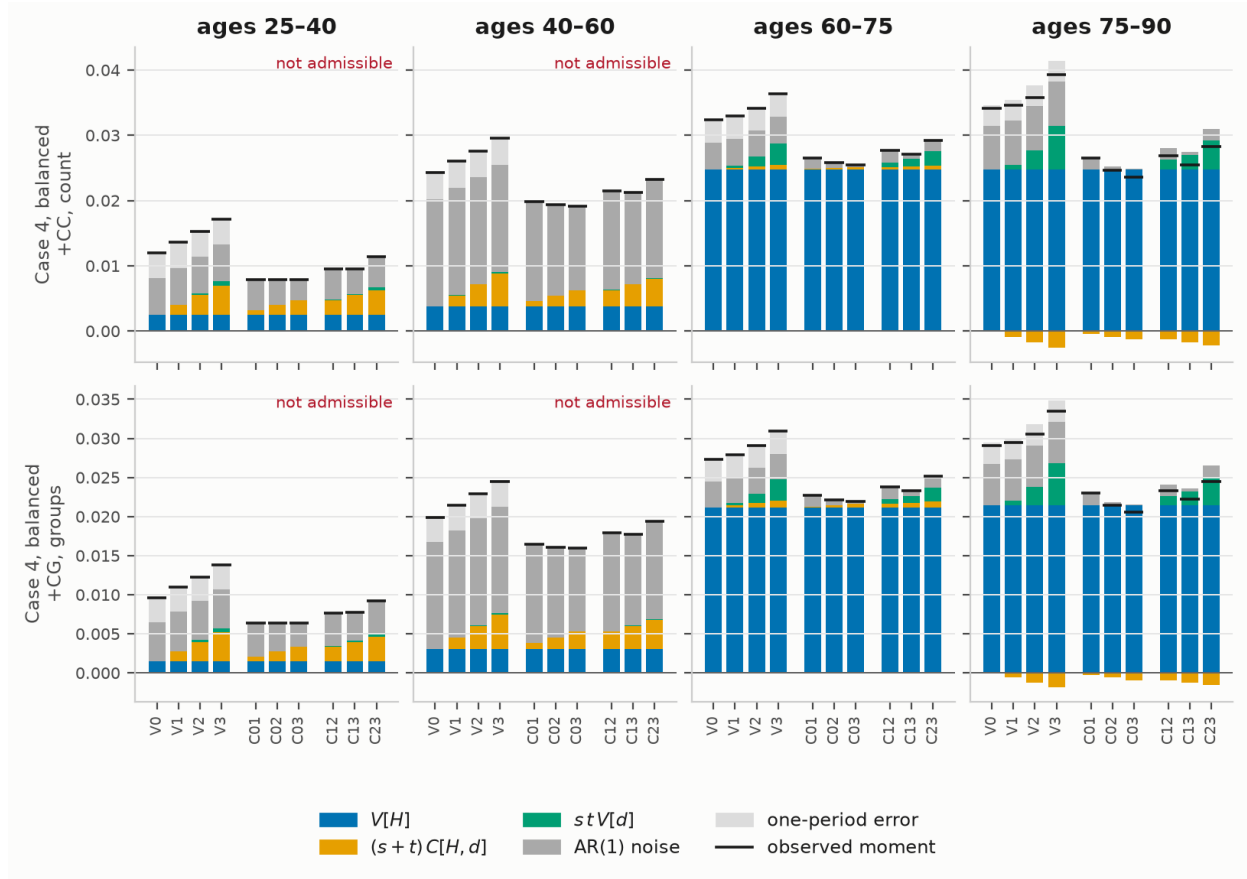


Figure 44: Case 4 barcharts for P-LIM1+CC and P-LIM1+CG on h , as Figure 20.

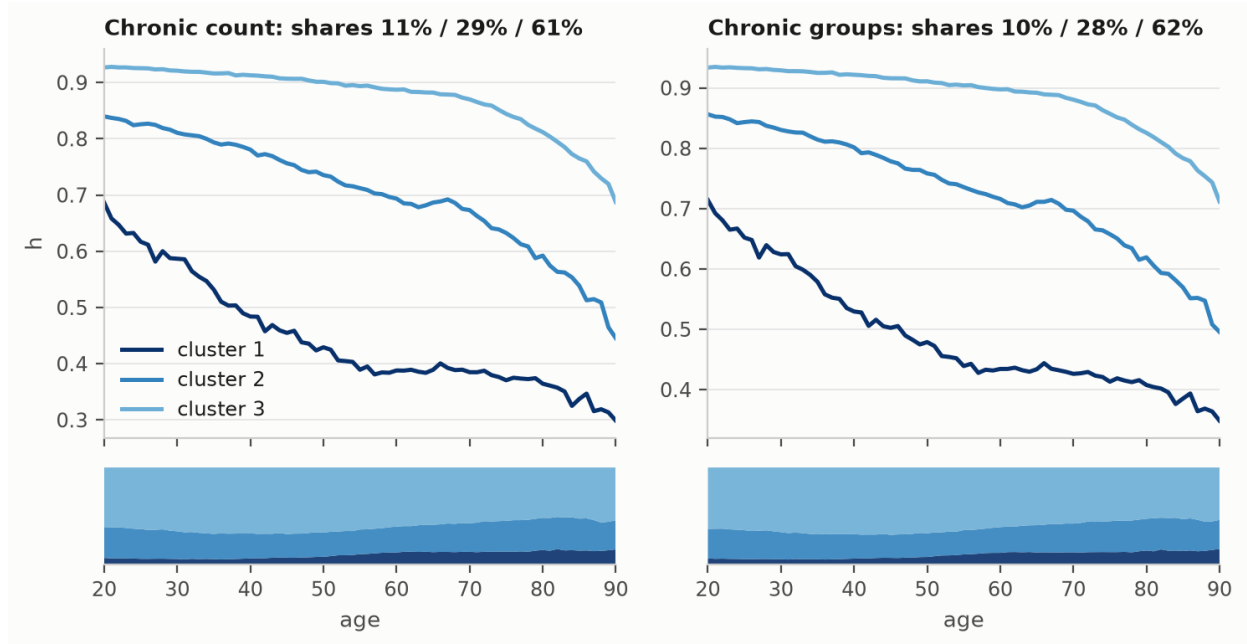


Figure 45: Partial K-means, $K = 3$, for P-LIM1+CC (left) and P-LIM1+CG (right) on h , as Figure 21.

16.5.2 P-LIM with conditions

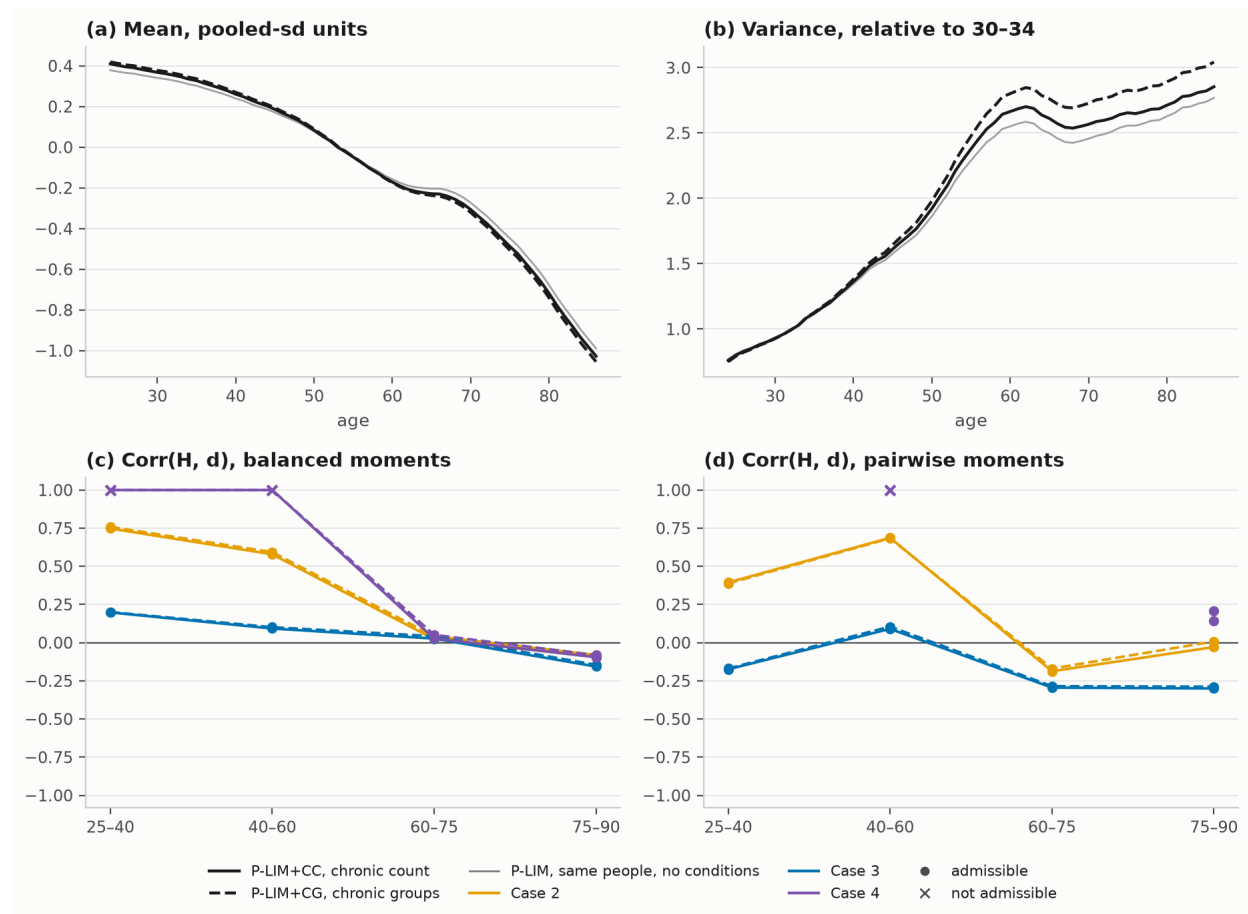


Figure 46: Exhibit 1 for P-LIM+CC (solid) and P-LIM+CG (dashed) on h , as Figure 18; grey lines are P-LIM without conditions on the same people.

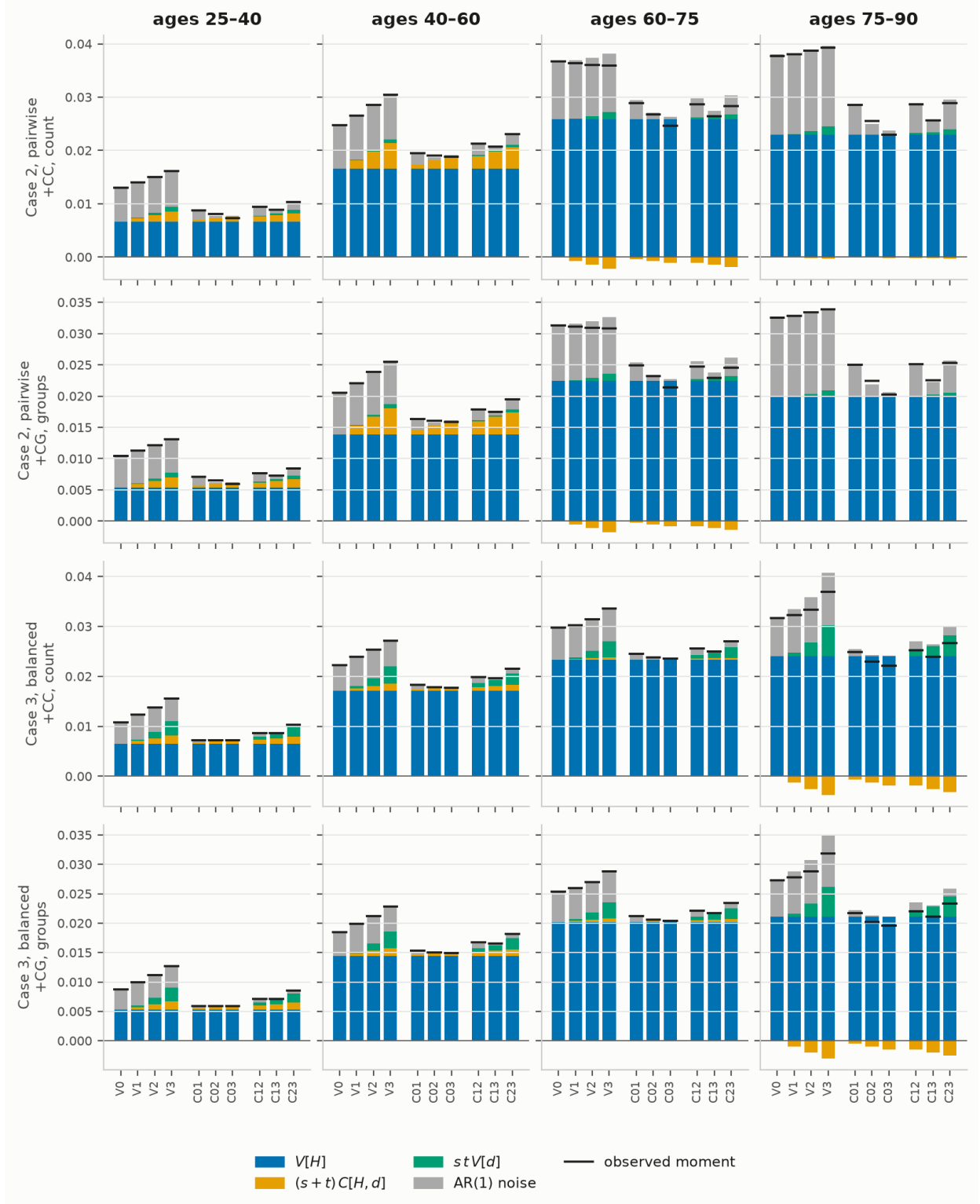


Figure 47: Identification barcharts for P-LIM+CC and P-LIM+CG on h , as Figure 19.

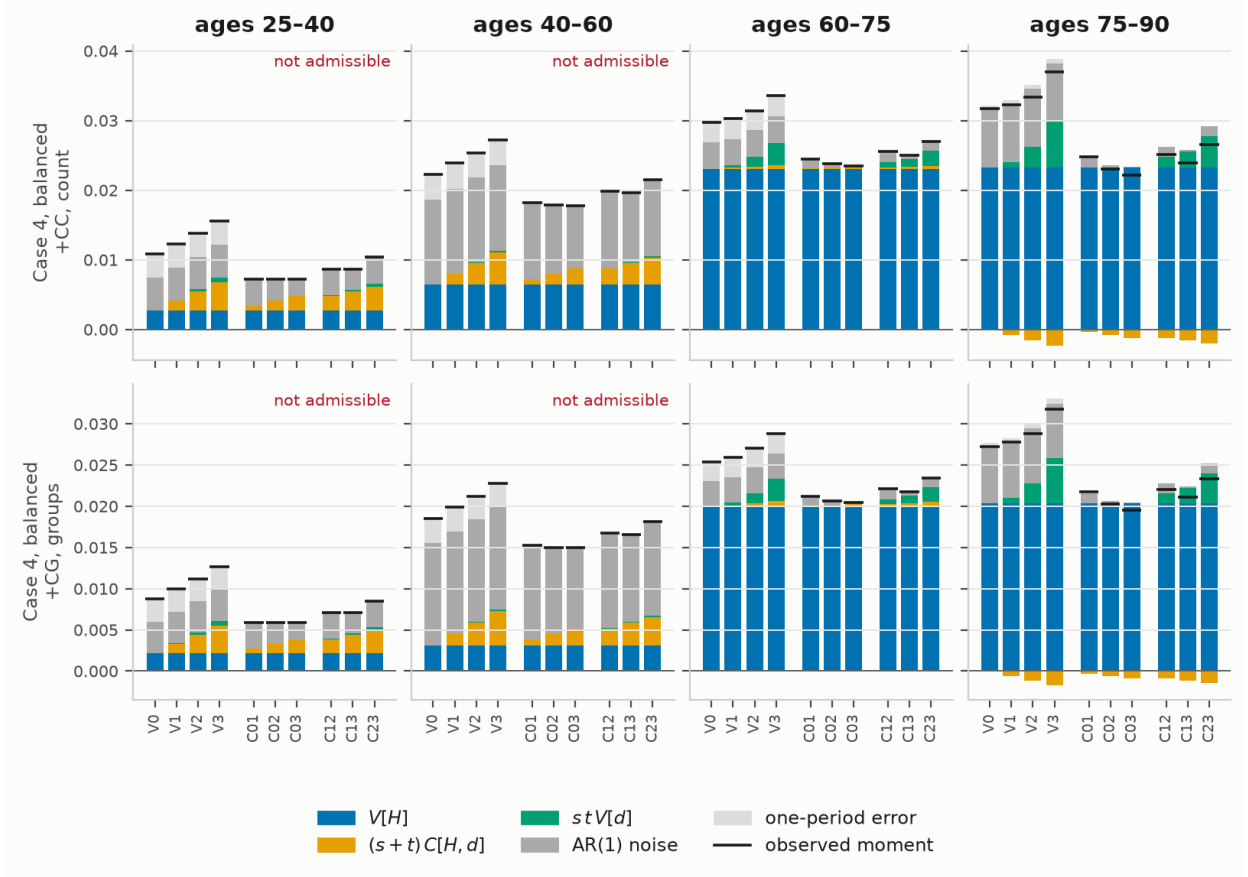


Figure 48: Case 4 barcharts for P-LIM+CC and P-LIM+CG on h , as Figure 20.

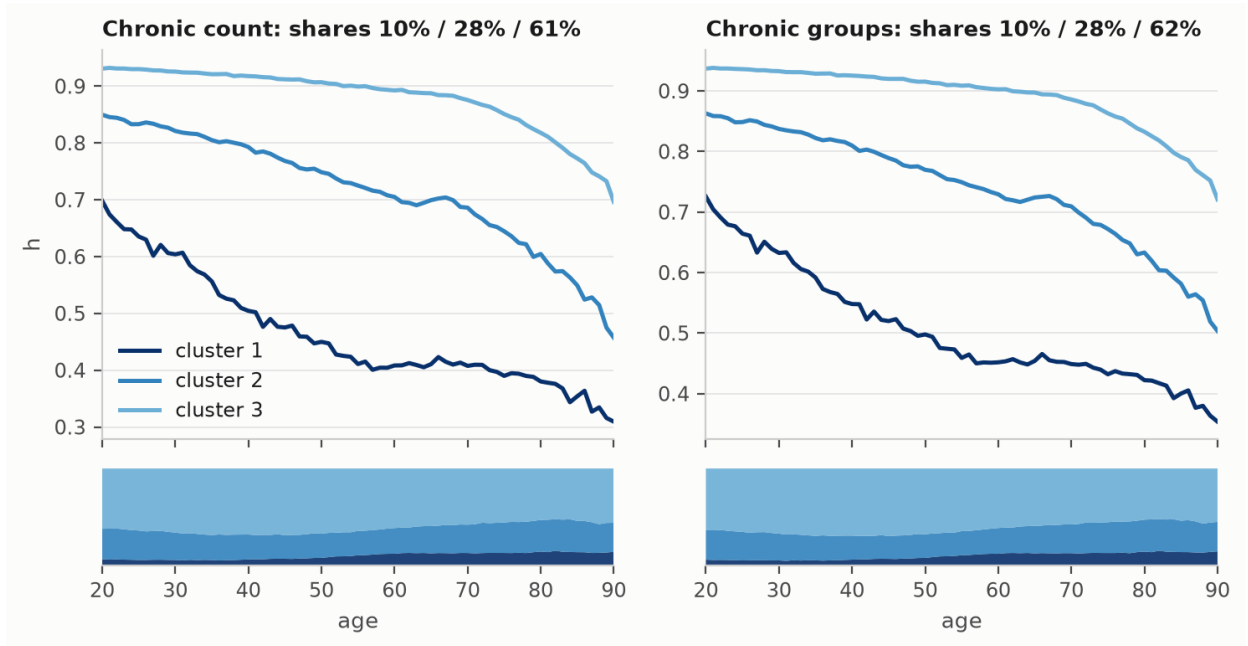


Figure 49: Partial K-means, $K = 3$, for P-LIM+CC (left) and P-LIM+CG (right) on h , as Figure 21.

16.5.3 P-LIM4 with conditions

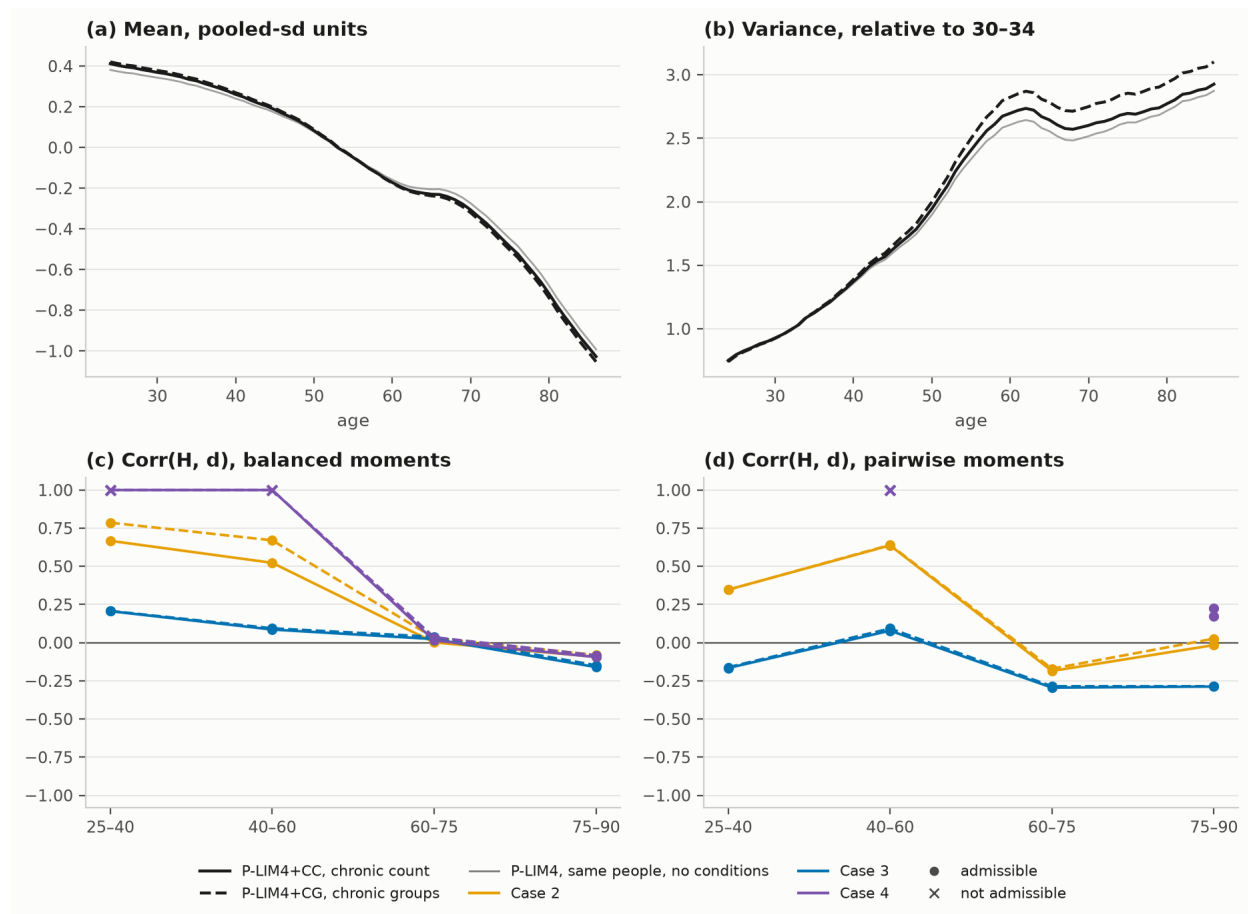


Figure 50: Exhibit 1 for P-LIM4+CC (solid) and P-LIM4+CG (dashed) on h , as Figure 18; grey lines are P-LIM4 without conditions on the same people.

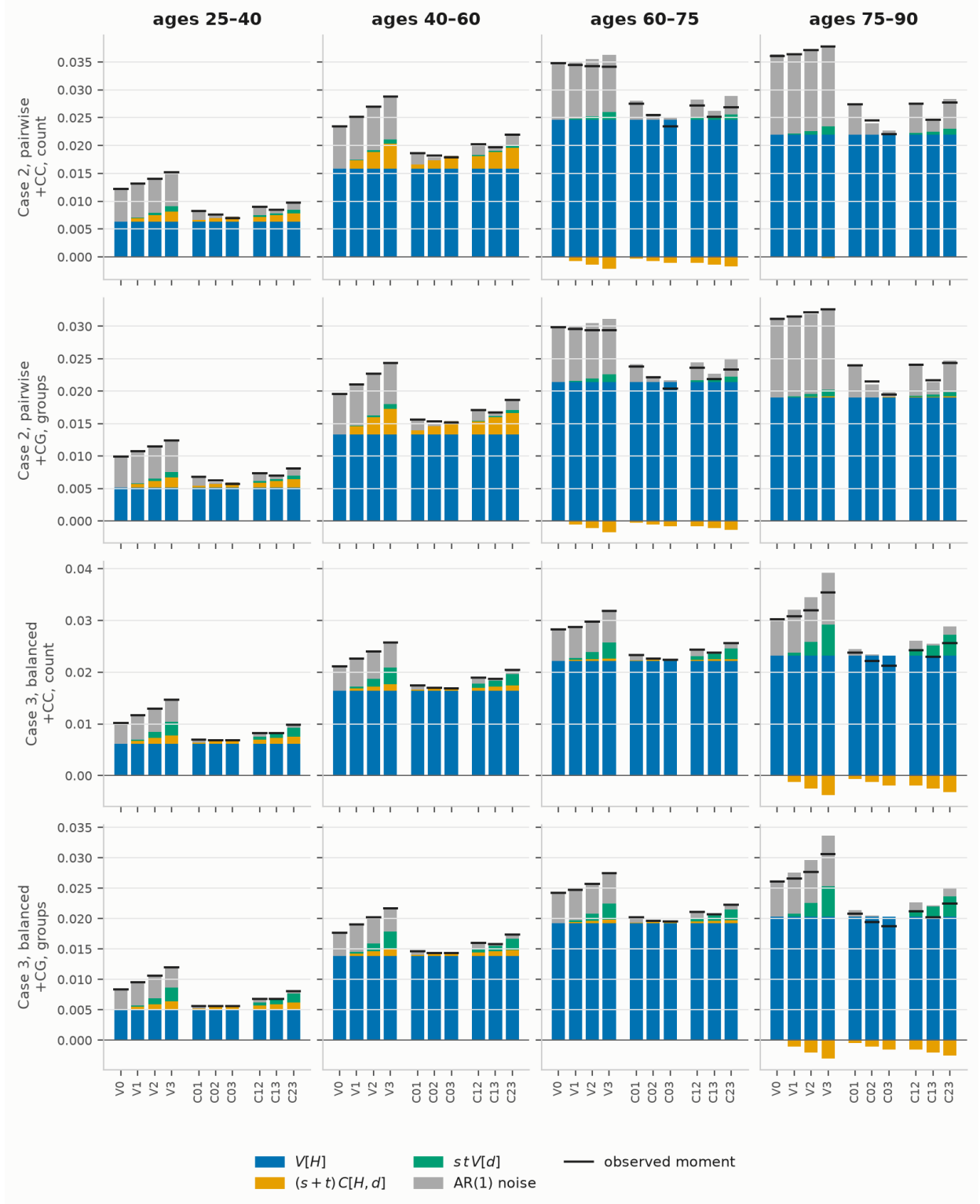


Figure 51: Identification barcharts for P-LIM4+CC and P-LIM4+CG on h , as Figure 19.

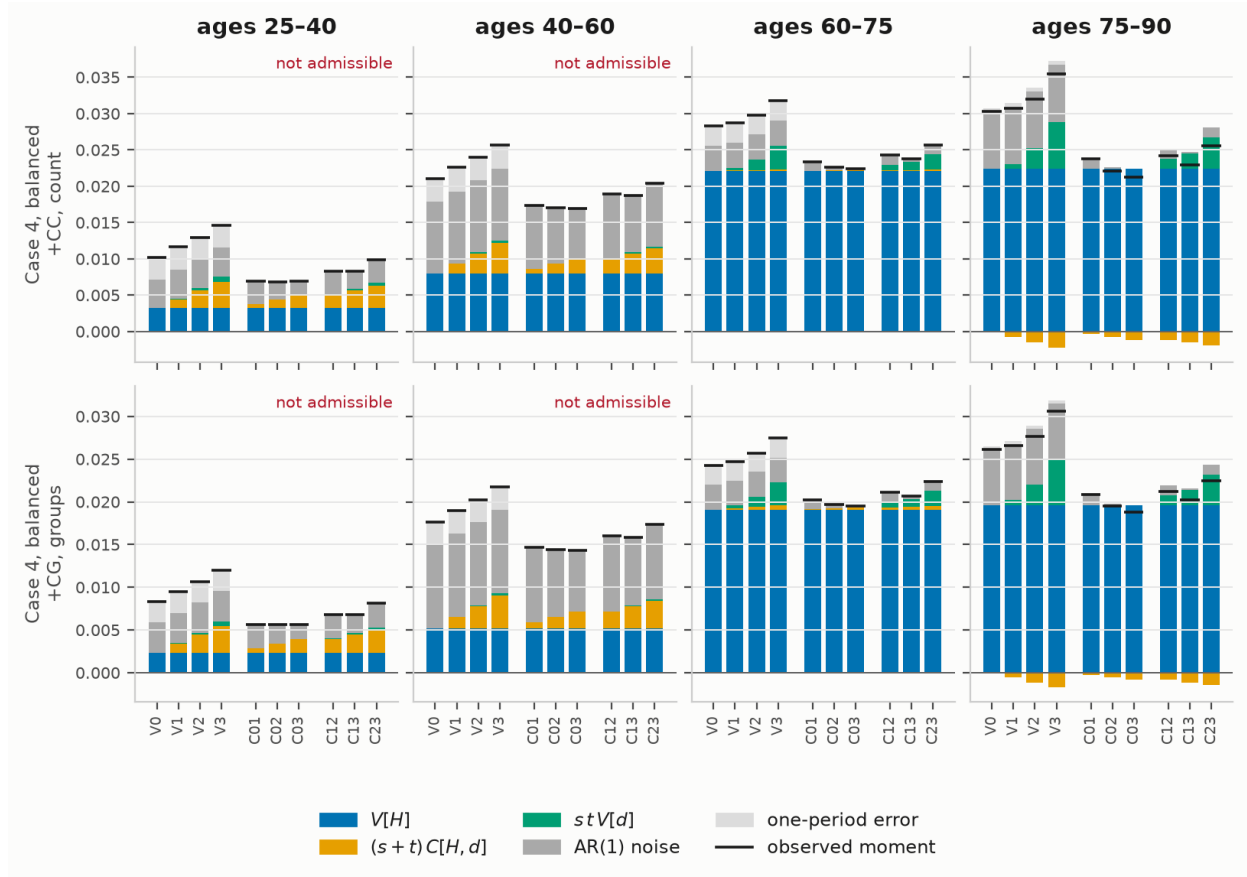


Figure 52: Case 4 barcharts for P-LIM4+CC and P-LIM4+CG on h , as Figure 20.

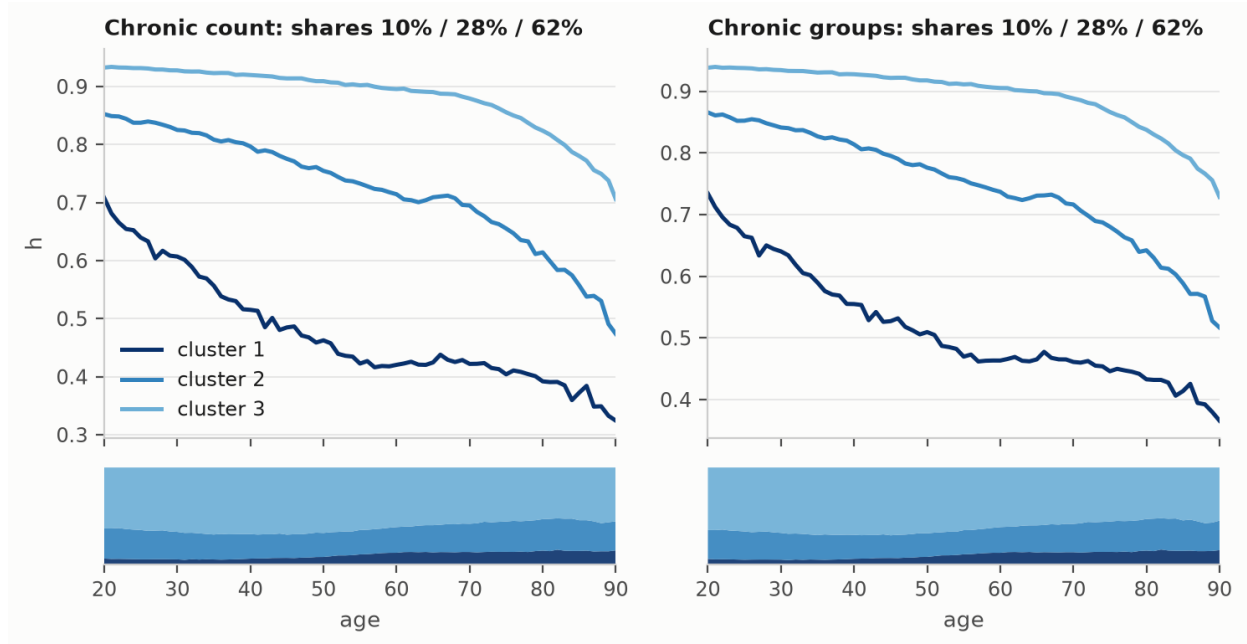


Figure 53: Partial K-means, $K = 3$, for P-LIM4+CC (left) and P-LIM4+CG (right) on h , as Figure 21.

16.5.4 P-LIM3 with conditions

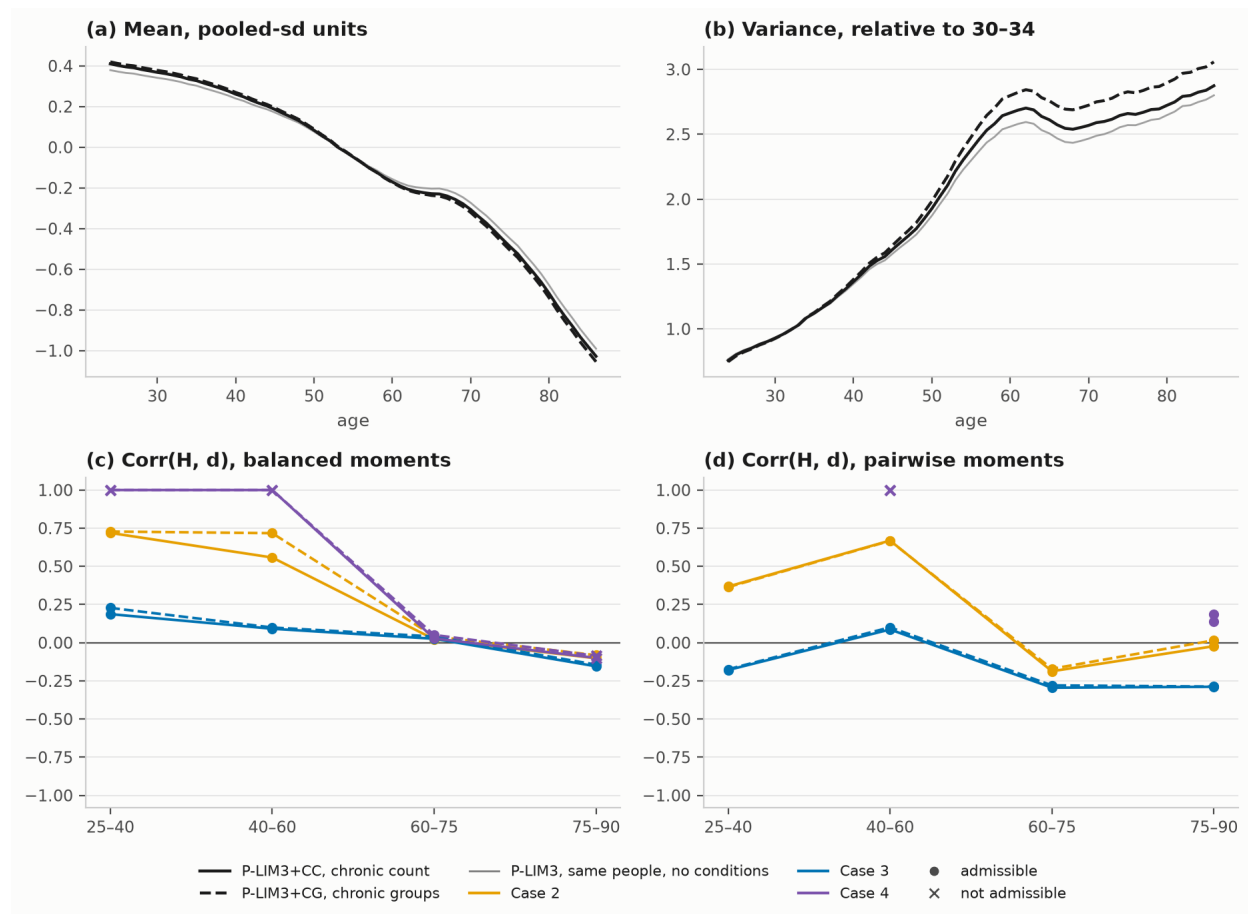


Figure 54: Exhibit 1 for P-LIM3+CC (solid) and P-LIM3+CG (dashed) on h , as Figure 18; grey lines are P-LIM3 without conditions on the same people.

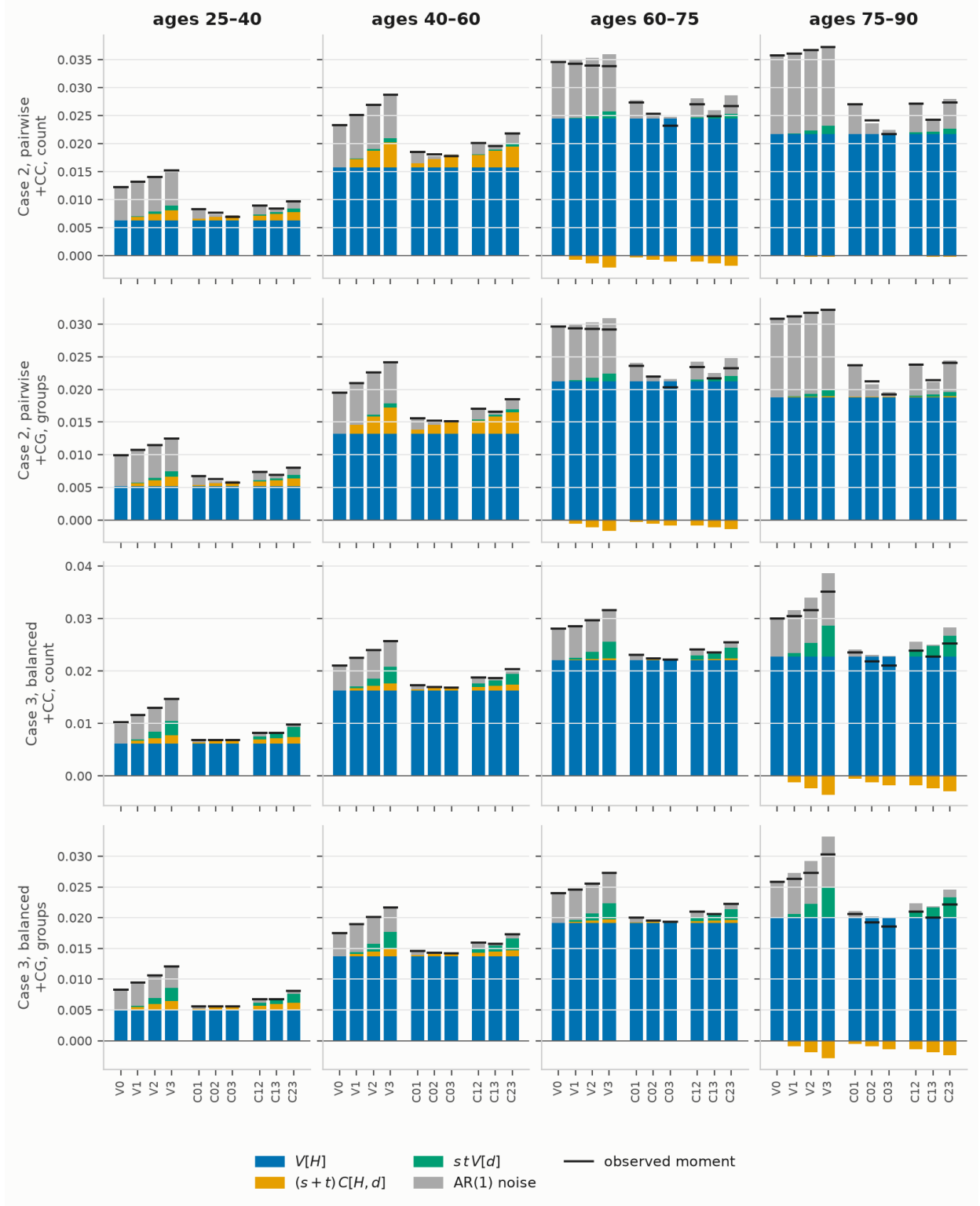


Figure 55: Identification barcharts for P-LIM3+CC and P-LIM3+CG on h , as Figure 19.

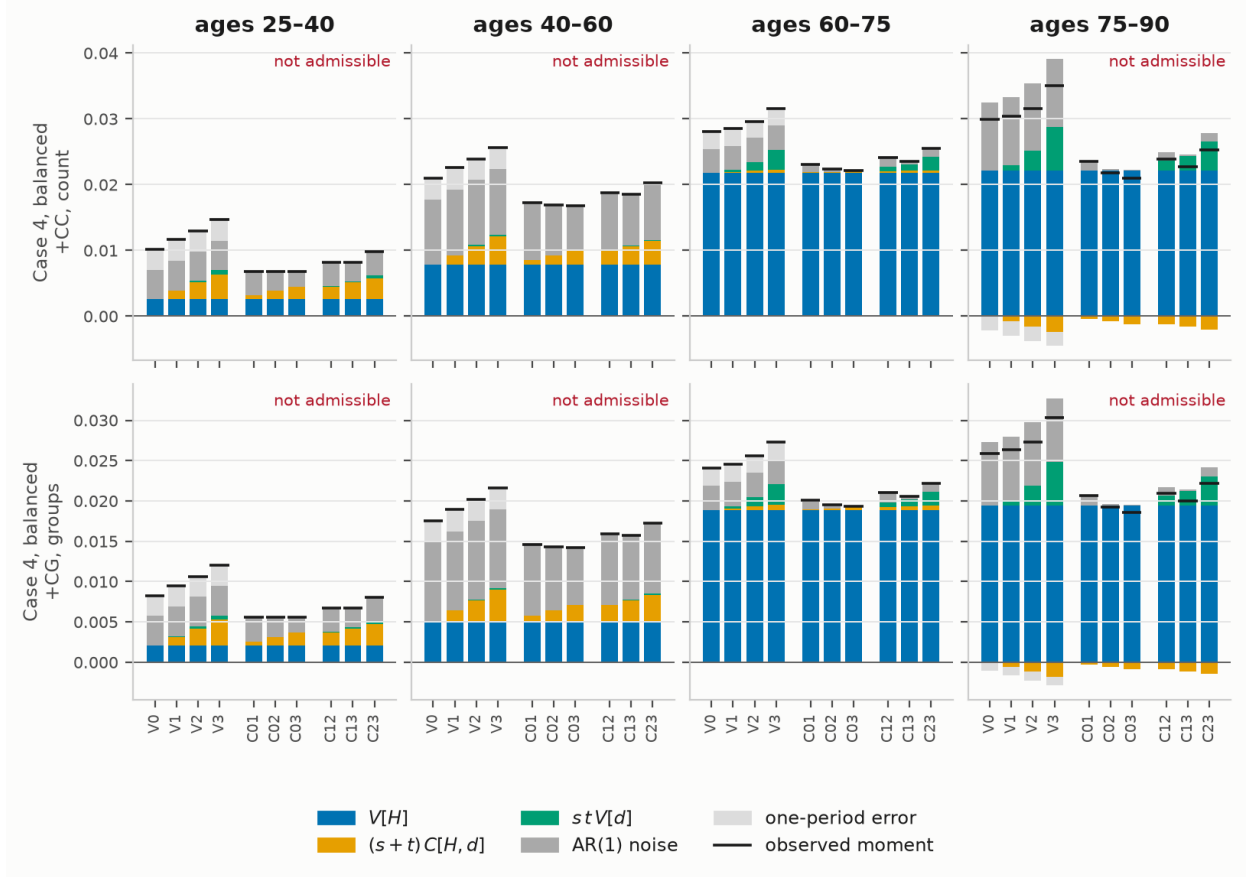


Figure 56: Case 4 barcharts for P-LIM3+CC and P-LIM3+CG on h , as Figure 20.

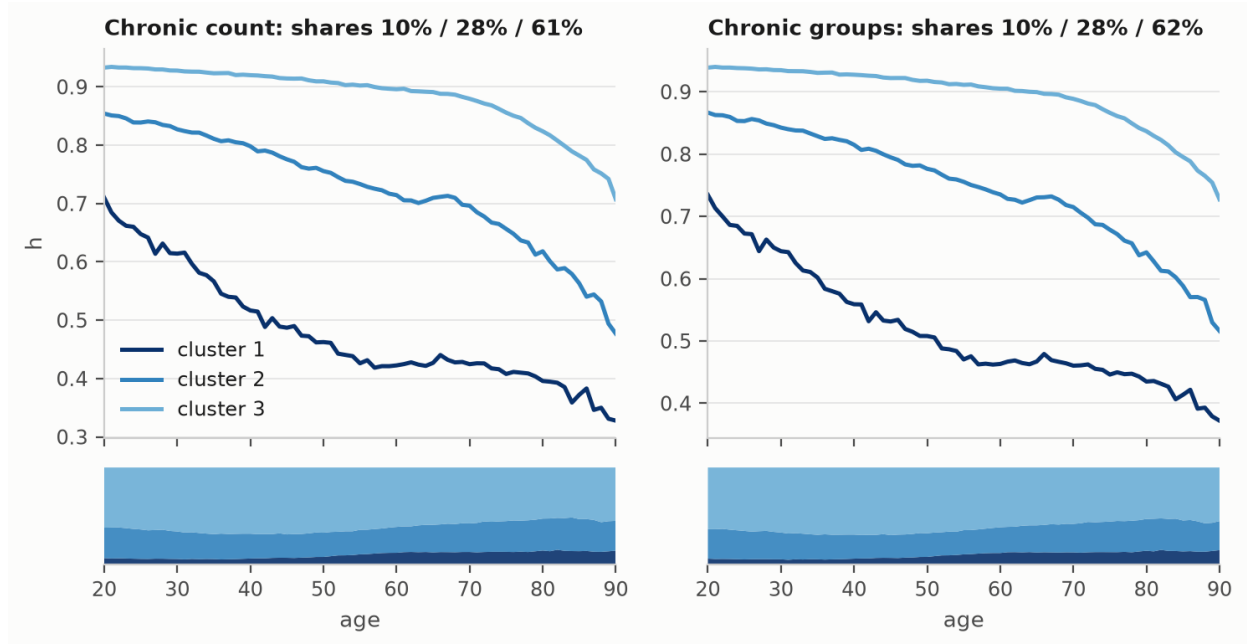


Figure 57: Partial K-means, $K = 3$, for P-LIM3+CC (left) and P-LIM3+CG (right) on h , as Figure 21.

16.5.5 P-LIM3+O with conditions

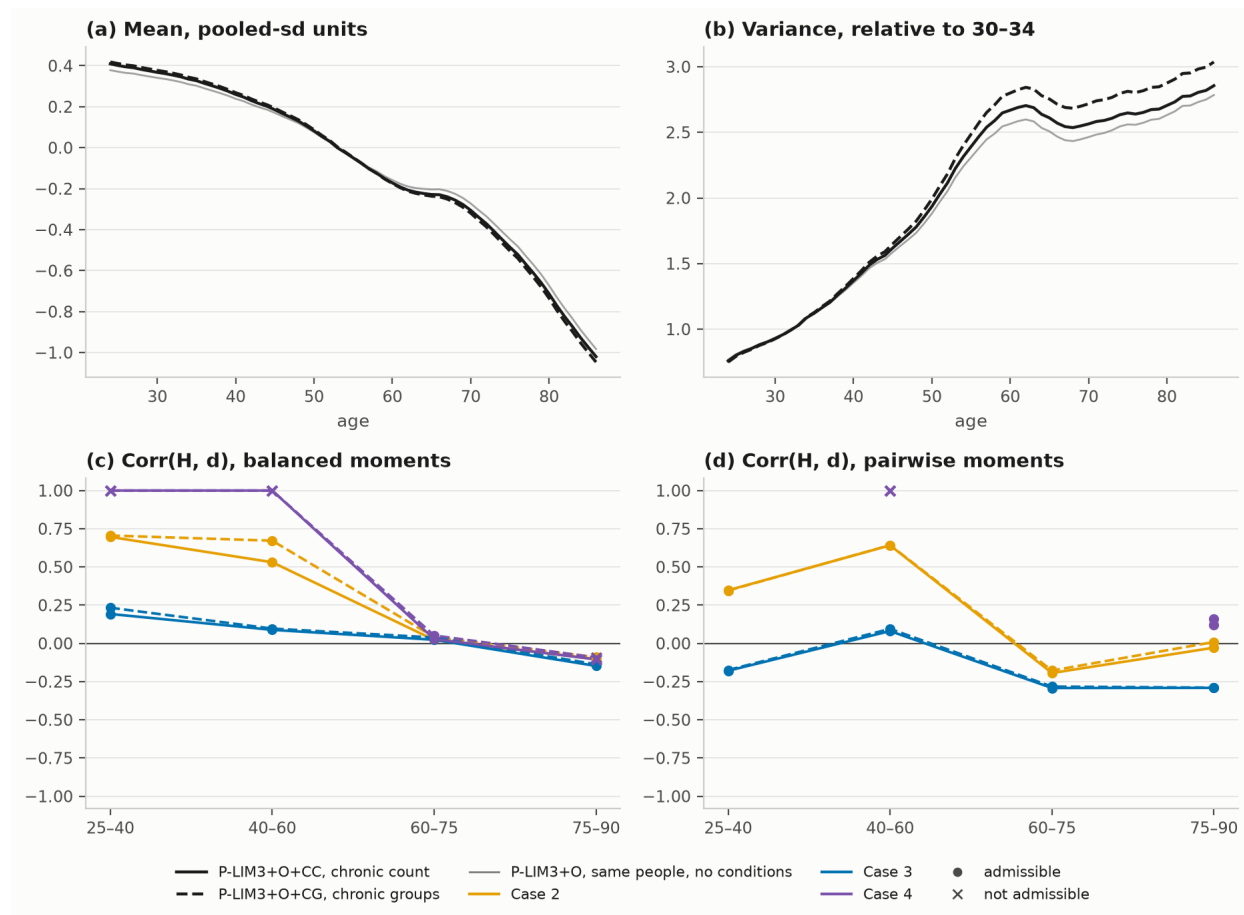


Figure 58: Exhibit 1 for P-LIM3+O+CC (solid) and P-LIM3+O+CG (dashed) on h , as Figure 18; grey lines are P-LIM3+O without conditions on the same people.

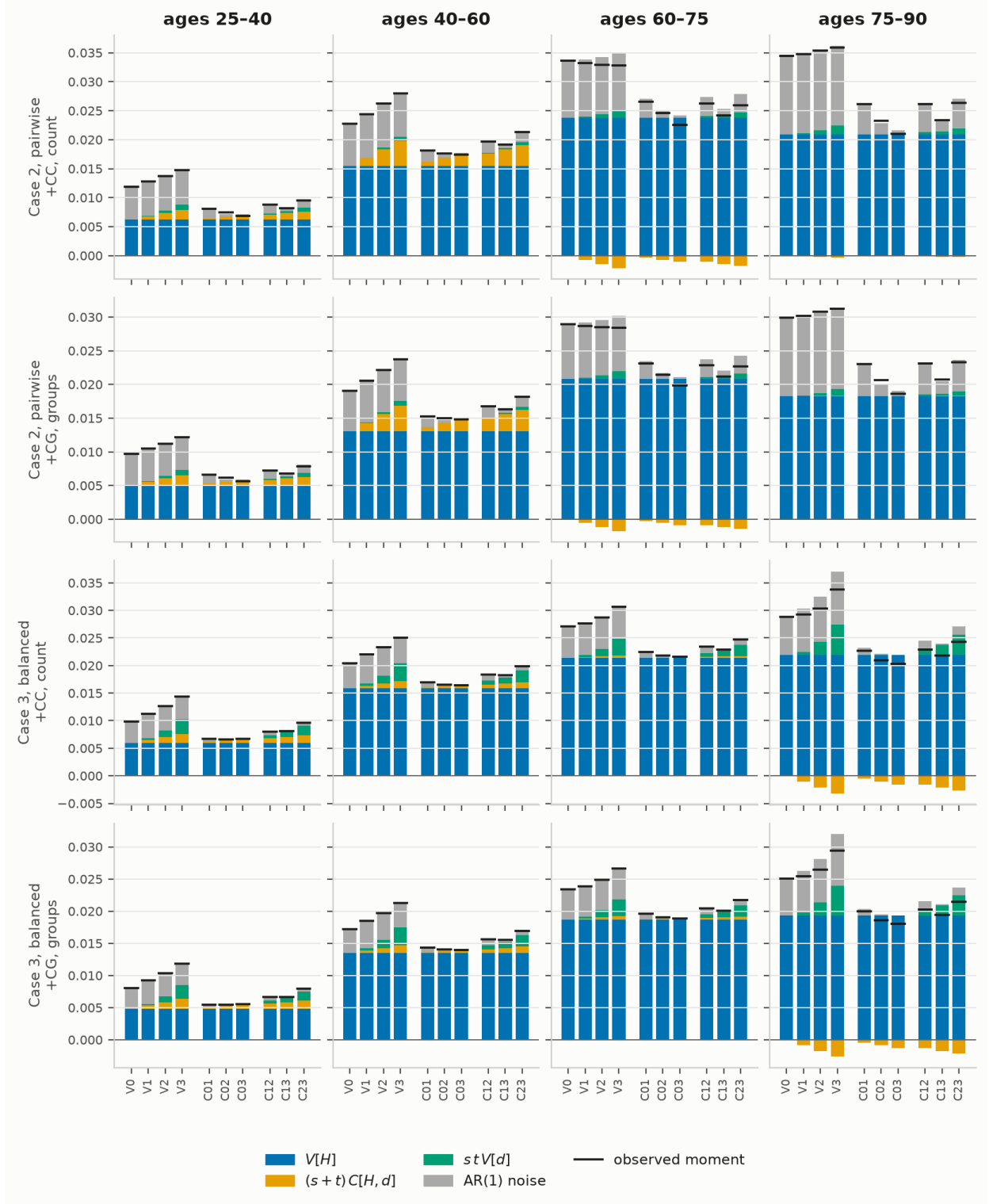


Figure 59: Identification barcharts for P-LIM3+O+CC and P-LIM3+O+CG on h , as Figure 19.

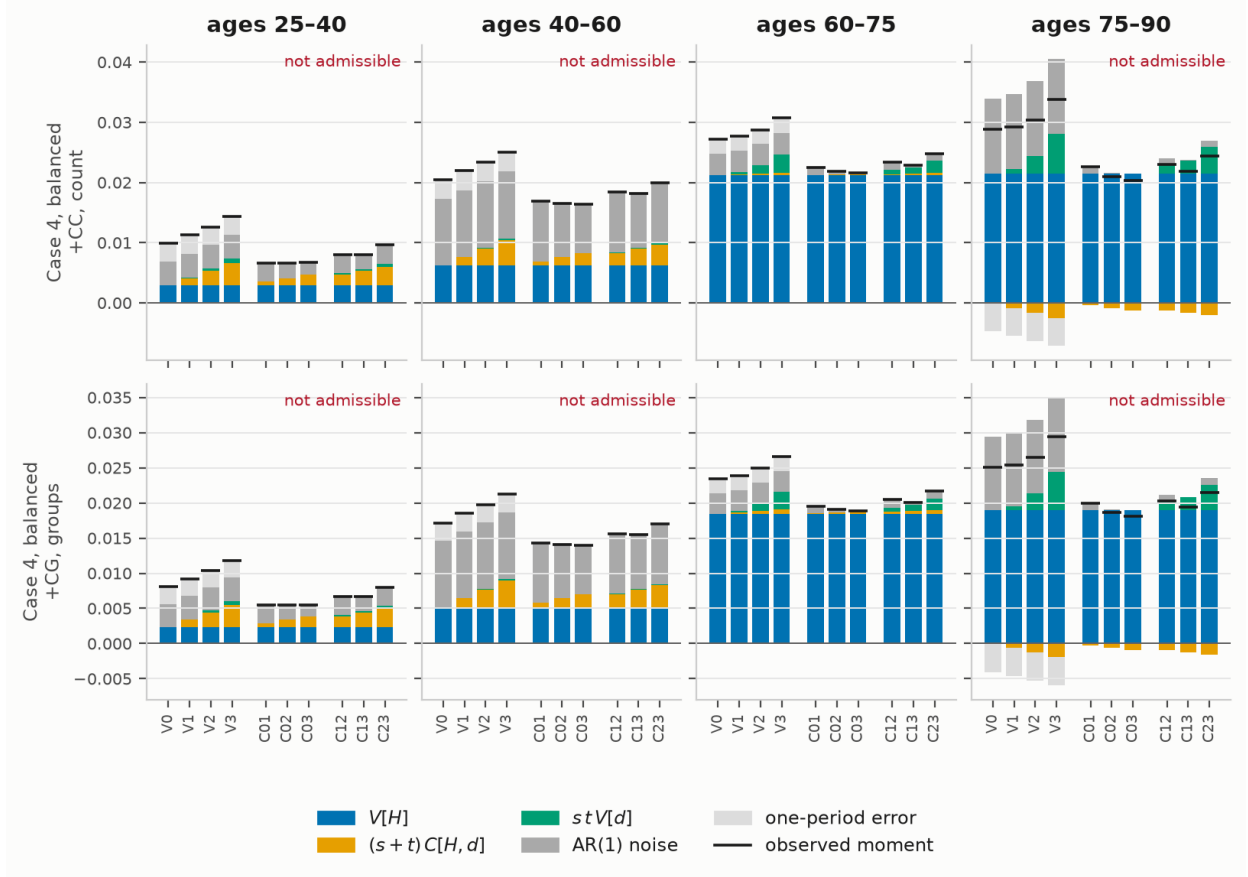


Figure 60: Case 4 barcharts for P-LIM3+O+CC and P-LIM3+O+CG on h , as Figure 20.

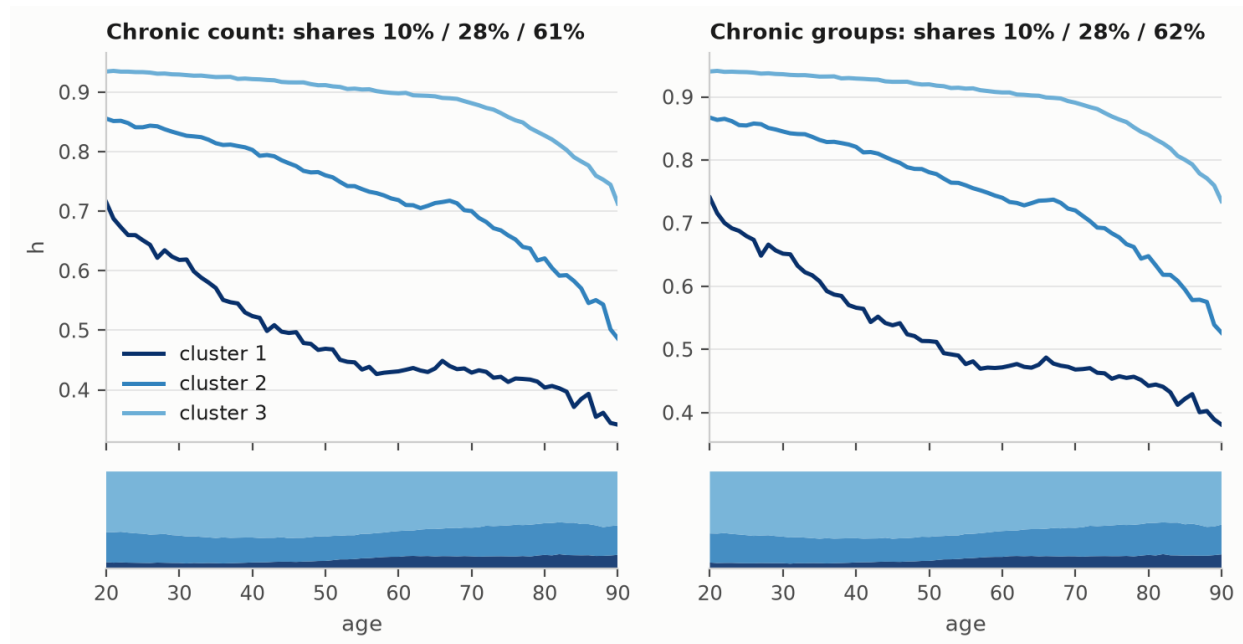


Figure 61: Partial K-means, $K = 3$, for P-LIM3+O+CC (left) and P-LIM3+O+CG (right) on h , as Figure 21.

16.6 What this suggests

- One condition count, added to any limitation bank, gives the pairwise Case 2 sign path on h , flattens the variance into the band, and leaves convexity where it was. Part of the pairwise improvement is the sample itself; the count completes it.
- The six groups carry two to three times the count's weight and do less for identification, and they take a little off convexity. On this evidence the count is the better way to add conditions.
- Either way the price is the panel: every condition bank keeps only 5% of the wave-2 entrants' person-years, and those are the longest panels the covariance surface has.

17 Recommendations

1. Physical health for lifecycle dynamics: **P-FUNC** (or the original GRM). Mental health: **MENT**. Report both; their correlation (0.457) is a finding, not an assumption.
2. **P-FULL** as the descriptive companion when condition burden should show in the score; P-REC/P-TIMED quantify its recency sensitivity. It is also the cleaner instrument: on identical people its measurement error is .418 against P-FUNC's .445 (§9). Prefer it for dynamics where losing the BHPS entrants is acceptable; P-FUNC keeps them at the cost of more noise.
3. Conditions in the Bayesian mixture as separate channels; “ever” is the right state variable there.
4. Avoid the combined score unless a single index is externally required; if it is, build the bifactor version first.

5. SF-6D: treat as a mixed physical–mental utility (its closest GRM cousin is the combined bank), never as a physical measure; licence before publishing utilities.
6. Report θ and `grmh` together when consequences matter: θ is the latent-normal unit the models want, `grmh` the near-linear-in-utilisation unit (§8); the choice of cardinalisation is substantive, not cosmetic.
7. **Estimate persistence with the state-space specification, not AR(1).** AR(1) reads the transient noise in a GRM score as fast mean reversion and attenuates persistence badly — on the same P-FULL sample, classes 2 and 3 fall from .90 and .86 to .27 and .39 — and between a quarter and a half of the year-to-year movement in these scores is measurement (§9). The state-space model also forecasts better once both models condition on a person’s history, though modestly: 0.084 nats per person (§9).
8. **Keep K=3.** A fourth class is reproducible across measures and the model can recover one in simulation, but it sits at the measurement ceiling and its defining dynamics are the ones a ceiling produces (§10).
9. **Move the multidimensional ordering constraint onto the chronic channel** before relying on the state-space multidimensional fits; on the physical channel it traps chains in an inferior mode (§12). Include mortality to describe survival by class — the worst class’s odds of dying at 55 are about seven times the healthiest’s — but not to sharpen the typology, which it leaves unchanged.

18 Reproducibility

<code>data_cleaning/03_build_measures.py</code>	SF-12/SF-6D layer; validates against the construction note (max err 0.0054, 528,485 rows, the 6.6-pt artifact, tariff checks)
<code>data_cleaning/04_build_grm.py</code>	original GRM replicated to 10^{-5} on all 498,424 person-waves
<code>data_cleaning/05_build_chronic.py</code>	chronic layer; 100% timing coverage; $\geq$ EIT count on 100.000% of rows
<code>data_cleaning/06_build_grm2.py</code>	all GRM-2 specs; Q3, EAP identity, invariance, recency weights
<code>descriptives/01--20</code>	measure panel, correlations, every figure, the ceiling/floor, convexity, class-composition, outcome-prediction and cost-validation tables
<code>clustering/run_fit.py,</code> <code>runs/overnight_queue.py</code>	the §9 batch: generic registry fits, last-two-observations holdout, per-person held-out densities
<code>clustering/run_multidim.py,</code> <code>runs/multidim_queue.py,</code> <code>multidim_channel_decomp.py</code>	the §12 batch and its per-channel held-out decomposition
<code>data_cleaning/:</code> <code>09_build_cost_proxy.py;</code> <code>descriptives/:</code> <code>15_cost_validation.py</code>	the cost proxy of <code>cost_proxy_plan.pdf</code> , stages 1–3: flat-rate and condition-weighted indices, and validation against published age, concentration and level profiles
<code>descriptives/16_cost_convexity.py,</code> <code>17_cost_age_interaction.py</code>	§14: convexity with pounds on the vertical axis, the twelve-variant costing sensitivity, and the age interaction with person-clustered standard errors
<code>descriptives/19_k4_state_space.py</code>	§10: K=3 against K=4 under the state-space specification — class trajectories and composition, the person-by-person class correspondence, persistence and signal share by class level, in-sample fit and classification certainty
<code>clustering/validation/:</code> <code>synthetic_recovery_ssm.py,</code> <code>synthetic_recovery_ssm_k4.py,</code> <code>ssm_holdout_check.py</code>	recovery checks for the state-space specification at K=3 and K=4, and the validation of its held-out generated quantities against an offline filter
<code>clustering/runs/ssm_queue.py,</code> <code>ssm_holdout_queue.py, k4_queue.py,</code> <code>multidim_ssm_queue.py</code> <code>clustering/holdout_matched.py</code>	the state-space, holdout-and-control, K=4 and multidimensional state-space batches
<code>data_cleaning/:</code> <code>07_build_limitation_banks.py;</code> <code>measures/gpcm.py;</code> <code>tests/test_gpcm.py</code>	§9's matched held-out comparison: AR(1) and AR(1) plus measurement error scored offline on the same people, rows, draws and estimands
<code>descriptives/:</code> <code>baseline_measures/</code> , <code>scripts 01–11</code>	§15: the six limitation banks under the graded-response and partial credit models; Q_3 , EAP identity, the weighted-sum check, and P-FUNC reproduced exactly; §16: the ten condition banks, and P-FULL reproduced exactly
<code>descriptives/:</code> <code>18_roster_difference.py,</code> <code>20_multidim_mortality.py</code>	the baseline criteria workbook and §15's criteria, weights, identification moments, clusters, figures and tables; scripts 01–06 reproduce the first version of the workbook exactly
	the P-FULL/P-FUNC roster diagnostic; the multidimensional state-space and mortality fits on their better-mode chains

Everything under `data/` derives from licensed UKHLS microdata and is never committed; all figures are aggregate statistics.

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